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Master Thesis

利用實驗與模擬監測體積成形機械與其產品品質的方法

Methodologies of Monitoring Bulk-Forming Machinery and Quality of its Products through Experiment and Simulation

指導教授：孫明宗 博士

Advisor: Ming-Tsung Sun, Ph.D.

Graduate Student: Gaurav Pandey

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Recommendation Letter from the Thesis Advisor

This thesis is by Gaurav Pandey (Author) of the graduate program in Department of Mechanical Engineering, entitled: Methodologies of Monitoring Bulk-Forming Machinery and Quality of its Products through Experiment and Simulation, which is written under my supervision, and I agree to propose it for examination.

Advisor 李和章 (Signature)

2019 Year 7 Month 12 Day

Chang Gung University
Thesis Oral Defense
Committee Certification

This thesis is by Gaurav Pandey of the graduate program in the Department of Mechanical Engineering,

entitled: Methodologies of Monitoring Bulk-Forming Machinery and Quality of its Products through Experiment and Simulation,

who is qualified for master degree through the verification of the committee.

Convener of the degree examination committee 廖駿偉 (Signature)

Committee members 孫明志 _____

王能治 _____

_____ _____

_____ _____

Department Chair/Program Director _____

2019 Year 7 Month 15 Day

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中文摘要

生產不同形式與尺寸的扣件需要可靠的製造機械。在本研究中，實驗數據與分析技術被用來分析這類製造機械在空車運轉或負載運轉的一般行為模式。分析的結果可被用來建立偵測製造機械與產品的異常情形之方法，從而設計出可靠的製造機械。偵測方法中的第一個步驟是在製造機械上安裝探測器，所得的訊號是一個加速規和定位系統的類比轉數位資料。加速規是安裝在主滑台上，由驅動器供應固定電流並將其加速度造成的電壓加以放大。定位系統是由安裝在主滑台上的齒條經由齒輪驅動固定在機台上的編碼器，編碼器所產生的 Gray Code 經由轉換電路變成二進位數位訊號再轉換成類比電壓。其次，將所得加速度與編碼器的電壓訊號經由運動學分析可以獲得主滑台的加速度、速度與位移的水平分量隨時間的變化關係。接著，使用加速度和位移的時間序列可產生在每一個週期內的(1)主滑台震動的均方根值、(2)主滑台震動的頻譜、以及(3)低頻震動強度隨主滑台位移的變化。速度和位移的時間序列可用來產生主滑台(1)被吸收的比動能和(2)所受水平分量的力隨位移的變化。然後，將前述所產生的曲線圖與由 Qform® 軟體所模擬的曲線圖比較，或將所有的曲線圖與正常操作下的曲線圖加以比較，可以找出不同之處，依此可預測製造機械或產品早期可能發生的問題。最後，依照本偵測方法可在最少增添設備與最低成本增加的原則下，用其他任何比較的技術，例如機器學習，來分析並找出製造機械與產品的異常並生產可靠的扣件製造機械。

關鍵字：冷鍛加工、震動分析、產品異常之診斷、機械故障之預測。

English Abstract

The manufacturing of fasteners with diverse shapes and sizes entitled with diverse specification requires credible and reliable bulk-forming machinery. In this research, both experimental data and simulation techniques are used to analyze the general behavior of the machinery that operates with or without fabricating parts. The results of the analysis are used to establish the monitoring methods that detect anomalies in the machinery and the products for building reliable and credible machineries. The methodology is firstly to equip the machinery with sensors. The data acquired are the A/D-converted signals from an accelerometer and a positioning mechanism, which consists of a set of rack and pinion and an optical encoder. Secondly, with the help of kinematics and the positioning signals, the voltage signals of acceleration were processed to obtain the horizontal components of acceleration, velocity, and displacement of the sliding block mechanism (SBM) with respect to time. Thirdly, the time series of acceleration and displacement were used to generate (1) vibration RMS value, (2) vibration spectrum, and (3) the power of low-frequency vibration with respect to displacement for each cycle. The time series of velocity and displacement were used to produce (1) the specific kinetic energy absorption of the SBM and (2) the horizontal force exerted by the SBM with respect to displacement. Fourthly, the resultant graphs and curves were compared with those resulted from simulation with Qform® software or with each other of normal and abnormal situations. The last step of the methodology is to use all kinds of comparative analysis techniques, such as machine learning, for the diagnosis or prognosis of the anomalies of the machinery or the manufactured products with the minimized additional equipment's and costs.

Keywords: Cold forging, Vibration analysis, Diagnosis of product anomalies, Prognosis of machinery failure.

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1. Introduction

1.1 Research background

This research is conceived with the fourth industrial revolution (IR) in mind, which have been updating from the past and is at a much higher pace at present. The growth in industries, mainly manufacturing sector, can be observed from the scenarios of the power source and production methods. The use of water and steam power characterized the first IR while the use of electricity and the invention of mass production in assembly lines unveiled the second IR. The development of computers gave birth to the third IR in which fully automated machineries and robots were widely used for mass productions. As for the fourth IR, all things are linked by networks through which they can talk to one another and make decisions. The schematic diagram in **Figure 1-1**[1] chronologically describes the history of the IRs.

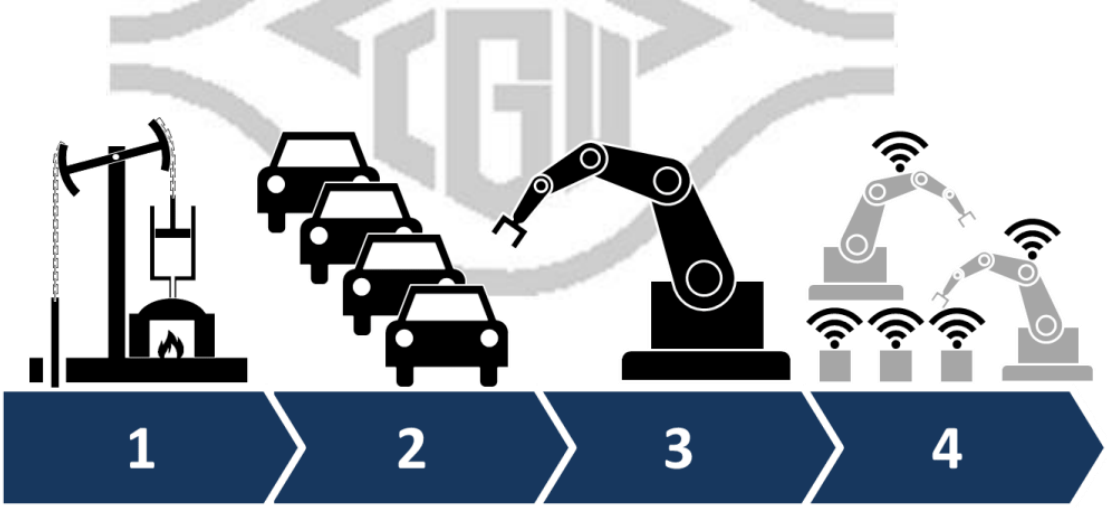


Figure 1-1 Industrial revolutions [1].

We are in the process of the fourth IR known as *smart manufacturing* in the United States, whereas Industry 4.0 in European countries and other competing industrial nations, notably Japan and Korea. Industry 4.0 or smart manufacturing comprises a paradigm shift from automated manufacturing toward an intelligent manufacturing concept. Every industrial revolution till now have led a decrement

in the involvement of the manpower and have tried to increase the production by enhancing the machinery and innovative methods for increasing productivity.

The driving objective of the Industry 4.0, activities in short can be called as **Cyber-Physical Systems** (CPS). It consists of physical entities like machineries and vehicles that are equipped with technologies such as radio-frequency identifying devices (RFID), sensors, microprocessors, telematics or complete embedded systems. They are characterized by their ability to collect data with the help of installed sensors positioned at various location and give feedback signal, which could be taken as a sign of precaution before failure. The prevention indication can be in the form of alarm or siren.

There are various types of industries having diverse range of products and needs, hence approaches of equipping CPS systems varies with their needs. Among various industry types, this research focuses on industrial research related to the fasteners manufacturing machinery. The development of fasteners and precision machineries has been one of the most developed and outraged industrial sectors in Taiwan and is at an accelerating pace with the time. Taiwan ranks among top five world's fastener suppliers, accounting for 13% of the global production and among them around 44% of firms in the industry are located in the area near Kaohsiung, southern region of Taiwan [2]. Due to Taiwan's such high fastener productivity it is necessary to keep updated with the developing technologies. The research concern of this thesis is an approach to develop a monitoring system and facilitate an industrial 4.0 environment for a fastener manufacturing machinery industry.

All types of fastener machineries being used by the industries making fasteners have a very common operating crank slider mechanism. The fastener machinery consists of motor, which drives the eccentric crankshaft connected by the connecting rod, which is guided by the main sliding table attached to the die

to make periodic reciprocating motion on the guide rail. The process of forming different parts may differ but mechanism by which the machines are operating remains the same and hence, the method of monitoring remains same for almost all the machineries with few exceptions indeed. This research focuses on developing methods to build a monitoring system to monitor the main factors affecting product quality and strictly make changes before the losses occurs.

In general practice many parts manufactured do not pass the quality test, the reason being improper forging. In order to understand the behavior and overcome the error before failure, there is an instant need of system similar to CPS. The system like CPS would help save lots of money, time and manpower with its attributes of feedback data and the cloud storage data in long term perspective. Taking into consideration all revolutionized techniques my motivation established for the technology is discussed in next section.

1.2 Research motivation

Globally, the manufacturing intelligence has become a crucial topic for researchers and industries. The monitoring systems for machineries have become the choice of manufactures not only to increase the quality of production but also to reach up the expectation of the customers. By building, a technology like monitoring system is considered “state of the art” for the company giving its competitors ignite to come to an era of 4th revolution, where the saturated growth in manufacturing research could be taken to the next level.

This research is motivated by smart manufacturing methods and revolutionized industrial environment practices. The research focuses on building a monitoring system to monitor the forging parts, along with monitoring the machinery that makes those parts both in the long and short run. It can be facilitated by creating an environment similar to CPS, where physical system like the machineries named as **CBF-64S**, **CBP-65S** (names of machinery) used for this research are equipped with the sensing entities like rotary encoder,

accelerometer and smart meter as a device to analyze and detect the dynamic vibrations. The company concerned with this research their R&D department has done several attempts in the past, to make a condition-based monitoring system among all the two most significant methods adopted by them are described below:

1. Embedding **Piezo sensor** in the rear of the mold, by which pressure can be measured directly in the forging process across varying forging stations. The pressure obtained then can be used in estimation of load variation across different forging stations.
2. Installation of the **Vibration sensor** on the base of machinery to measure the vibration signal during machining mainly to note the variation of frequencies at different conditions.

The above two monitoring methods are good for monitoring, but their drawbacks are more comparative to the advantageous. For illustration, the piezo sensor being able to directly measure the load when equipped at each sub-station but has very expensive setup installation and due to its poor design setup cannot be considered a well-planned system for the company. Whereas, in the case of vibrational sensor setup with simple installation, low price, and low maintenance has the drawback of insensitive detection of actual vibration and requires a high range amplifier to detect the vibration reaching the base.

Along with above failures in ideas and setup described, there has been progressive research done on the setup. The setup described in this paper, uses an accelerometer installed on the machine paired with different types encoding sensors to collect the data. The setup objective is to analyze the dynamic vibrations, caused due the motion of the crank and frictional wear inside the mold cavity. There have been various trials done on sensor selection and location of installation for temporary sensor installation. The temporary setup has been

accustomed on the machinery for obtaining the desired data without disturbing any of the components of the machinery.

This paper uses a better sensor installation comparatively as discussed above leading to low cost and easy installation. With ongoing monitoring data through sensors, an engineer could get an experience of machinery fault detection with help of the periodically obtained data. My next section will deal with the objectives to setup a monitoring system for machinery.

1.3 Research objectives

The main objective is to overcome factors causing error and notify the operator before the failure with help of alarm or other emergency signals. According to the industrial working experts the main causes of failure are - The changes in the size of the forging die stations, operation of the machine as a cold or warm run and the internal mechanism of machinery causing dynamic vibration issues about which there is not much evidence yet found.

The die cavity enlarges due to wears by long-term impact or friction between the workpiece and the die. In the meantime, the size of the punch decreases also resulting from wears. The size of the screw or the nut produced is incorrect and considered wastes. The fault like this goes unnoticed by the company, which lead to increased production costs and loss in productivity until the error is solved. The above dealt problem is physical but other internal phenomenon like dynamic vibration caused by the change in bearing clearance, which may affect very little in the short run but affects the machinery and the formed parts greatly in a long run scenario. The other natural phenomena like deterioration of lubricants due to high temperature, mechanical wear and tear damage can hinder the production process and affect the quality of the product in a long run.

With a perspective to above problems, the best contribution could be a technology that helps to reduce errors and notify errors beforehand. Methods or

technology facilitating the above discussed problem can prevent the loss of time and money. Hence, my effort and objective through this research would be a contribution in manufacturing sector by defining a methodology for building a monitoring system.

The scenarios, which are causing loss of capital and productivity in manufacturing industry, can be improved by installing highly sensitive sensors, data acquisition systems (DAS) and computing devices. The indication of the warning can be in form alarm easily recognized as a form of signal of danger before breakdown or degradation of any element of the machinery.

Numerical simulation like FEM and FDM plays a vital role in establishing methods with various intrinsic conditions that could be a tough task to analyze experimentally. In this research both real data from machinery and numerical simulation has been considered side by side for better predictions and estimations of methods to build a correct methodology required for monitoring system. With an objective to make a device like this requires a plan and well-developed methods which can help to get desired results. Hence the next chapter will focus and discuss on methodology that would be followed in order to make a setup required for monitoring.

2. Methodology

The follow up guide in process to achieve or establish the methods for monitoring the bulk forming machinery with the motives as discussed above is shown with the chart description in Figure 2-1.

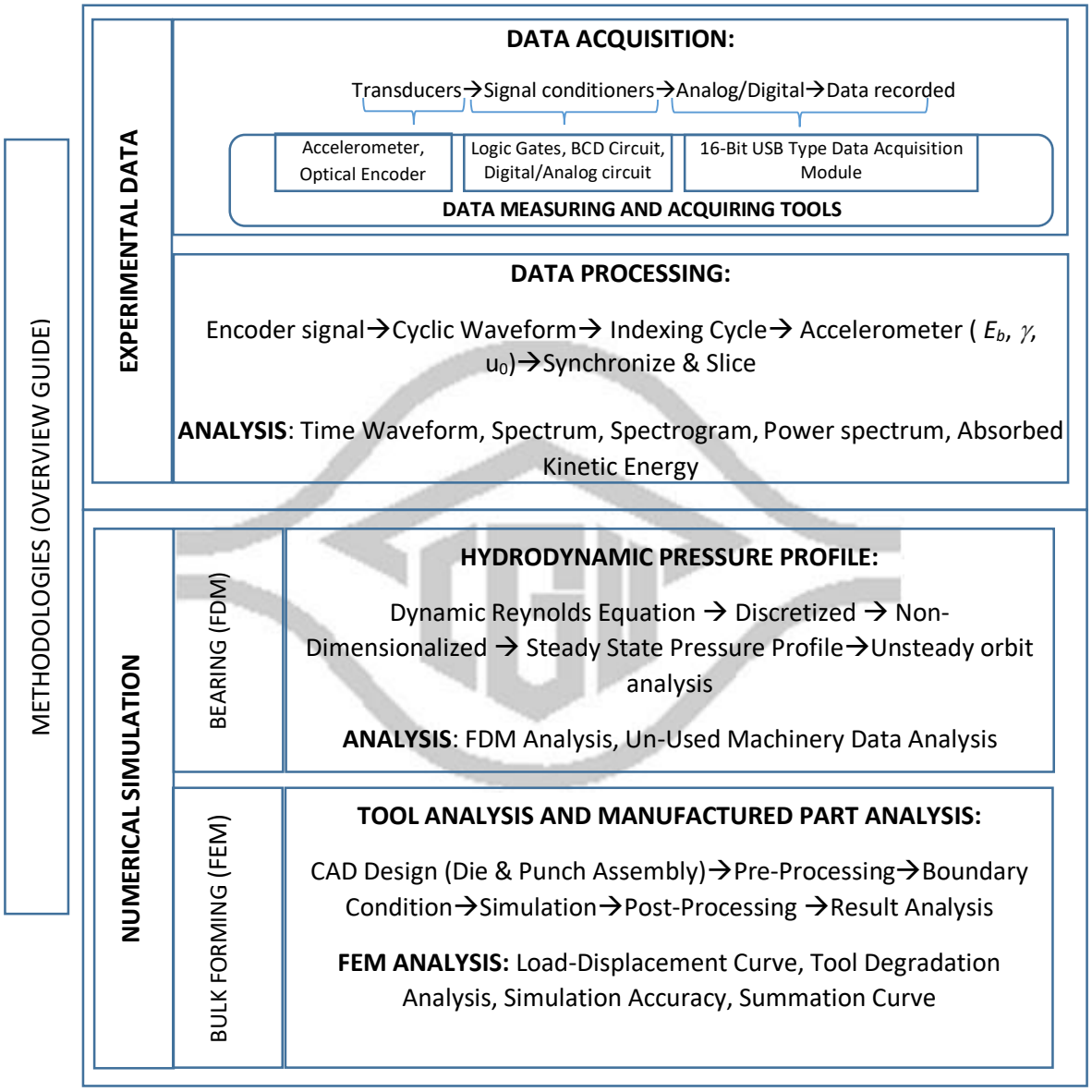


Figure 2-1 Methodology chart.

2.1 Theoretical and simulation modeling

The theoretical modeling consists of all the terminologies and parameters required in order to understand the machinery for which the monitoring system is developed. The study consists of kinematics of sliding block in relation with

dynamic working of machinery, analysis of bulk forming manufacturing significant to cold forming process and study of its oscillations simulated by FDM modeling of Reynolds dynamic equation for the hydrodynamic bearing located at eccentric main shaft. The machineries used in this paper for acquiring data has names assigned by the company according to their types such as CBF-64S and CBP-65S along with CBF-83S for which only the engineering drawing of multi-station parts was obtained as form of data which is used for simulation along with others machineries. The machine actual view along with its function in detail is given with the help of Figure 2-2 taken from the company's catalog [3].



機械型番 M/C TYPE		CBF-64S	CBP-65S
沖程數(段) NO. OF FORGING STATION 圧造段数		4	5
壓造力 FORGING LOAD 圧造力	Kg	40,000	40,000
最大剪斷直徑 MAX. CUT-OFF DIA. 最大切断径	mm	ø8	ø8
最大剪斷長度 MAX. CUT-OFF LENGTH. 最大切断長さ	mm	95	95
最高速度 MAX. OUT PUT. (PCS/MIN) 最高生産速度	PCS/ MIN.	240	200
沖具通出衝程 P.K.O. STROKE PKOストローク	mm	12	15
打模通出衝程 K.O. STROKE KOストローク	mm	80	80
主馬達 MAIN MOTOR. メインモータ	HP	20	20
剪模直徑 CUT-OFF DIE DIA. クイル直径	mm	ø30x45L	ø30x45L
沖具直徑 PUNCH DIA. パンチ直径	mm	ø40x90L	ø40x90L
打模直徑 MAIN DIE DIA. ダイス直径	mm	ø50x110L	ø50x110L
重量 MACHINE WT. 機械重量	Kg	10,000	11,000
螺絲最小與最大尺寸 NORMAL SIZE OF BOLT. 製造ネジ最小最大	mm	3-6	3-6
最短最長頸下長度 SHANK LENGTH OF BLANK. 製造ネジ最短最長	mm	10-75	10-75

Figure 2-2 Machinery view with its type description.

2.1.1. Slider crank mechanism

Fastener machinery operates on the principal of simple eccentric drive mechanism which is comparatively very similar to the general crank slider mechanism. In order to understand the mechanism better the model of the machinery is described with help of Figure 2-3 (Top) machinery schematic and CAD model (Bottom), which is very similar to the realistic machinery model.

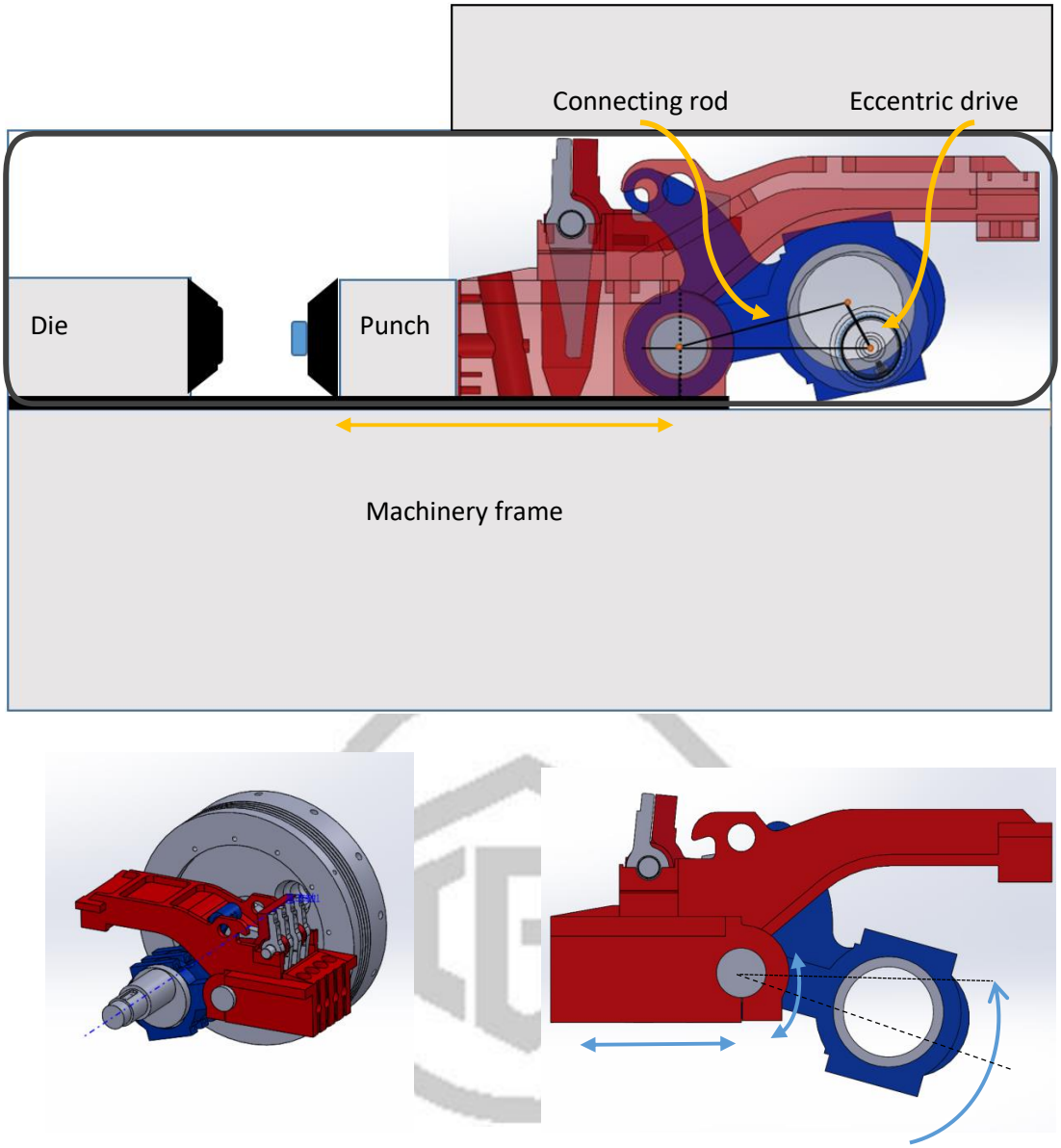


Figure 2-3 Schematic representation of machinery (top) and CAD models (bottom).

For its detailed kinematic study of its drive in specific, it's drive is simplified and represented by a schematic diagram shown in Figure 2-4. Where R is the eccentric radius of the crankshaft, L is the length of the connecting rod between the crankshaft and the slider, x is the length from slider block to the center of the crank radius, γ is the angle between the connecting rod and horizontal axis and θ is the rotation angle of the crankshaft.

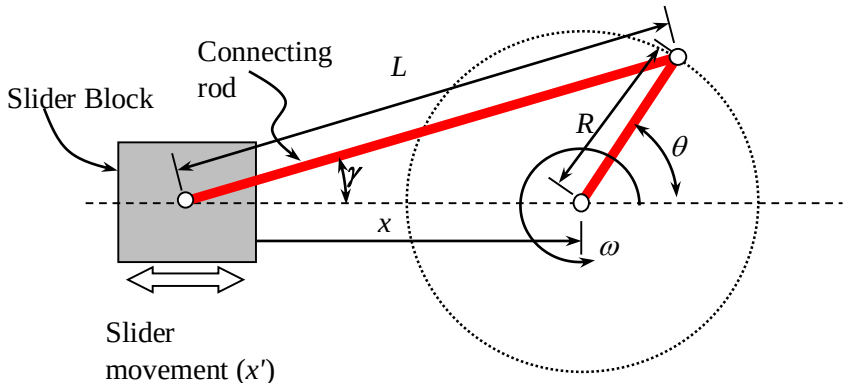


Figure 2-4 Crank slider mechanism.

In Figure 2-4, x is the horizontal distance between the crank radius center and the slider block and can be express as Equation (1).

$$x = L\cos\gamma - R\cos\theta \tag{1}$$

The relation between θ and γ is given in Equation (2).

$$R\sin\theta = L\sin\gamma \tag{2}$$

Rearranging Equation (2) we can have

$$\begin{aligned} \sin\gamma &= \eta\sin\theta \\ \cos\gamma &= \sqrt{1-(\eta\sin\theta)^2} \end{aligned} \tag{3}$$

where

$$\eta = \frac{R}{L} \tag{4}$$

Substituting Equation (3) into (1) gives

$$x = L\sqrt{1-(\eta\sin\theta)^2} - R\cos\theta \tag{5}$$

If the post dead point of the slider-block is defined as the origin of a new coordinate x', also considered as the displacement of the slider block, then x' is given by

$$x' = x - L + R. \tag{6}$$

Since the crankshaft is rotating at a constant speed, the rotation angle θ can be expressed as

$$\theta = \omega(t + p) \tag{7}$$

where t is the time (sec), p is the phase time (sec), ω is the angular velocity (rad/s) of the crankshaft and can be expressed as

$$\omega = \frac{2\pi}{T} \tag{8}$$

where, T is the cycle of crankshaft rotation, is also equal to the main slip forging cycle. General machine forging speed is N pieces per minute to that T as given by the equation (9).

$$T = \frac{60}{N} \tag{9}$$

Then, after substituting equation (7) into equation (6) and then differentiating the time, we can get the optimal velocity u (m/s).

$$u = R\omega\left\{\frac{-\eta\sin\theta\cos\theta}{\sqrt{1-(\eta\sin\theta)^2}} + \sin\theta\right\} \tag{10}$$

And then (10) of the velocity differential on time, you can get the main slide empty car in the ideal acceleration a (m/s^2), can be expressed as in equation (11).

$$a = R\omega^2\left[\frac{-\eta\left(\cos^2\theta - \sin^2\theta + \eta^2\sin^4\theta\right)}{\left[1-(\eta\sin\theta)^2\right]^{3/2}} + \cos\theta\right] \tag{11}$$

If R is 68 mm, L is 270 mm, N is 100 pcs/min and p is –0.096 s, the ideal acceleration curve calculated using Eq. (11) is shown in Figure 2-5. In the figure we can see that the acceleration is not symmetrical between positive and negative regions. The forging process is located at a short period of time before the minimum acceleration as circled by the dashed circle in the figure.

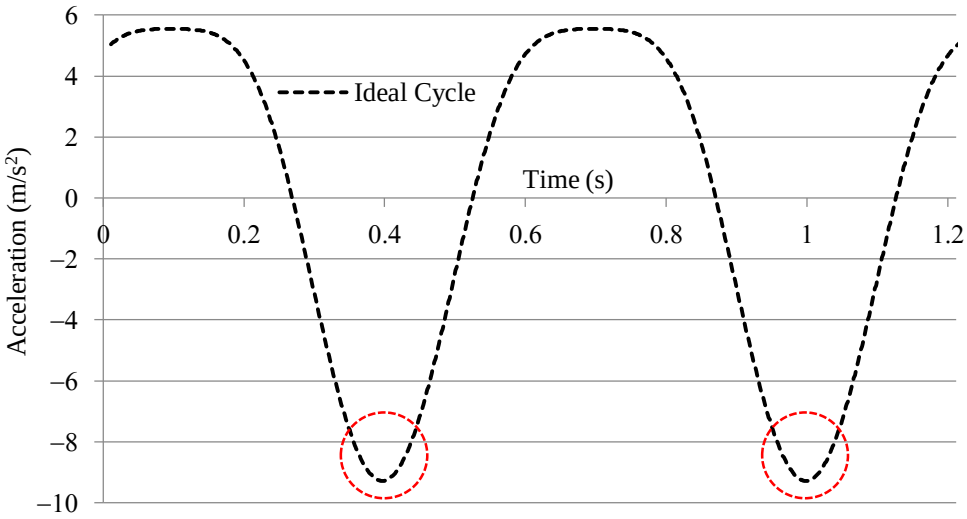


Figure 2-5 Graph of the ideal acceleration versus time for a slider-free slider crank mechanism.

2.1.2. Bulk forming analysis

Forging is one of the manufacturing processes in which a compressive force at a localized place is applied to shape the metal into a desired structure with the help of heavy tools and dies. If forging is done by heating the metal above room temperature is called as warm forging, whereas the forging when done at or below room temperature is called as cold forging. Most of the fasteners are manufactured by cold forging process, as it allows products with high-fidelity having surface quality that doesn't require any machining. The forging stations mostly lie in range of 2-5, with no exact range defined theoretically. As there are infinite conditions with various intrinsic shapes and size of fastener being used to full fill necessities that can be from the miniature bolts used in circuit boards and surgery of knee to the biggest bolts and nuts being used in the marine and aviation industry.

The manufacturing of fasteners is done with help of a machine of a varying die cavity having multi-processing sub-stations with varying range, working simultaneously to form parts as shown in Figure 2-6 (a) and (b).

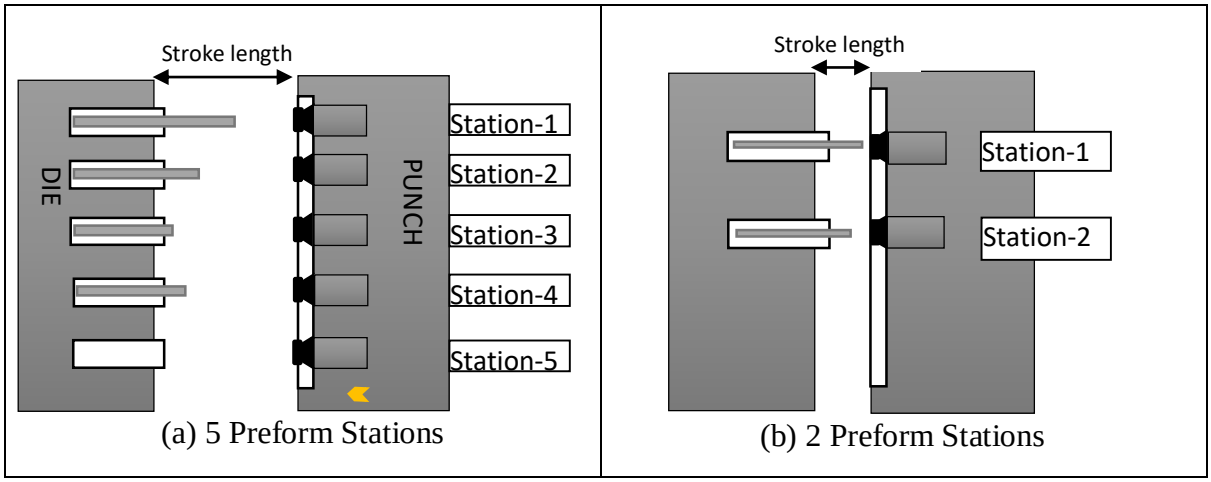


Figure 2-6 Top view of forging stations.

The operation steps involved in the manufacturing process of any shape and size of fastener having substations and frame as discussed above are described in the followings:

(1) The material to be forged especially steel or low carbon steel alloy used in cold forging, before going through forging machinery is kept in a furnace for 25-30 hours to make it soft enough to be worked.

(2) The steel rods are then dipped inside sulphuric acid and rinsed with water to make it free from rust or foreign particles and then coated with the phosphate in order to prevent it from rusting before its made into bolt, screw or any parts.

(3) The forming machine makes the steel wire straight and then with the shearing operation cuts the wire into pieces. The steel wire of length much more than the actual bolt size is sheared and passed through the progressive dies in order to make a fastener.

(4) The procedure is similarly followed for various intrinsic shapes and sizes of fasteners with the changes in their cavity impressions of the dies and punch assembly designed differently for different preform sub-stations.

The manufacturing of a fastener is a very fast process, which include various sub-stations to produce a single formed product. The machineries producing at a very fast production rate need to be precise, as any small error can led to rejection

of the whole lot produced. The fastener producing machineries works in a various range of production rate like 100 pcs/min, 150 pcs/min or 200 pcs/min. Any fault in machine gone unnoticed by the operator or maintenance department for even an hour can lead to losses in production and capital.

Taking in consideration the dynamic flaws as discussed above, it becomes an essential task to monitor the details of in line machinery. The sensors with data acquisition system is equipped to estimate the running real time details of machine. Besides taking the information through sensor, a relative data for comparison study and to understand various fundamental behavior of different process the simulation analysis is done in this section, which deals with the simulation of forging process to predict the load, power and energy consumed at each subsequent station. Many companies nowadays use finite element techniques with help of the commercial software packages for simulation. The tools used in this paper for simulation is quite popular among industries and in academic area for the simulations of various forming process.

The analysis of the cold forging process is done with the help of simulating and designing tools like SolidWorks® [4], ANSYS® [5] and QForm® [6]. ANSYS, being one of the powerful CAE tools used in research and industry, could predict the stresses in both the tool and workpiece but was unable to model the significant plastic deformation [7]. Hence there was a requirement of simulating tool like QForm, designed especially for Simulating forming process. It helps to simulate and predict various detailed features of tool and workpiece interaction that could be taken in account for designing and analysis of the tool and workpiece. The SolidWorks was used as a CAD platform to make dies and punch. Various illustrative simulations were done to study the various tool and workpiece of different fasteners types. The various types of simulation done on varying categories of fasteners are studied comparatively to obtain the common patterns and fundamentals which could help analyze the tools and parts behavior

of the manufacturing process. Table 2-1 describes the product name with their CAD model type along with their stations of machinery on which there are produced.

Table 2-1 Parts and machinery specifications

S.NO	Product	Machinery	Station	Material (W/p) DIS/AISI	Material (Tool)	Model
1	Hex socket bolt M8*20L **Hex-head screw**	**CBF-83S**	3	Stainless steel (X20Cr13 Cold)	D2	3-D
2	M5*60mm P "ASSY" ACS **ASSY head screw**	**CBP-65S**	2	Carbon steel (1020 Cold)	D2	2-D & 3-D
3	M4.5*35mm/21 **Plus head screw**	**CBP-65S**	4	Carbon steel (1020 Cold)	D2	2-D & 3-D

The analysis of all types of parts listed above is done in a procedure described below:

(1) CAD modeling: The forming tools such as punch and die of the progressive sub-stations used in cold forging operation is modeled using Solidworks. The dies and punch required for the stations are made by keeping in mind the law of volume constancy. During the process, the loss or gain of extra material can lead to many defects and error during simulation and in real production process. The whole process and arrangement is designed in optimum mold and die design tolerance, so that with the least amount of material and force the product could be produced along with a tool design having long life and durability. The designer may design the best and most economical tools but still there is always a room for improvements, and there comes the role of routine analysis and monitoring technology. The aim of simulating different process is to obtain a set of ideal fundamental behavior which could be compared and checked with the real systems data which could help build better monitoring methods for a system.

(2) Material modeling: The material model used in bulk forming simulation consists of rigid-plastic, elasto-plastic and visco-plastic. The elasto-plastic

is one of the most used with its two variants being hypo-elastic and hyper-elastic. The hypo-elastic is having material with elastic and isotropic property, whereas the hyper being with the change in material property being anisotropic. The model simulated for analysis of the tool and formed parts is the **elasto-plastic** model with assumption of material type being isotropic.

(3) Tool parameters: It includes the selection of material, drive mechanism, lubricant and temperature required while forging process to make a realistic effect. The drive type used is a mechanical press with the parameters from actual machineries. Figure 2-7 shows the view of simulation control parameters which is set according to different drive types.

After setting up the parameters required for simulation its modeled to depict the various characteristics. The cold forming of fasteners is done in several passes with subsequent sub stations at room temperature, designed according to the preforms. The reasons of designing preform is to reduce the amount of sudden impact loads on the bulk or material of workpiece, to prevent accumulation and breakage. The increase in force at once above the range can exceed the limit of yield value of the material hence, maximum reduction in each pass is decided according to yield criteria of the material, while designing the process. Few methods have been studied along with few case studies for estimation of load and power curve in time and distance reference.

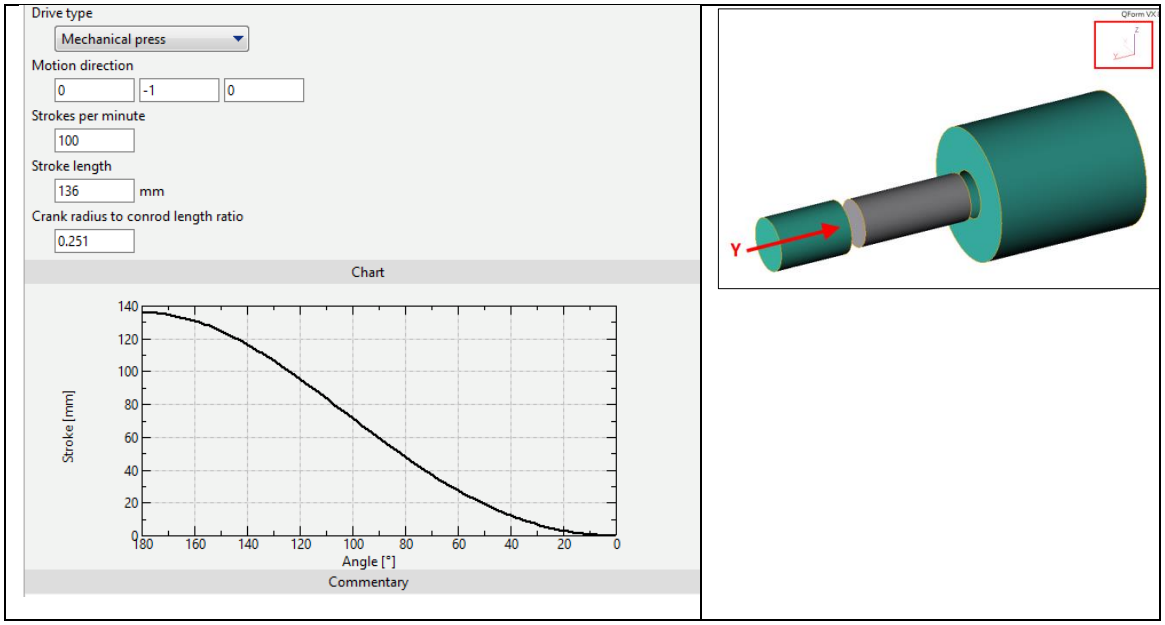


Figure. 2-7 Drive type selection in QForm.

The ways for the analysis can be categorized in four methods: (1) Signature curves of formed shapes, (2) Forming tool and formed parts analysis, (3) Simulation design model and boundary condition check, (4) Curve summation analysis for estimations of peaks and falls. They are described as followings.

(1) Signature curves for formed shapes

Many simulations of different forming shapes were carried out and few signature curves related to each shape is analyzed with help of load curves obtained from simulations. The load pattern obtained from these load curve could help in estimation of the load pattern which would be very similar, if load vs time or displacement curve is analyzed with piezometer sensors installed on each substation. The result of the simulation with signature curves obtained is discussed in section of result and discussion ahead in Chapter 3.

(2) Forming tool and formed Parts analysis

The few of the reasons for rejection of parts in quality test after forming could be, not matching the designated designed standards related to parameters such as:

1. Dimensions: The Dimensions may be length, radius, width or height may be above or below tolerance range.

2. Surface finish: The surface finish of the manufactured parts like rough, shiny, lap formation, grain boundary formation etc.
3. Mechanical property: The mechanical property of the formed part fails to pass the standards of various destructive or non-destructive tests designed, due to improper flow of micro- grain structures caused by dynamic error in manufacturing process.

The forming tools being die and punch needs to be cared and analyzed as special regions, hence few methods with simulating tool have been studied to analyze its behavior during the various wear conditions. Any defect either in form of cracks or the worn-out geometry due to high frictional and cyclic loading in tool indirectly or directly leads to inappropriate formed part. The analysis of die wear caused due to long and extreme working condition of machinery goes unnoticed and cost loss of money due to its catastrophic failure. The collection of data for analysis, when the machine dies actually worn out depends on the luck and convenience offered by the industry. Hence, to avoid the external problems and to study the fundamental nature of die and punch along with the quality of the products formed, FEM simulation is performed. Few case studies have been developed as described ahead chapter for understanding the nature of various causes of tool wear and its influence on parameters like load.

(3) Simulation design model and boundary condition check.

This method aims to develop a perspective to simulate a model with most idealistic approach with few additional boundary conditions and time indeed. The simulation is done both on 2-D and 3-D model with coupled and uncoupled conditions to check for the differences and find the optimum way to simulate and analyze the behavior of the forming process.

(4) Curve summation analysis for estimations of peaks and falls.

In this method all the simulated formed process with various preform stations are added to analyze the overall maximum load reached with displacement of the

slider block. All the methods discussed above with the concluding results after simulation is been discussed ahead in results and discussion section.

2.2 Experimental analysis

The experimental setup consists of an efficient vibration signal acquiring sensor like accelerometer connected with a power supplier and a signal amplifier. A positioning mechanism, which consists of a set of rack and pinion connected in assembly with optical encoding transducer. A circuit to convert logical signal to analog signal is designed for optical encoder in order to synchronize and manage with data to pass through USB-type Data Acquisition System (DAQ). The schematic view of the setup with sensors and their connections is shown in Figure 2-8.

In data analysis locating of the sensors according to their best suited place in machinery, analyzing the real time signal and storing it for future use is a long and tedious task. On the counterpart the task to interpret the obtained data correctly and accurately plays a very crucial role in building of a monitoring system for acquiring a good and desired feedbacks.

2.2.1 Instruments

The instruments selected to build a monitoring system, specific to desire task is discussed with their names and specific details:

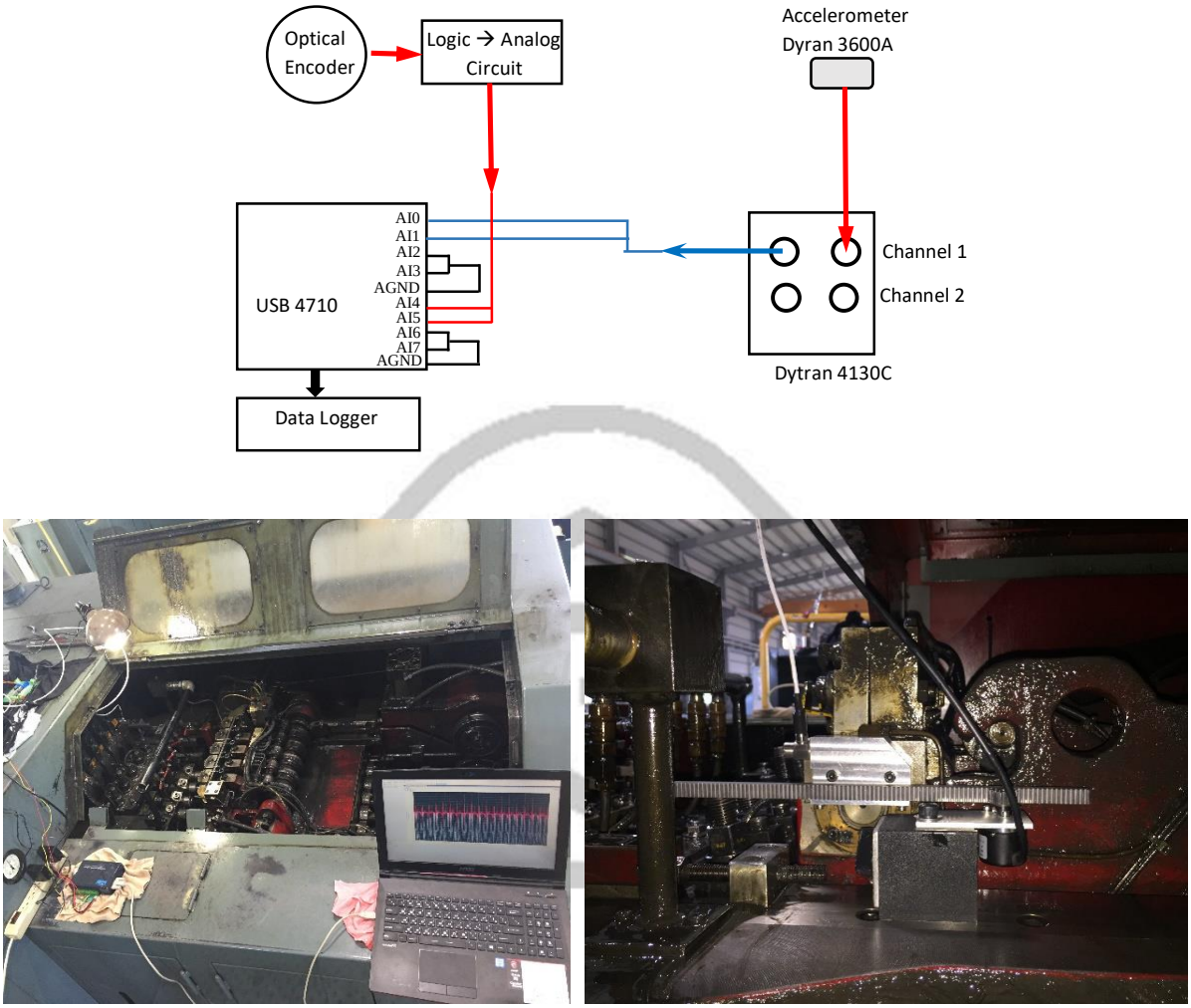


Figure 2-8 Schematic view of the setup (top) and actual installed experimental setup (bottom).

2.2.1.1 Accelerometer-Accelerometer is a sensing device which works on the principle of newton second law of motion which in general defines the force as product of mass and acceleration. The accelerometer device consists of seismic mass attached on the top of a piezoelectric material which when accelerated displaces the mass as the result of the generated force, which is sensed and a digital value in forms of volts is obtained as a feedback depicted in Figure 2-10.

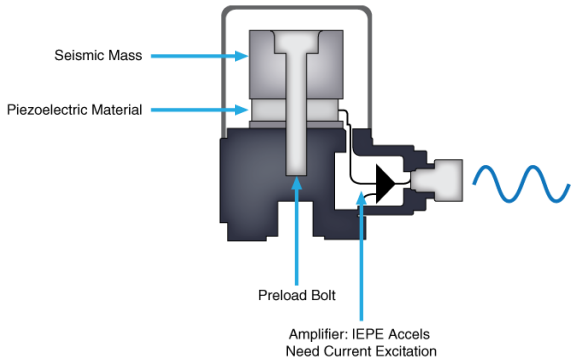


Figure 2-10 Schematics of accelerometer [8].

Sensitivity is an parameter which describes a conversion relation between the vibration and voltage with reference frequency. Sensitivity of the accelerometer, also known as the “scale factor” of the accelerometer, is the ratio of the sensor’s electrical output to mechanical input. In Figure 2-10, the graph represents the trend followed by the accelerometer frequency with respect to the sensitivity whose inverse could be used as conversion factor to obtain actual acceleration value.

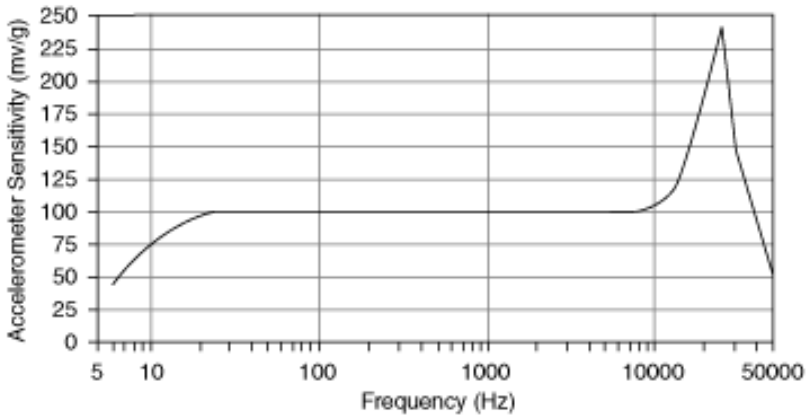


Figure 2-11 Sensitivity [8].

The accelerometer used in this research is a piezoelectric sensitive type show in the Figure 2-11. The Dytran 3006A follows a defined frequency range and sensitivity, which is used in the data analysis to obtain the conversion factor required for converting of the voltage signal to acceleration discussed in the topic of data analysis in detail. It is a single-axis accelerometer with an acceleration

measurement range of ±50 g, which is ±490 m/s^2. The accelerometer's resonant frequency is 35 kHz, it is much higher than the shock frequency (about 10 kHz) caused by the forged impact.



Figure 2-12 Dytran 3006A acceleration gauge.

2.2.1.2 Current supply and signal amplifier

The instrument shown in Figure 2-12 is a 3-channel battery powered current source used to provide the accelerometer with the power to operate, along with a pre-installed amplifier to amplify the accelerometer signal.



Figure 2-12 Dytran 4103C current supply and signal amplifier.

2.2.1.3 Optical rotary Encoder

It is an absolute encoder with 4-bit encoding channels, which means 2 with a power of 4 which is 16 divisions of the 360-degree rotation. The 16 divisions in a rotation having a unit change of 22.5 degree of encoding region each step as

show in the Figure 2-13 of the optical encoder. The value in the interior most of the rotor is called the most significant bit (MSB) and the value at the outer most path is denoted as least significant bit (LSB). The output given by the rotor is in form of gray codes.

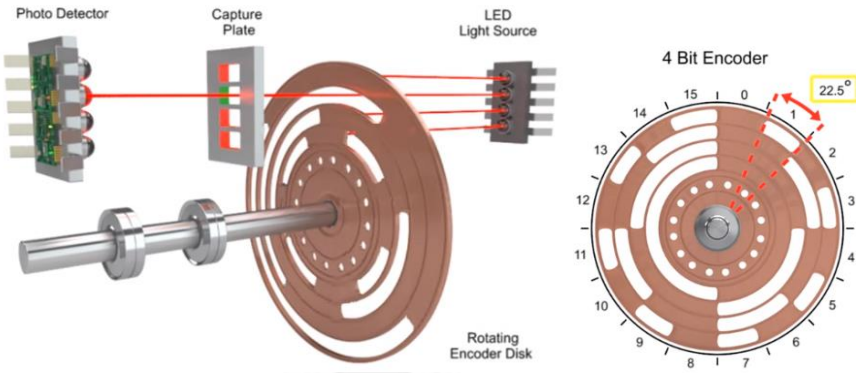
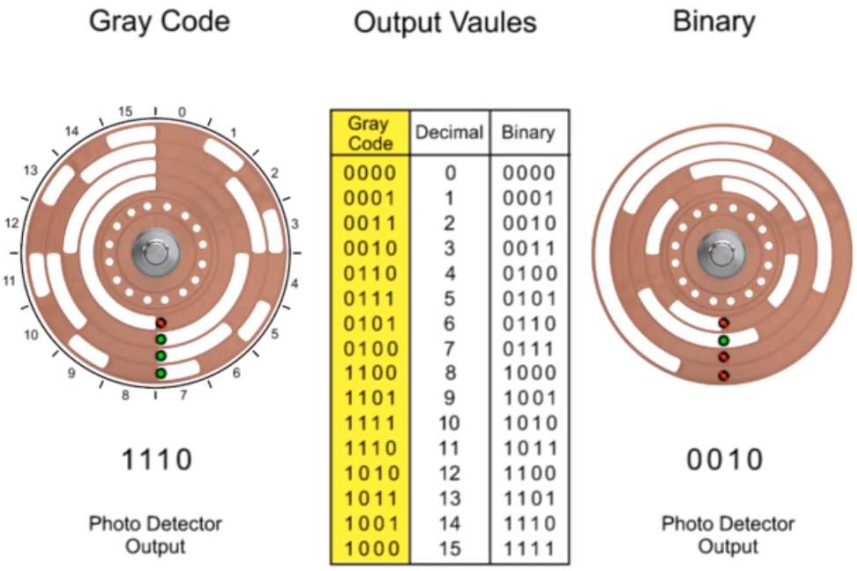


Figure 2-13 Optical encoder [9].

The truth table for gray and binary is very similar as both works on the binary number 0 and 1, but the gray code is considered to be better because of its uniqueness to show the results having a single bit change at each instance as shown in Figure 2-14. In the case of binary system two- or three-bit changes significantly at an instance, which makes decoding bit comparatively difficult.



Gray Code	Decimal	Binary
0000	0	0000
0001	1	0001
0011	2	0010
0010	3	0011
0110	4	0100
0111	5	0101
0101	6	0110
0100	7	0111
1100	8	1000
1101	9	1001
1111	10	1010
1110	11	1011
1010	12	1100
1011	13	1101
1001	14	1110
1000	15	1111

Figure 2-14 Truth table [9].

The encoder used in research experiment for data collection is the rotary encoder shown by Figure 2-15.

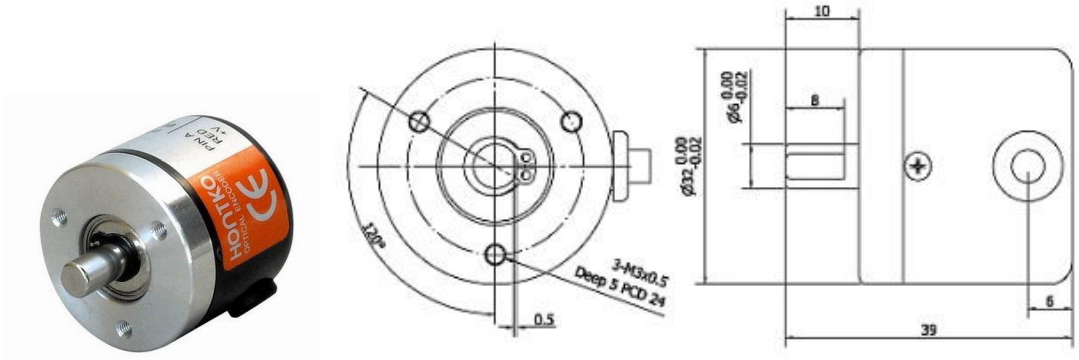


Figure 2-15 Dimensions of A1 absolute light load shaft encoder.

The encoder installed on the machine is an A1 absolute light-duty shaft type encoder manufactured by Hontko. The specifications are 6 mm shaft diameter, shaft load 2 kg radial, 1 kg axial. The maximum mechanical permissible speed is 3000 rpm and the weight is 100gm. The other specifications are shown in Table 2-2.

Table 2-2 A1 Absolute light load shaft encoder specifications

Detection System	Absolute Encoder
Output Code	Gray Code
Standard Number of Pulse Per Revolution	16
Electronics	NPN Voltage
Power Supply	DC 5~12V
Current Consumption	≤ 70 mA
Logic	Negative Logic (H = 0, L = 1)
Max. Response	10 kHz

As the value obtained by the encoder has to be amplified and converted from logical signal to analog signal, hence require a simple circuit designed for the purpose which is described below.

2.2.1.4 IC (Integrated Circuit) board

IC circuit consist of various elements such as NOT gate (IC 7404), XOR gate (IC 7486), Resistors, Operation amplifier (NE5230). As the signals given by

the encoder is in negative logic, hence the output Gray code is first inverted by NOT gate and then converted into binary by the XOR gate combination. Secondly, the binary weighted resistor circuit consisting of the low voltage operational amplifier NE5230 digitizes signal in the form of voltage signal that is conversion of digital to analog signal, because the signal needs to go through acquisition system along with other sensor for comparative data capturing. The designed circuit components were mounted first on a breadboard for testing and then finally soldered on the circuit board, as shown in Figure 2-16 [7].

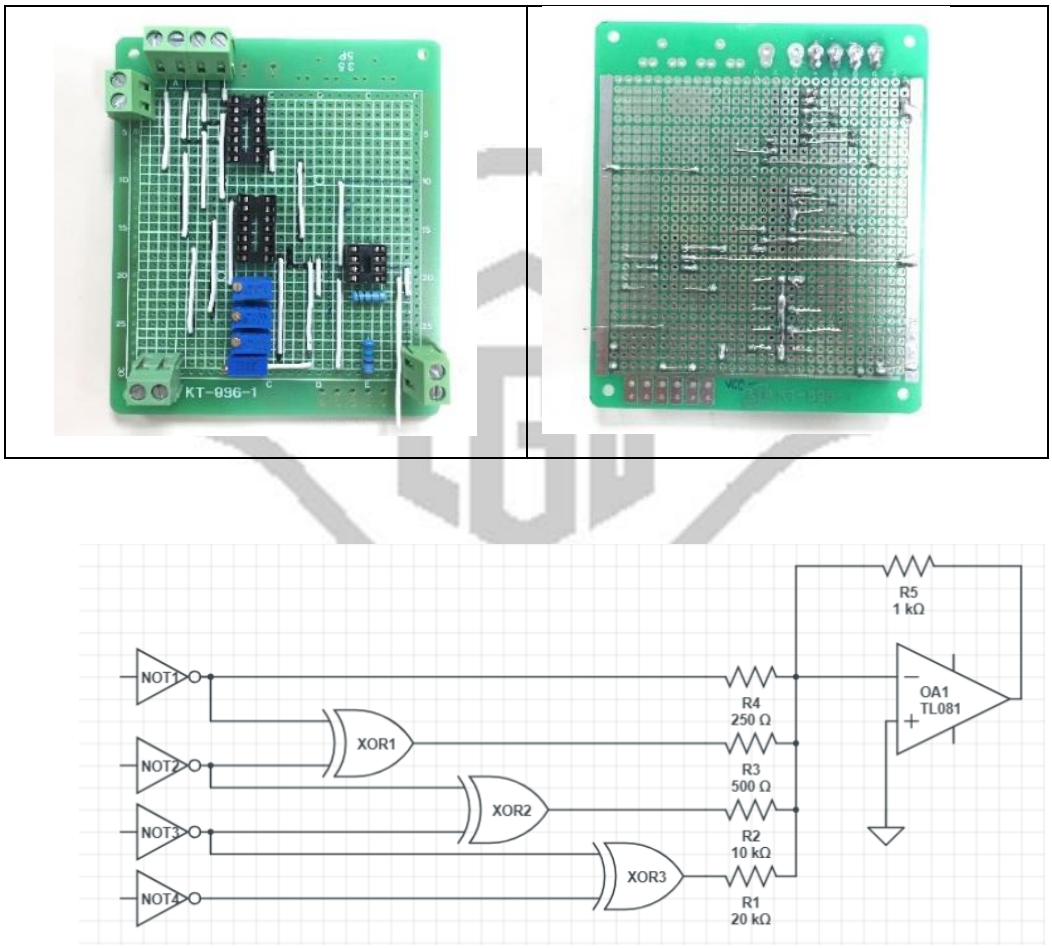


Figure 2-17 Designed circuit board (Top), Circuit diagram (Bottom).

During the test, the encoder is rotated by hand, and the output analog signal is read via the USB-4716 data extractor and the Data Logger.

2.2.1.5 Data Acquisition System (DAS)

The hardware shown in Figure 2-17 is used to sample and record the analog signal by converting to digital signal as a part of data acquisition system. The data extracting hardware for data acquisition used in this paper is USB-4716 from Advantech. Data logger is a software which comes with this hardware that is installed in the desktop or any portable system like laptop to track and analyze the signals as per the requirements.



Figure 2-17 Advantech USB-4716 data capturing device.

It has 16 analog-digital channels and 8 digital-analog channels. The USB-4716 is operated by the inbuilt software application that can be easily accessed and used without any programming software requirements. As there are two sensors accelerometer and optical encoder, by which data has to acquired, hence 2 input channels are provided for each sensor and wired according the grounds provided in hardware. After wiring and switching the USB into the laptop the data logger is logged in to create an empty project with defined inputs. In order to add the acquisition, the analog input with hardware timer is selected which then connects to our USB to access the data on laptop screen. The AI signal and its conversion is asked in the next subsequent window where the selection of no.

of channel their connection type with the sampling frequency and start and stop cannel is entered. In the end the screen opens where the display of the signal could be accessed, remoted and saved by enabling recoding before starting the acquisition.

2.2.1.6 Rack and pinion

Rack and pinion are used as a supporting tool used for the converting the linear motion of the mechanism to the rotary motion.

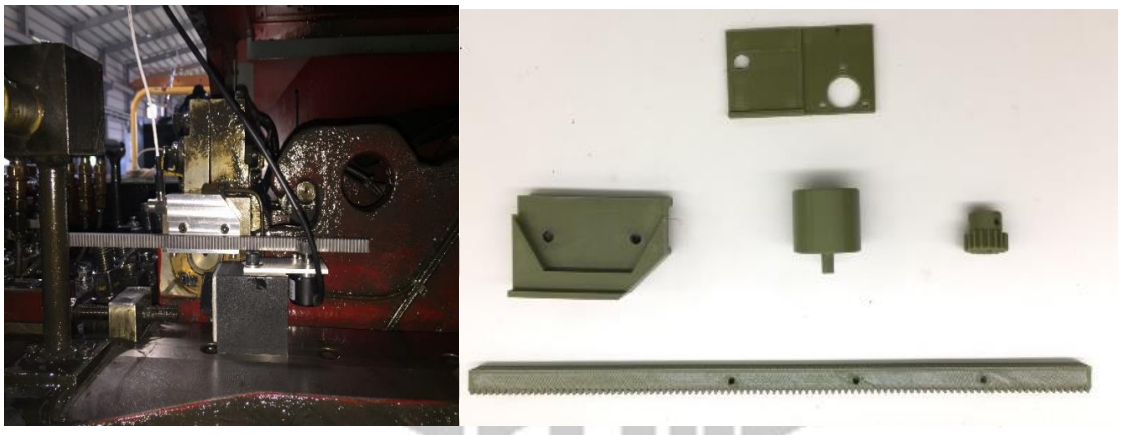


Figure 2-18 (a) Rack and pinion arrangement with, (b) prototype made in 3-D printer.

The rack and pinion arrangement are made according the mechanism of the sliding block of the driving machine as shown in Figure 2-19. The encoder is equipped with a pinion on its shaft and meshed with a linear toothed gear bar, where linear toothed gear bar is known as rack and the circular tooth gear is called pinion. The rack is of the 298.5mm length having modulus of one having 95 teeth designed to mesh with a pinion of 18 teeth having the same modulus. The resolution of the encoder signal as described: circumference of the encoder gear circle is $18 \times 1.0 \times \pi = 56.54867$ mm per revolution of the encoder, and 16 signals are obtained in one revolution of the encoder. Therefore, the line spacing between two adjacent signal changes is 56.5487/16 = 3.5343 mm. The value of the known step size helps in counting the number of steps and predict the motion

of the accelerometer which is described in section of data processing explained ahead.

2.2.2 Data processing

The section deals with the processing of the data obtained by data logger through various sensors installed on machine. The data once acquired in form of digital signals are processed and synchronized for better prediction and interpretation of signals. The data processing consists of conversion of the signals from the accelerometer which are in the form of voltage signals to actual acceleration signals and conditioning the encoder data obtained from the optical encoder for acquiring reasonable position for comparison.

The data processing starts with acquiring of voltage signals from the digitized encoder signals. The encoder signals in actual before processing repeats with indefinite steps. These indefinite steps while processing is synchronized logically to give a sine or cosine waveforms, that could help estimate the cycle length and period. After estimation of the cycle length with encoder data the data followed with time of acceleration is tracked and sliced to get the data of acceleration for each cycle. The acceleration data of cycle is then computed to get the bias value and the factor that could be used as product to get the actual acceleration.

The above short description of the method adopted in signal processing is discussed in detail for each sensor data.

(1) **Encoder signal processing:** The data obtained by encoder varies with respect to time, without any defined notable variation in cycle as can be seen from the Figure 2-19. In order to make it meaningful and comparable it's constrained to vary like a cyclic waveform for easy comparison and computing and slicing the cycles with respect to ending and starting data point. The processing of the encoder signals is done by the following steps and the Python codes are listed in Appendix A:

(i) **Digitize:** As the signal has irregular noisy peaks which in mathematical terms can be termed as float values. In first step of conditioning of the obtained signal is to digitize the signal by making the float values to integer value which will remove the glitches from the steps.

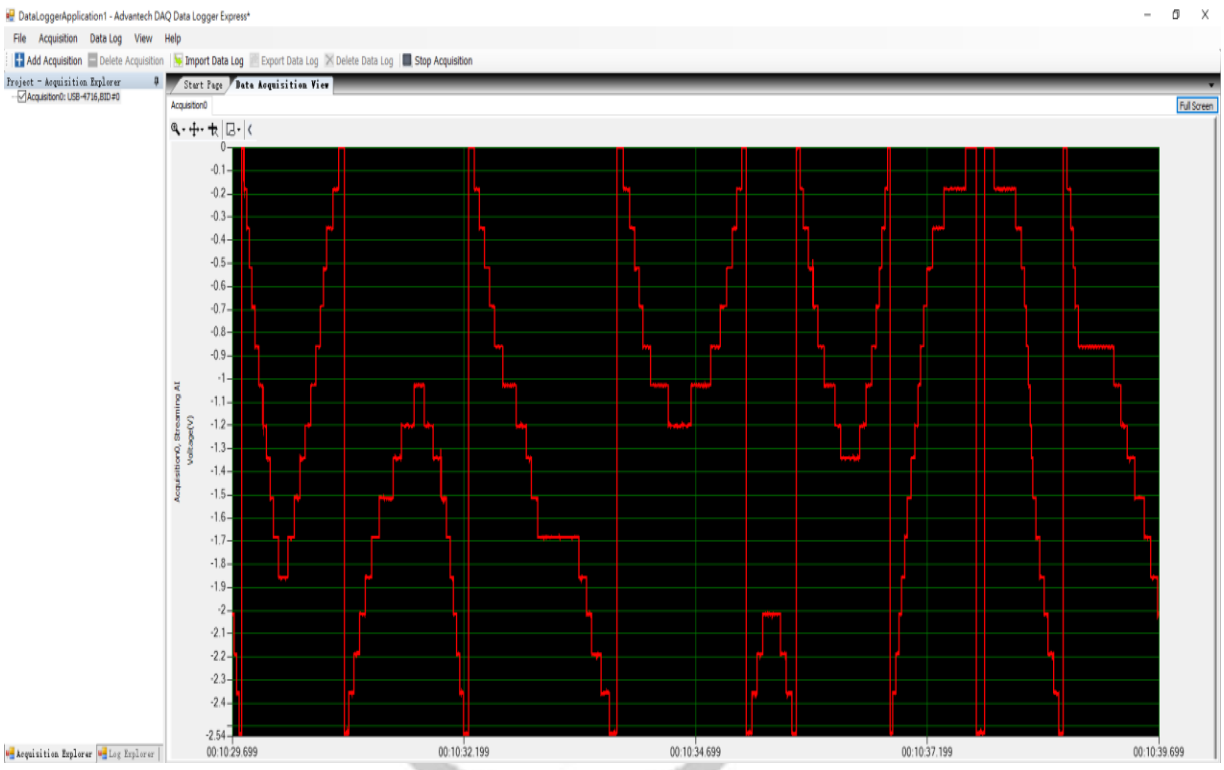


Figure 2-20 Encoder signals.

(ii) **Manage jumping steps**: The next step includes the incrementing or decrementing of the steps in with help of logics that give cyclic waveforms. This type of cyclic waveform gives better intuition to estimate the position with respect to time.

(iii) **Shifting:** The final step includes the shifting of the waveform obtained and giving a least value referenced to minimum peak value at zero as can be seen in the cycle of the obtained waveform in the Figure 2-20**.**

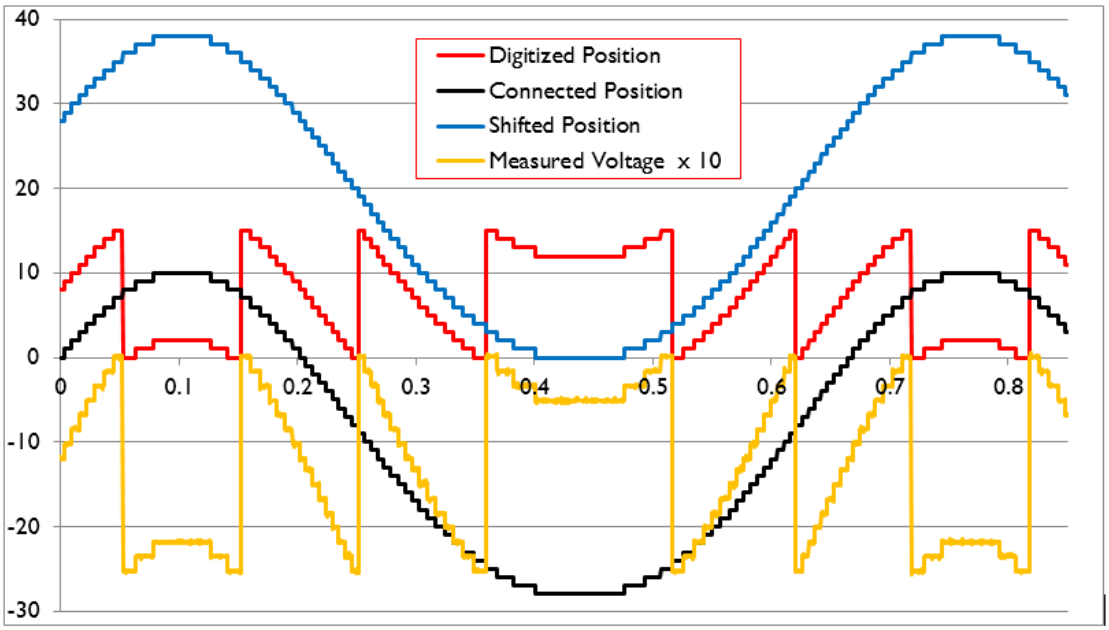


Figure 2-20 Conditioning of the encoder data.

(2) **Accelerometer signal processing:** The data from the accelerometer are in Volts and have to be converted into m/s^2 by multiplying with the conversion factor, γ.The conversion factor is the reciprocal of the sensitivity factor which is 100 mV/g as given in the specification of the accelerometer used for analysis. However, sensitivity factor varies with the frequency of the signals and is denoted by ± 3dB in the frequency range from 1 Hz to 10 kHz. The experimental data consists of signals with varying frequencies and bias therefore the sensitivity factor in the specification cannot be used directly.

To estimate the parameters like conversion factor (γ) and bias voltage (E_b) following methods are adopted. The acceleration is product of conversion factor γ and the summation of biased and varying voltages in general as shown in equation (12).

$$a_i = \gamma\left(E_b + E_i\right) \tag{12}$$

The value of the bias voltage can be calculated by averaging the voltage data obtain from the machinery in reference to defined time range. The value of E_b varies with sampling rate so it has to be excluded.

The varying velocity with time is calculated with the help of equation having an initial velocity u_0 and E_b equal to zero and value of acceleration replaced by equation (12).

$$u_i = u_0 + \sum_{j=1}^{i} \frac{a_j + a_{j-1}}{2}\left(t_j - t_{j-1}\right) \tag{13}$$

E_j and E_{j-1} are series of voltage data points obtained by sensors corresponding to their time referred as t_j and $t_{j-1.}$

$$u_i = u_0 + \sum_{j=1}^{i}\left(\gamma E_b + \gamma \frac{E_j + E_{j-1}}{2}\right)\left(t_j - t_{j-1}\right) \tag{14}$$

For initial condition the value of E_b and u_0 being zero the value in terms of $\gamma u_{b,i}$ is calculated for u_i.

$$u_i = u_0 + \gamma E_b u_{a,i} + \gamma u_{b,i} \tag{15}$$

Similarly, the value for displacement x_k is calculated with series of velocities obtained by the equation (15).

$$\begin{aligned} x_k &= \sum_{j=1}^{i_k} \frac{u_j + u_{j-1}}{2}\left(t_j - t_{j-1}\right) \\ &= u_0 \sum_{j=1}^{i_k}\left(t_j - t_{j-1}\right) + \gamma E_b \sum_{j=1}^{i_k} \frac{u_{a,j} + u_{a,j-1}}{2}\left(t_j - t_{j-1}\right) + \gamma \sum_{j=1}^{i_k} \frac{u_{b,j} + u_{b,j-1}}{2}\left(t_j - t_{j-1}\right) \\ &= u_0 x_{a,k} + \gamma E_b x_{b,k} + \gamma x_{c,k} \end{aligned} \tag{16}$$

The displacement at the end of 1 cycle equal to zero by the help of which the equation formed gives the ratio as q as described below in equations.

$$x_{N+1} = u_0 x_{a,N+1} + \gamma x_{c,N+1} = 0 \tag{17}$$

$$q = \frac{u_0}{\gamma} = -\frac{x_{c,N+1}}{x_{a,N+1}} = -\frac{x_{c,N+1}}{T} \tag{18}$$

After calculating the value of q the value of x' minimum and maximum is calculated in the varying time interval. The value of conversion factor γ (m/s^2/V) is then calculated with help of x' and radius R as shown in equation (6).

$$x_i = \gamma\left[q\left(t_i - t_1\right) + x_{c,i}\right] = \gamma x_i' \tag{19}$$

$$\gamma = \frac{2R}{x'_{\max} - x'_{\min}} \tag{20}$$

The conversion factor can then be further used in determining u_0 from equation (7). The actual value of acceleration will then be obtained by multiplying with conversion factor. Similarly, the parameters like displacement and velocity can be calculated by the actual acceleration obtained and then compared with the encoder position. The value of obtained position and velocity values is plotted as shown in Figure 2-21. The Python code is listed in Appendix C.

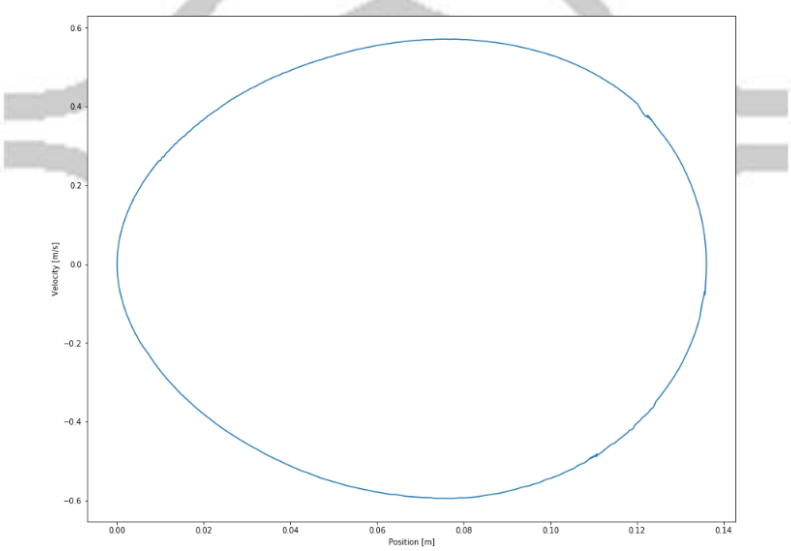


Figure 2-21 Plot of displacement and velocity.

From the force diagram shown in Figure 2-22, the forging force experienced by the slider block cannot be derived directly from the linear acceleration acquired with the help of Newton's 2nd law. However, if we know the angular acceleration of the crankshaft assembly, we can reversely find the torque experienced by the assembly and thus the horizontal force experienced by the slider block. Since the data we have are the linear acceleration of the slider block

and its calculated velocity and position, we need some more derivation to find the angular position, velocity and acceleration of the assembly. Firstly, referring to Equation (3) and Figure 2-22, we have

$$\cos^2\gamma = \left(\frac{x + R\cos\theta}{L}\right)^2 = 1 - \eta^2\sin^2\theta\,. \tag{21}$$

Then manipulating Equation (21), we can relate θ to the calculated x as

$$\cos\theta = \frac{1 - \eta^2 - (x/L)^2}{2\eta(x/L)} \tag{22}$$

Referring to Equations (10) and (21) we can relate the angular velocity ω of the crankshaft assembly with the calculated u as

$$\omega = \frac{u}{R\left(\dfrac{-\eta\sin\theta\cos\theta}{\sqrt{1-\eta^2\sin^2\theta}} + \sin\theta\right)} = \frac{u}{R\left(\dfrac{-\eta\sin\theta\cos\theta}{x/L - \eta\cos\theta} + \sin\theta\right)} \tag{23}$$

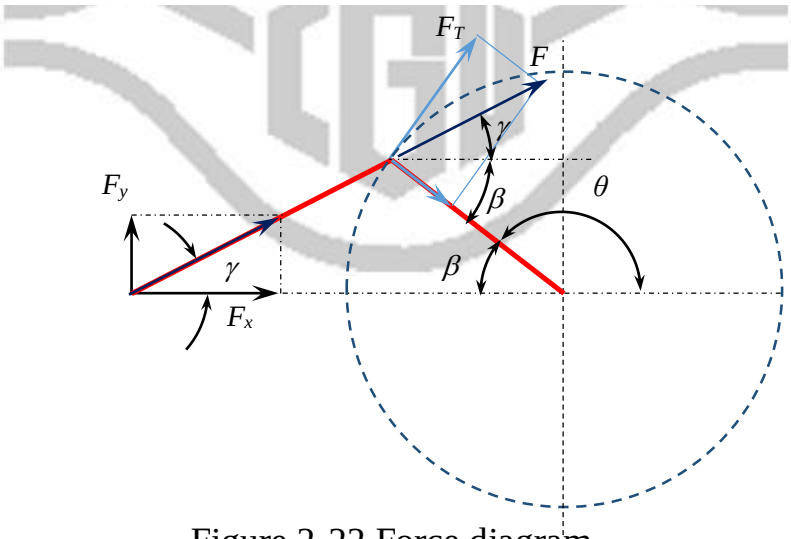


Figure 2-22 Force diagram.

The angular acceleration of the crankshaft assembly can be expressed as

$$\frac{d\omega}{dt} = \frac{F_T R}{I} \tag{24}$$

where I is the moment of inertia of the crankshaft assembly, F_T is the tangential force acting on the crankshaft assembly by the connecting rod. F_T is then related to the force F exerted on the connecting rod by

$$F_T = F\sin(\gamma+\beta) = F\sin(\gamma+\pi-\theta) = F\sin(\theta-\gamma) \tag{25}$$

The horizontal force component F_x exerts on the slider block can be written as

$$F_x = F\cos\gamma = F\sqrt{1-(\eta\sin\theta)^2}\ . \tag{26}$$

Substituting Equations (24) and (25)

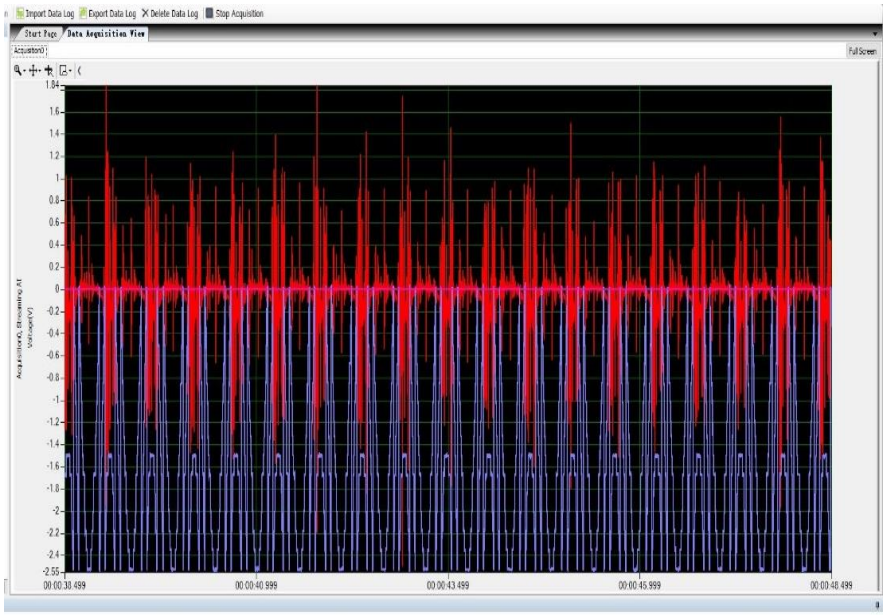
$$F_x = \frac{F_T}{\sin(\theta-\gamma)}\sqrt{1-(\eta\sin\theta)^2} = \frac{R}{I}\frac{\sqrt{1-(\eta\sin\theta)^2}}{\sin\theta\cos\gamma-\cos\theta\sin\gamma}\frac{d\omega}{dt} \tag{27}$$

where $\cos\gamma$ can be evaluated with Equation (21) and $\sin\gamma$ can be easily evaluated with a trigonometric formula.

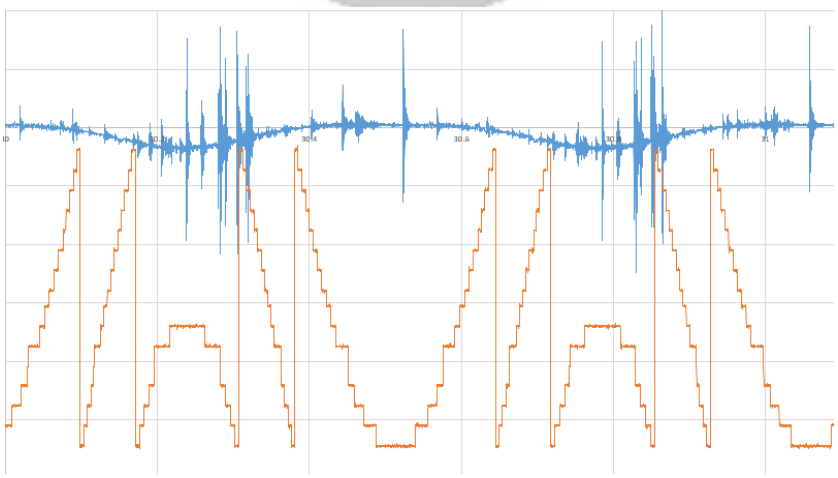
2.2.3 Data analysis

Time waveform: Time waveform signals obtained from transducers consists of the combination of sine waves evolved due to the dynamic vibration caused by the various elements on which the transducer is mounted. The data obtained through sensor consists of voltage signals with respect to time, peaking up and down. In order to understand the messy and unreasonable data requires few parameters analysis like RMS, mean, Kurtosis, Skewness which emphasize many things about the signal embedded inside the time waveform data. The other things besides parametric study includes the insight knowledge of every little detail of the machine. The time waveform of the machinery before processing looks like as shown in Figure 2-23. Time waveform analysis is little troublesome task, but if analyzed properly with certain processing techniques could help in estimation of various data associated at each crank motion of the machinery. In the Figure 2-24 the processed data as discussed above is used to show the variation in signals at each progressive stroke with comparisons between machinery working with and without workpiece. The data helps in predicting most of the behaviors of the

machinery at each instance or step of the sliding block. The red stepped waveform and red phase waveforms with combination of noise and actual signal from accelerometer represent the data from the empty cycle of forging machinery. The black stepped waveform and peak waveforms designate the loaded cycle of the forging machinery. The time waveform data helps in prediction of the location of forging operation with respect to time by various comparative data study with working of free and loaded run.



(a)



(b)

Figure 2-23 (a) Actual view of accelerometer and encoder signals in data logger [10], (b) Plot of data recorded by data logger.

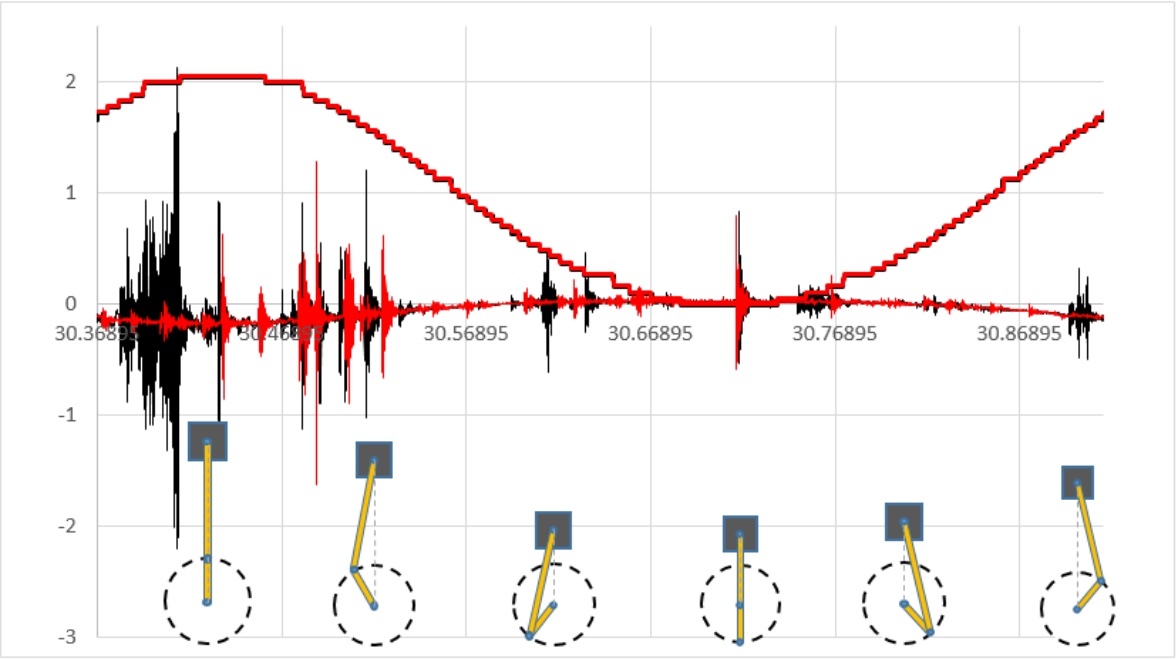


Figure 2-24 Comparisons of waveforms of machinery working with and without workpiece.

Spectrum analysis: This method includes the analysis of the spectrum or spectra which is 1-D and can be obtained through FFT (Fast Fourier Transform) of time-waveform. The uniqueness of spectrum analysis is that, it could give better estimation of noise peaks, harmonics and sidebands of frequencies generated in whole cycle. It decomposes the time wave signals which are mostly the combination of sine waves and noises stacked in with different phase and amplitude to form a typical waveform. In order to compute the FFT of time waveform signals the maximum frequency of the machinery or sliding block in our case where the accelerometer has been installed is approximated. The maximum frequency for the sliding block in our case was taken to be 10KHz. The maximum frequency (f_{max}) of the process is then used for deciding the sampling frequency (f_s). The selection of correct sampling frequency is very important in data processing as any error could led to misinterpretation of information leading to aliasing. In order to avoid aliasing, **Nyquist theorem is used** to sample the

digital data. According to Nyquist theorem the sampling frequency should be twice or more than the maximum frequency. The estimated FFT of empty run and loaded run obtained from the machinery can be seen in the Figure 2-25.

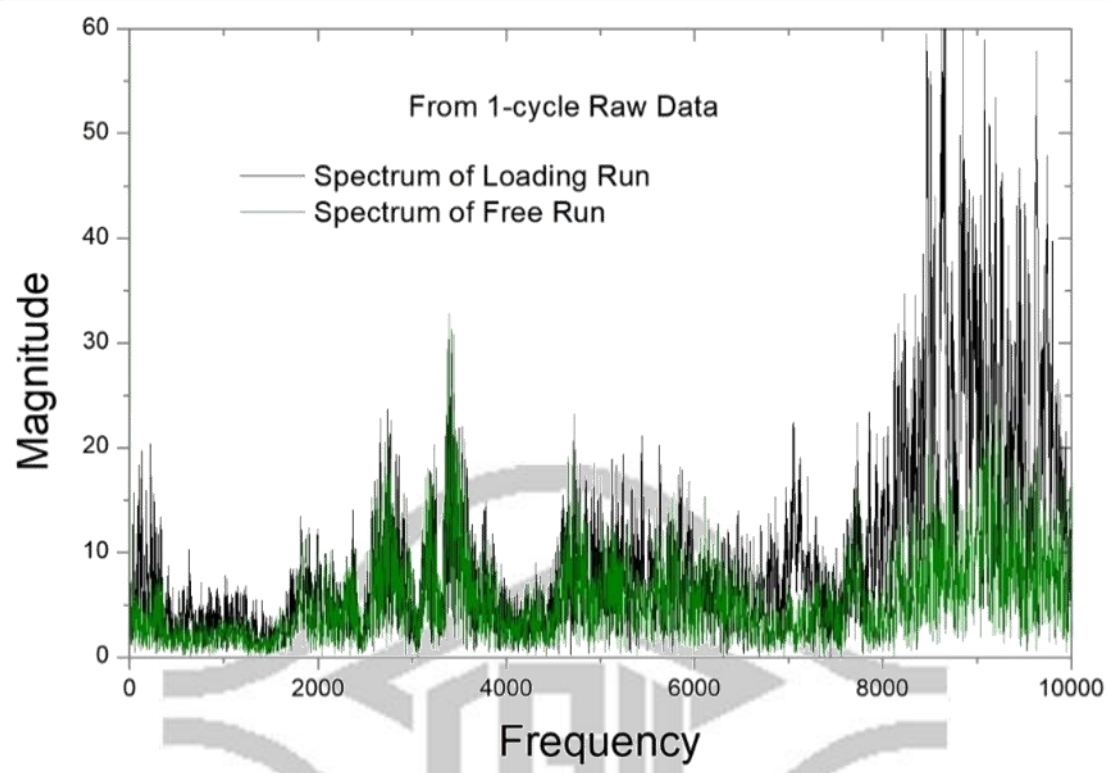


Figure 2-25 Comparison of peaks by FFT [10].

Spectrogram: Spectrogram is a tool for analyzing signals as a function of time and frequency. Spectrogram is a tool for analyzing signals as a function of time and frequency. Spectrogram also called short Fourier transform, is generated by applying FFT on the signals from consecutive short periods of time centered at the times of interest. The period of time in which data extracted is called a window. The wider the window is, the finer the frequency resolution is but the coarser the time resolution becomes. The step size of the moving sampling window determines the continuity of signal in time span. The larger the step size is, the more distinct the signals appear. It is intuitive to show a spectrogram in grey scale or color for its decibel value of power with time as the axis of abscissa and frequency as the direct axis. In a spectrogram, one can easily see the time of

the occurrence and the duration of a vibration incidence and the power and the frequency band of the vibration. The spectrogram of the data from the cycle of producing a screw is shown in Figure 2-26 as an example.

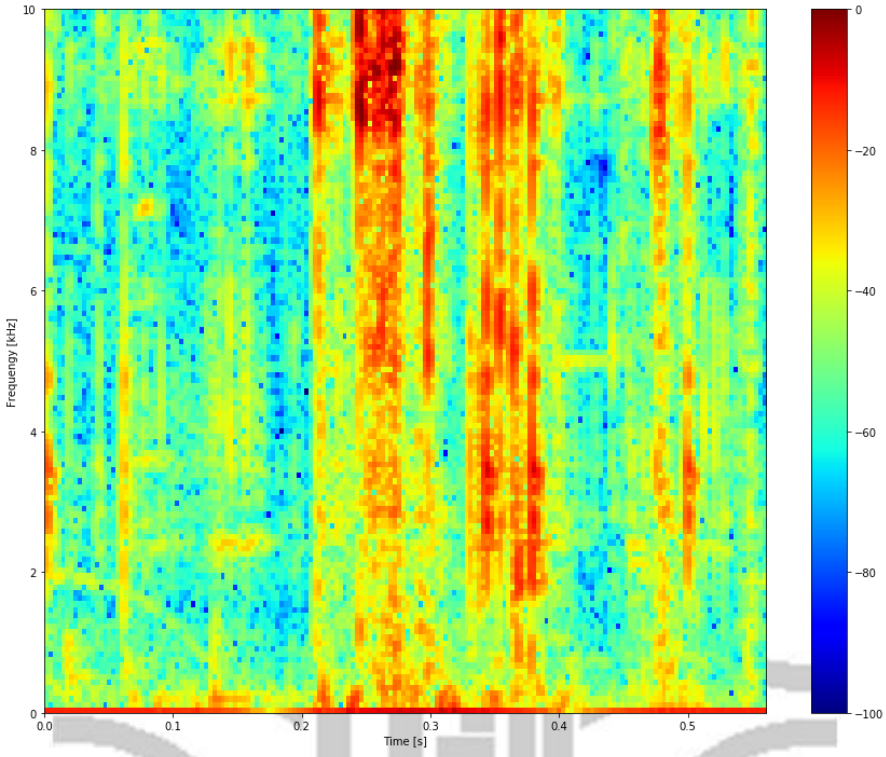


Figure 2-26 The spectrogram of a working cycle.

The analysis described above such as spectrogram, FFT and time series are used in the paper for various data analysis and discussed in detail with its uses in recognizing the signals from machinery in ahead sections.

3. Results and Discussion

3.1 Methods established through bulk forming simulation

There are few methods which shows the characteristics features and patterns that can be used for monitoring is discussed below:

3.1.1 Signature curves for formed shapes

In this method signal associated with various common forming process has been tried to observe through a pattern, which could be associated as their signature curve. The height and diameter are taken as a parameter to evaluate the pattern w.r.t load curve. The diameter denoted by ***i*** denotes the preform number and ***b*** denotes the base end diameter which is inside and towards the stationary die Figure 3-1.

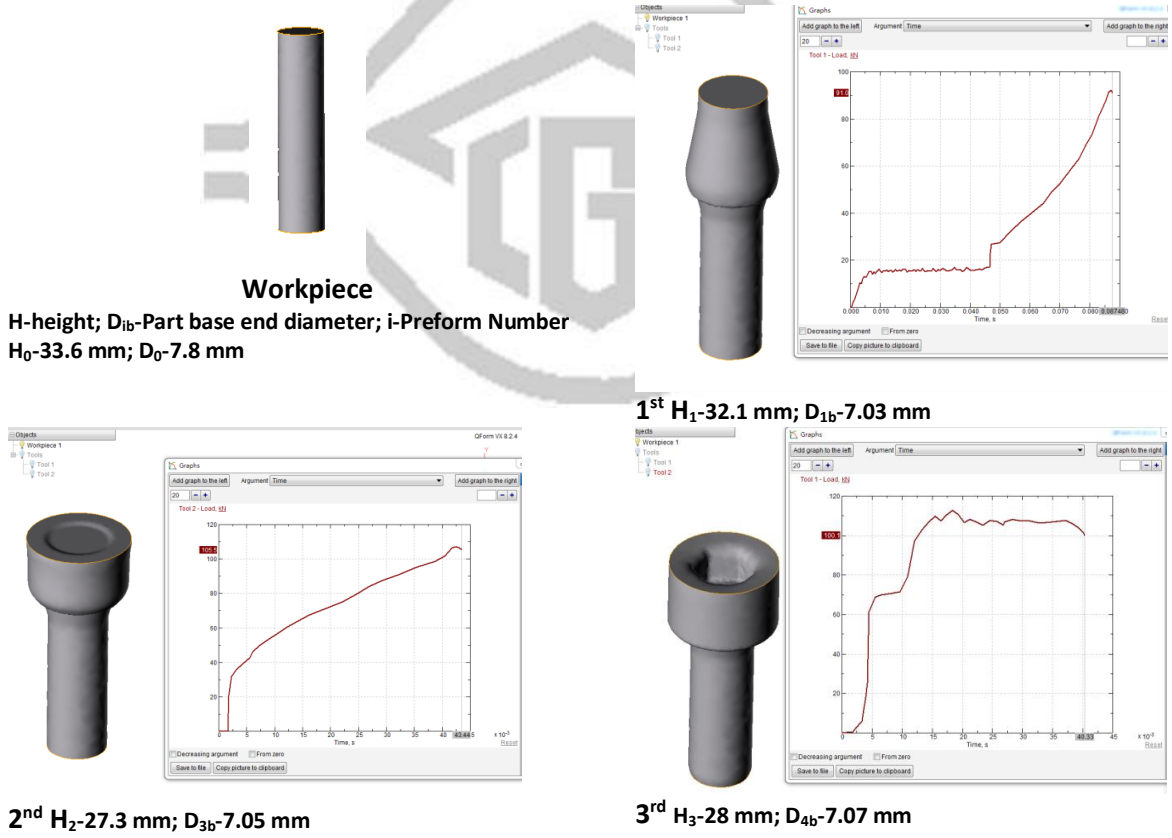
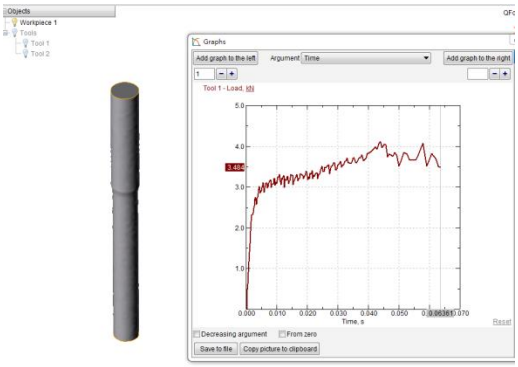
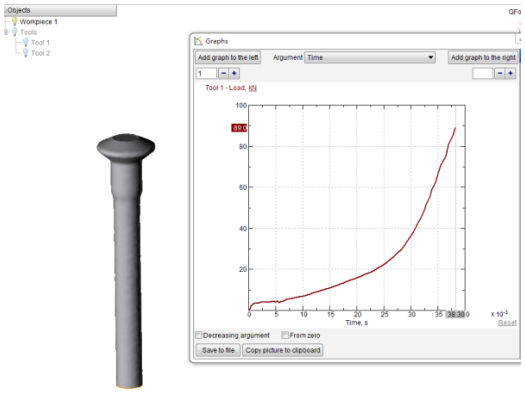


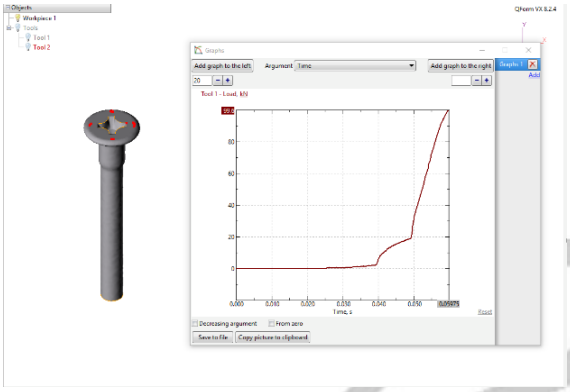
Figure 3-1 Three Hex-head process station load curves.



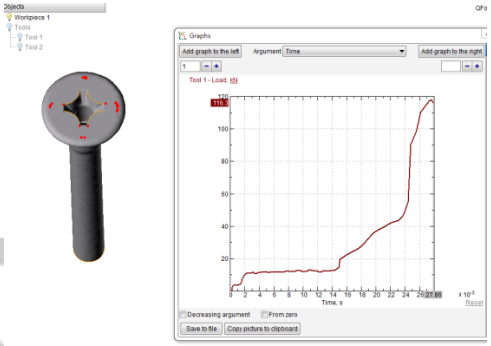
1st H_0-33.53 mm; D_{0b}-3.50 mm; H_1-38.4 mm, D_{1b}-3.12 mm



2nd H_2-31.7 mm; D_{2b}-3.15 mm

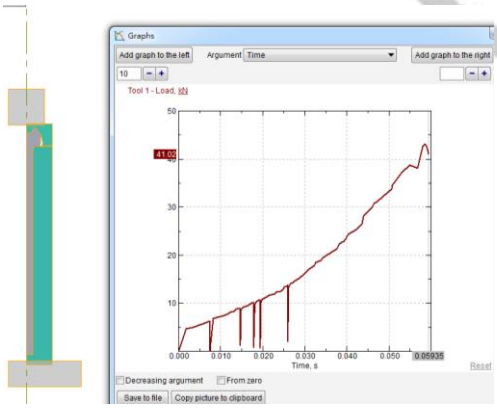


3rd H_3-31.3 mm; D_{3b}-3.15 mm



4th H_4-32.7 mm; D_{4b}-3.20 mm

Figure 3-2 Plus head process stations load curves.



(2-D) 1st H_0-78.06 mm; H_1-62.4 mm; D_{1b}-3.6mm



(3-D) 2nd H_2-61.5 mm; D_{2b}-3.6mm

Figure 3-3 ASSY process station load curves.

The result that could be obtained from the method discussed are as follows:

(1) With the help of simulation various types of common forging process of different shapes and profile has been studied. Every part manufactured possess forward, reverse and pure forms of upsetting, and extrusion process. All these processes are analyzed with signature or patterns associated with these forms.

(2) The common pattern established by simulation describes the features in the process of various manufacturing part from workpiece to actual desired part.

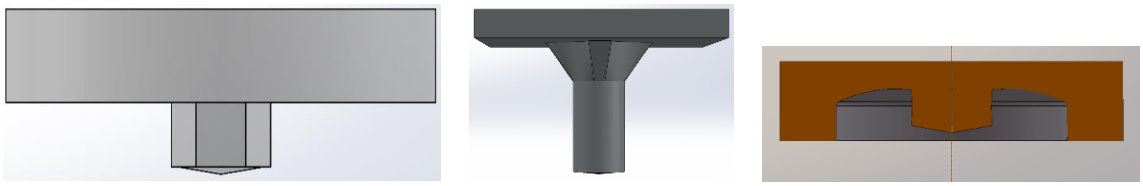
(3) The pattern observed could help us in knowing the load patterns associated with various types of shape and size of fasteners, making it one of the methods for monitoring.

3.1.2 Forming tool and formed parts analysis

This method consists of three cases on the bases of which the formed parts quality and the tool health has been analyzed and methods to monitor it has been illustrated.

Case 1**:** Analysis of the punch with and without tip common to all punches.

The tip of a die is very beneficial in any type of forming punch, for impinging the impression of die on workpiece. Either it may be screws, bolts or any type of fastener until or unless the tip depth on it is not as per design, the proper torque required for fastening would be hard to indulge. The tip is a necessary part of tool design in fastener and its present and common in almost all type of dies irrespective of the part design. Few of the cone tips of die design for varying die design shape is shown in Figure 3-4.



(a) (b) (c)

Figure 3-4 (a) Hex-head tool tip, (b) Plus head tool tip, (c) Internal ASSY head tool tip (Section view).

During forging process the chances of tip of the punch undergoing excessive wear is very high due to high repetitive impact loading. The other sever nature of the cone shape tip is that its first to hit, so any abrasive present in the coolant can lead to severe fracture or led to change in its dimensions in long repetitive cycles. Hence the cone shape tip becomes the necessary part of analysis while analyzing for the quality of manufactured parts. The models to be analyzed with and without tip is shown in Figure 3-5.



Figure 3-5 Tool tip types.

The simulation is performed on the models shown above with parameters like stroke length, production rate obtained from the actual machinery. Before, actually simulating for the conditional process, preforms of the stations are designed and simulated. As the hex-heading being the end of the process cycle and the finished part, it should possess high credibility in its design. In order to check the load variation, which is mostly measured by various piezometer

sensors, the motive was to establish the relationship of load with tip and without tip to check the dependency of load variation due to wear shown in Figure 3-6.

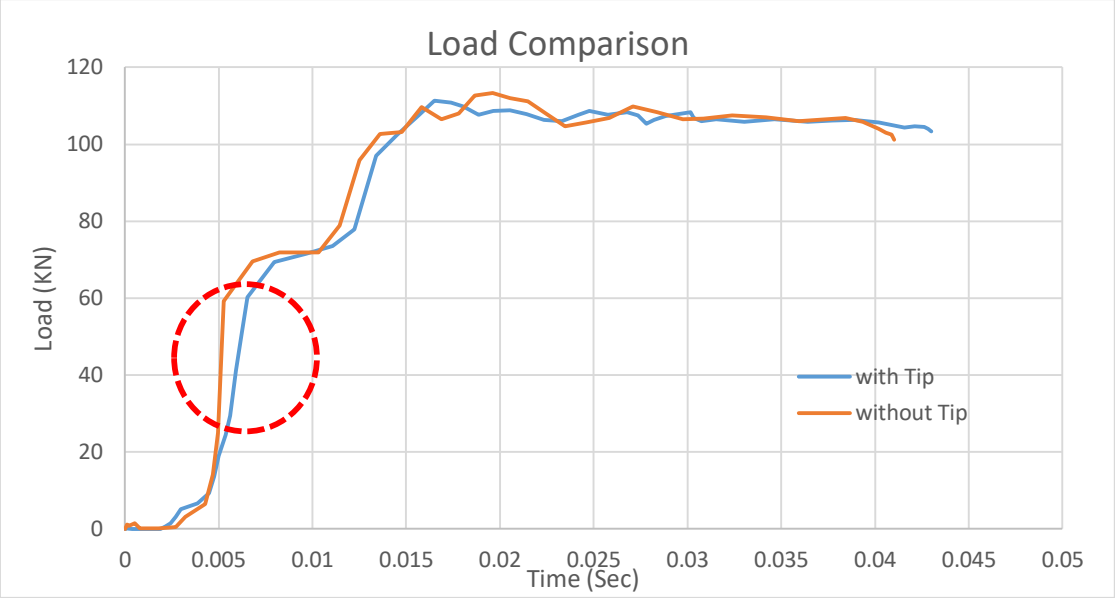


Figure 3-6 Load-time curve.

The variation of the load due to the change in eroded dimension shows changes in the load at specific duration but the pattern is very similar in both the conditions. But it can be seen that due to the increase in load at sudden there is no change in slope in case of without tip, it can be concluded that with diminishing tip the slope will divert or approach to linear curve specific to the portion described with red dot circle.

As the load pattern with respect to time doesn't show any high amount of load variation, giving conclusion that parameter like load can't always help us judge the quality of material manufactured or die wear until or unless it's too severe.

Case 2: Analysis to find the tolerance range, before the degradation

In this **case study** the tolerance range of punch would be decided for estimation of good or bad product. This defined method is to monitor the quality of parts on the bases of tolerance limits for alarming monitoring system defining when is the requirement of tool change on (Hex head) Part type:

(1) In this method, the multi-station formed part station of Hex-Head type with **high intense load** is analyzed with the help of data curve obtained from the simulation of each process stations.

(2) The hexagonal head shape is applied to the conditional test, due to its intrinsic complicated shape undergoing highest amount of load among the subsequent processing stations and therefore has chances of high wear.

(3) The analysis of last forged sequence (Hex-Head) is done by pre-assuming the decrease in dimensions by 0.002 mm due to its high repetitive frictional contact with workpiece.

(4) The hexagon with inner circle diameter is varied from 6.08mm to 6.0mm with a change of negative tolerance (0.002 mm).

By the load displacement graph Figure 3-10 simulated in next case, it can be seen and observed that the 3rd process reaches to high peak comparative to other process, so the test of diminishing tolerance wear on the last sequence is performed with the help of simulation tool with varying tolerances as discussed above. For the first part of study different dimensions are selected with changes of 0.002 mm, which is actually done to **imitate the scenario of decrease in the dimensions of die to repetitive friction** caused by the interaction with the workpiece shown by the load curve in Figure 3-7.

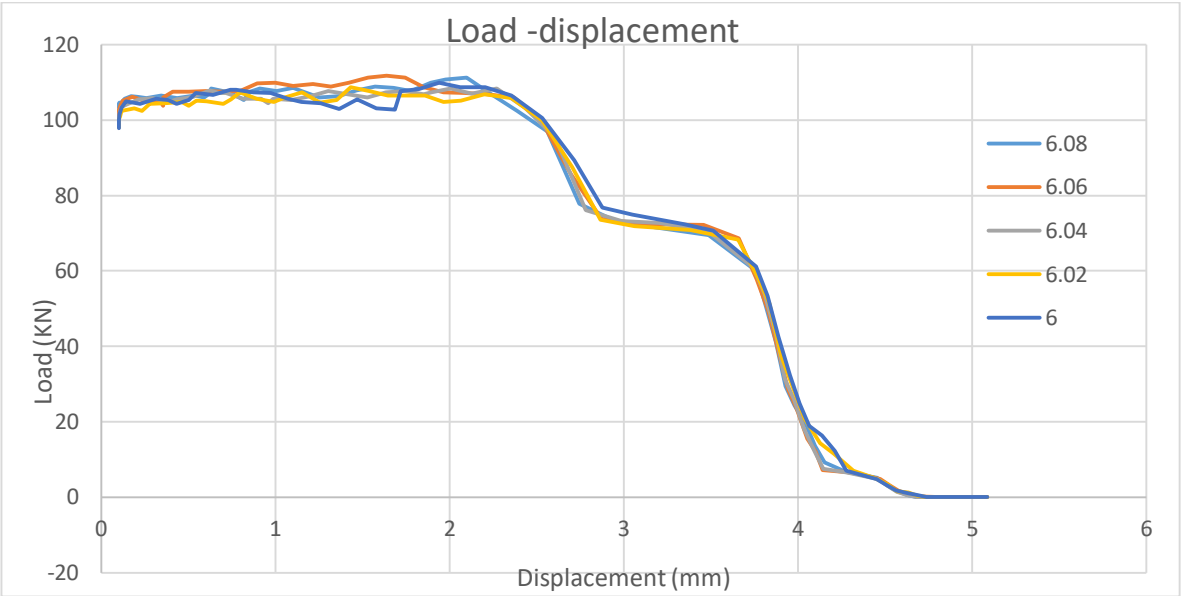
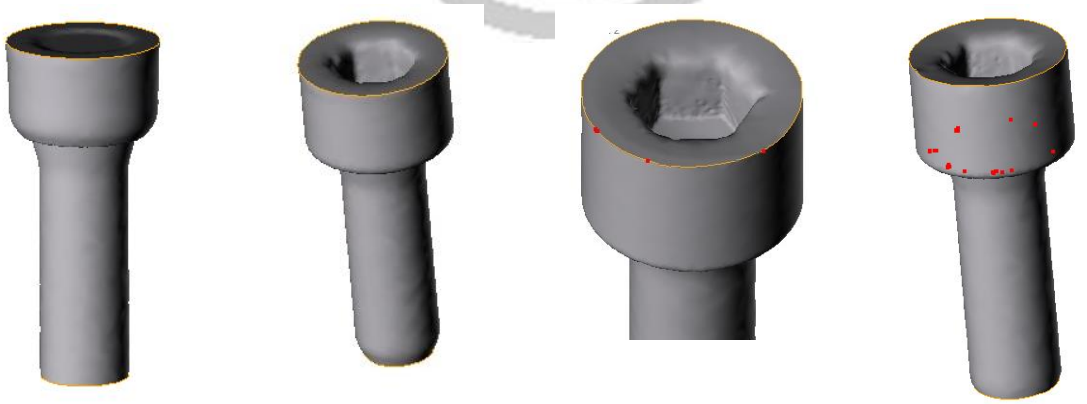


Figure 3-7 Tolerance range curve.

The variation of size in range 6.08 (actual die size) - 6.06 is tolerable but over the range of 6.06 the dimensional changes led to considerable amount of damage to the material, which can't be analyzed through the load analysis but shows formation of laps on the formed product. The part formed with dimensions 6 (d) and 6.02 (c) shows red dot formation which predicts the possibility of lap formation in the parts as shown in Figure 3-8.



(a) **Before Forming**

(b) **No lap formation**

(c) **Lap formation (less)**

(d) **Lap formation(High)**

Figure 3-8 Variation in parts with tolerance.

Case 3: Analysis of multi-station forming parts load curve

In this case study three different parts with varying intrinsic design from machineries as mentioned in Table 2-2 would be studied.



Figure 3-9 Hex head formed parts.

In order to simulate for defined load curve subsequent stations are designed and modeled to obtain the exact results required to check for the defined condition. The Figure 3-9 shows the layout of the process station parts, formed by the machinery. Figure 3-10, Shows the Final Die and punch designed for simulation to achieve formed part as show in the Figure 3-11, the dies are designed with tolerances, which in result give the simulated part as manufactured by machinery.

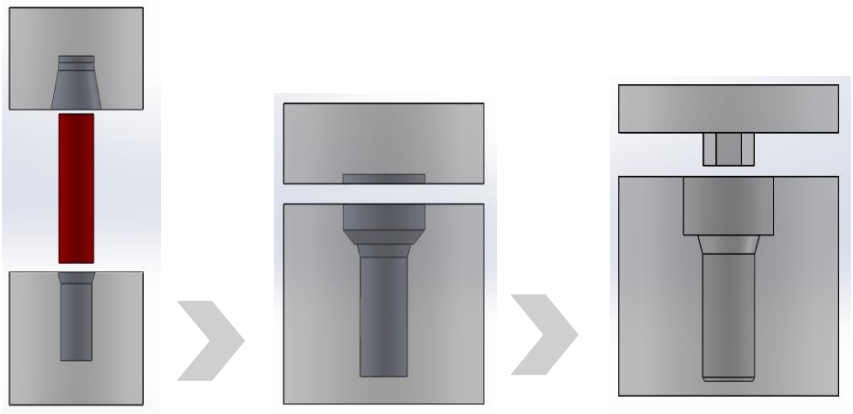


Figure 3-10 Hex head punch and die molds

The Simulation results for each process stations are described with the load-displacement curve Figure 3-11.

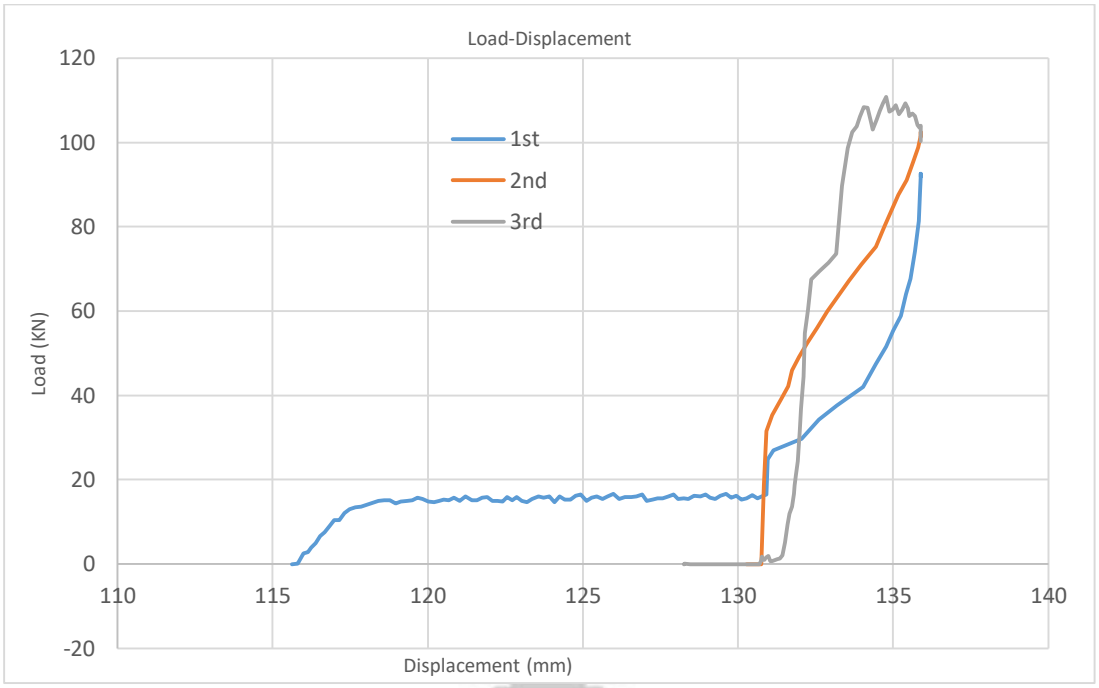


Figure 3-11 Load-displacement curve for Hex-head screw.



(a) Actual part (b) CAD model (c) CAD model (Hidden lines)

Figure 3-12 Plus head (a) Actual part, (b) CAD model, (c) CAD model (hidden lines).

The parts obtained after the actual forming practice is taken as sample from the company to simulate for the process and obtain various data with help of simulation which could help us obtain some characteristics of the actual forming process.

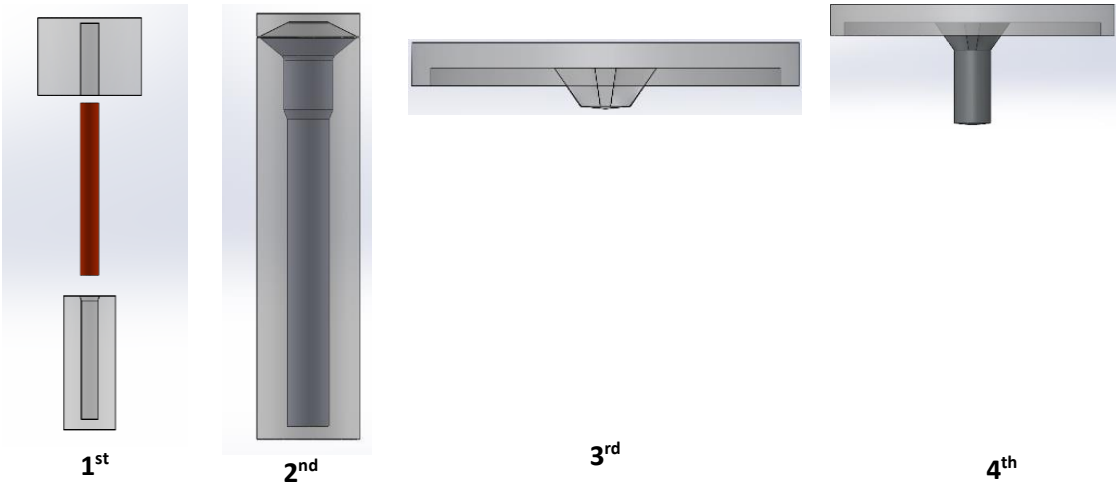


Figure 3-13 Plus head punch and die design.

The final die designed after considering the tolerances to obtain the desired shape is then put in conditions to simulate for various conditions.

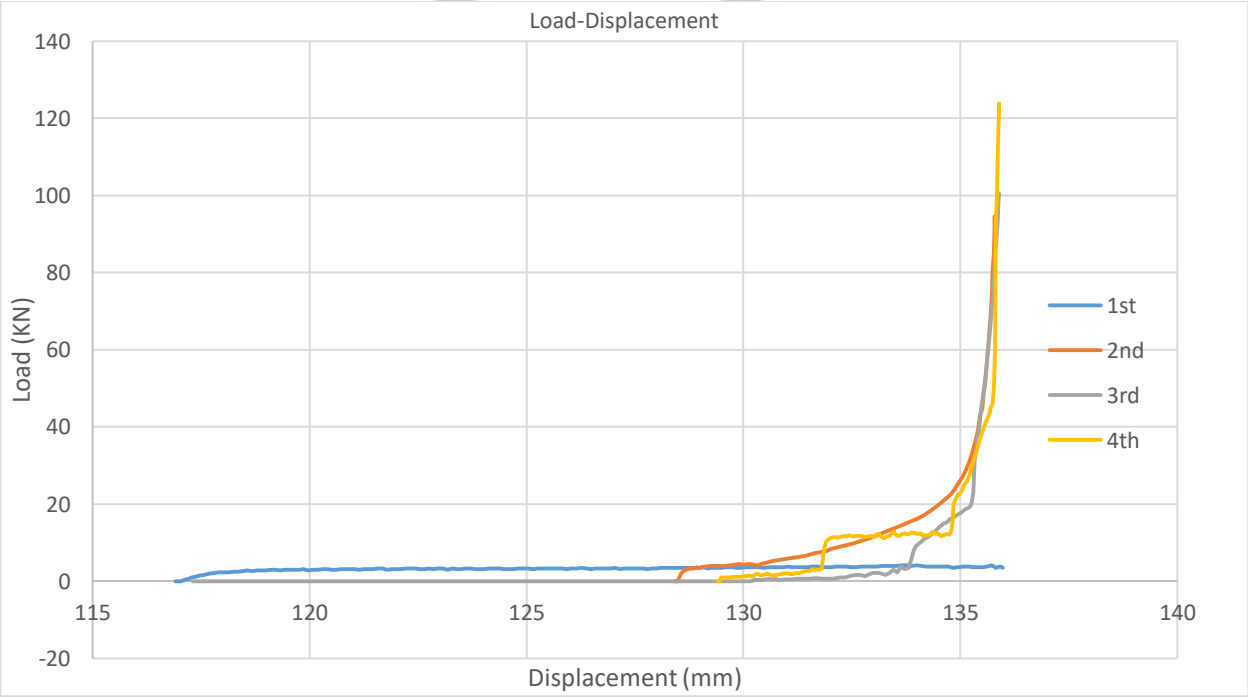
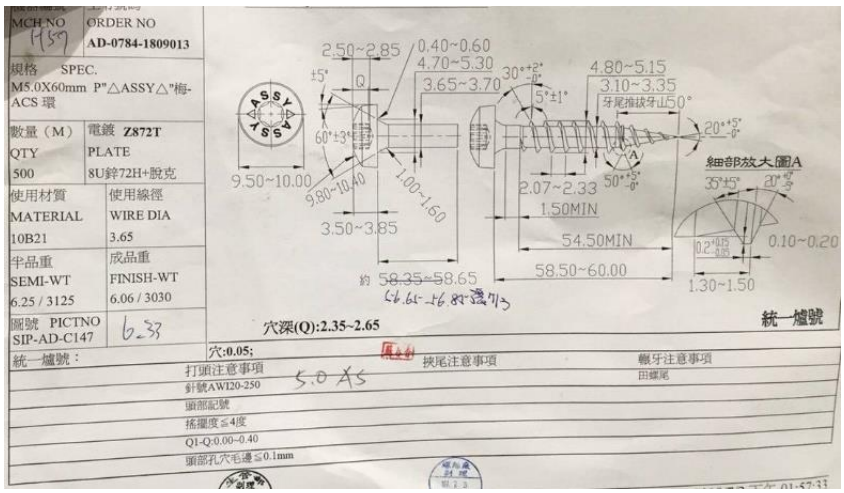


Figure 3-14 Load-displacement Plus head screw

The ASSY Screw consists of only two process stations with 2-D Modeled simulation followed by 3-D model as can be seen in Figure 3-15.



(a) Actual parts



(b) Final Part drawing



(2-D) Analysis 1st **Formed Part** **(3-D) Analysis 2nd** **Formed Part**

Figure 3-15 ASSY Head (a) and (b) actual parts and below are the simulated parts corresponding to dies.

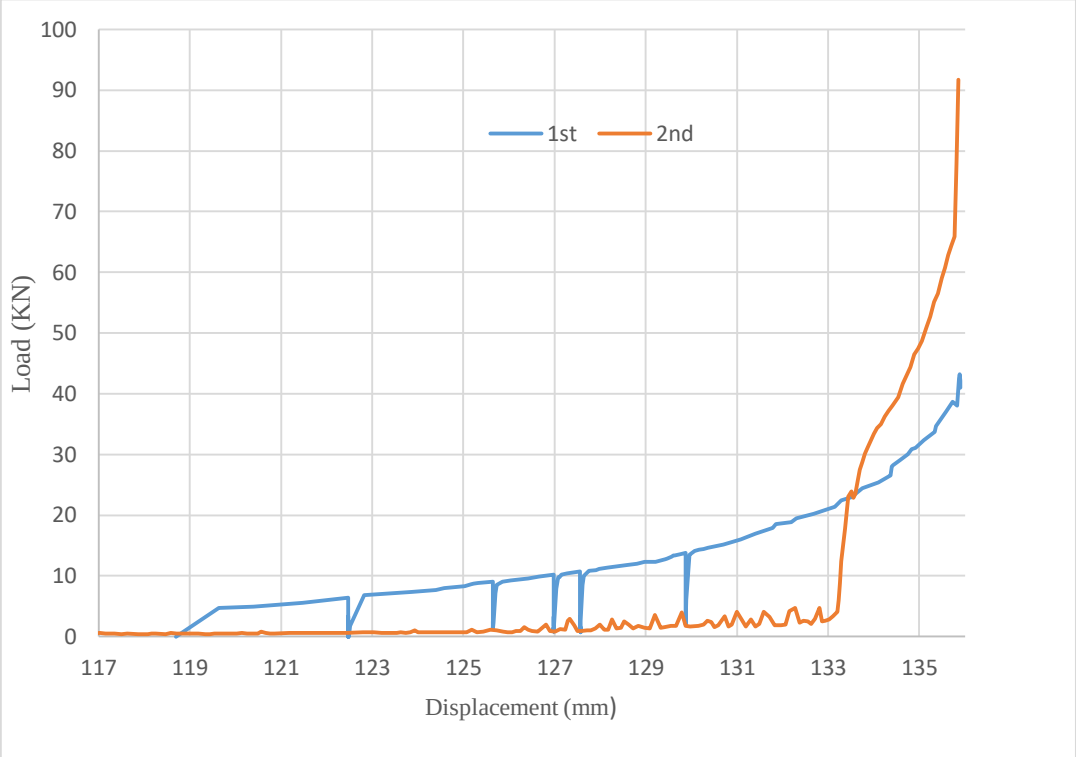


Figure 3-16 Load displacement curve ASSY head screw.

The results that could be obtained from the analyzed method are as follows:

(1) The Hex-head simulated to analyze the load behavior under the conditions of **with and without tip** showed the load behavior of the tool in relation to formed part could be used as the signature to define the variations of similar kind of wear.

(2) Similarly, the results obtained from the same Hex -head bolt to analyze the tolerance range could use, by taking in conclusion form it that the load varying curves can't be chosen as one of the parameters to describe the health of the tool. As load curves for all most all dimensions don't show any significant changes which could describe the quality of workpiece, but the simulated formed parts could be seen with lap formation.

(3) The other simulated formed parts like Plus screw head and ASSY head shows the load pattern stating its variation with displacement, which could be used for the comparison and study with raw data from sensors.

(4) The load curves simulated for each part gives the detail description about the machinery working with respect to the various preforms that could be used as fundamental behavior of the load when sensed with transducer to obtain the load pattern of the forging.

3.1.3 Simulation designed model and boundary condition check

Case 1: Analysis of 2-D and 3-D parts with uncoupled and coupled

The 2-D and 3-D curve formed are very similar as can be seen in Figure 3-17.

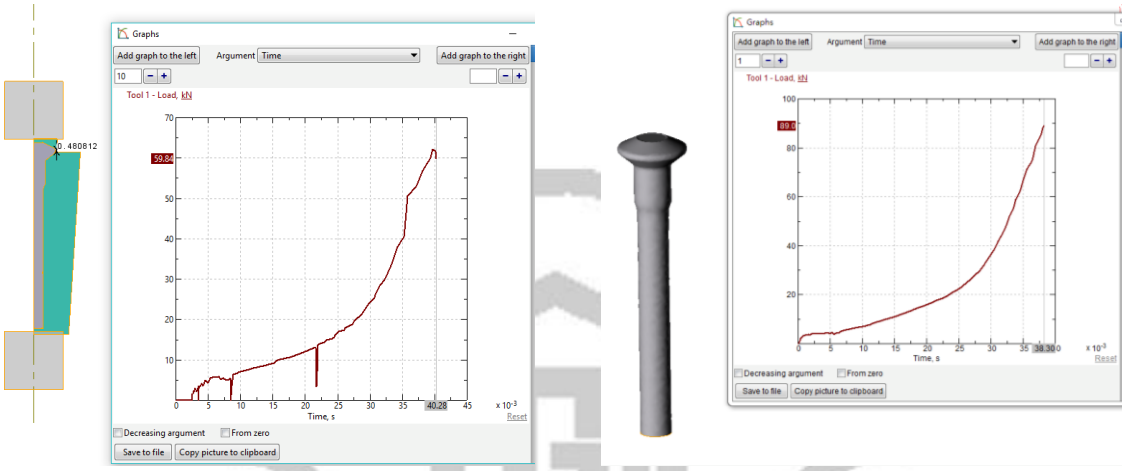


Figure 3-17 Comparison of (a) 2-D and (b) 3-D curve for uncoupled conditions.

Plot in Figure 3-18 shows a relation in between two case of coupled and uncoupled plus head type shows a similar varying slope, which confirms that not much variation would be observed when the simulations are done considering either as uncoupled or coupled condition. The coupled condition gives very realistic results but with few drawbacks such as time consuming with 3-D models and comes with many additional boundary conditions to get desired result.

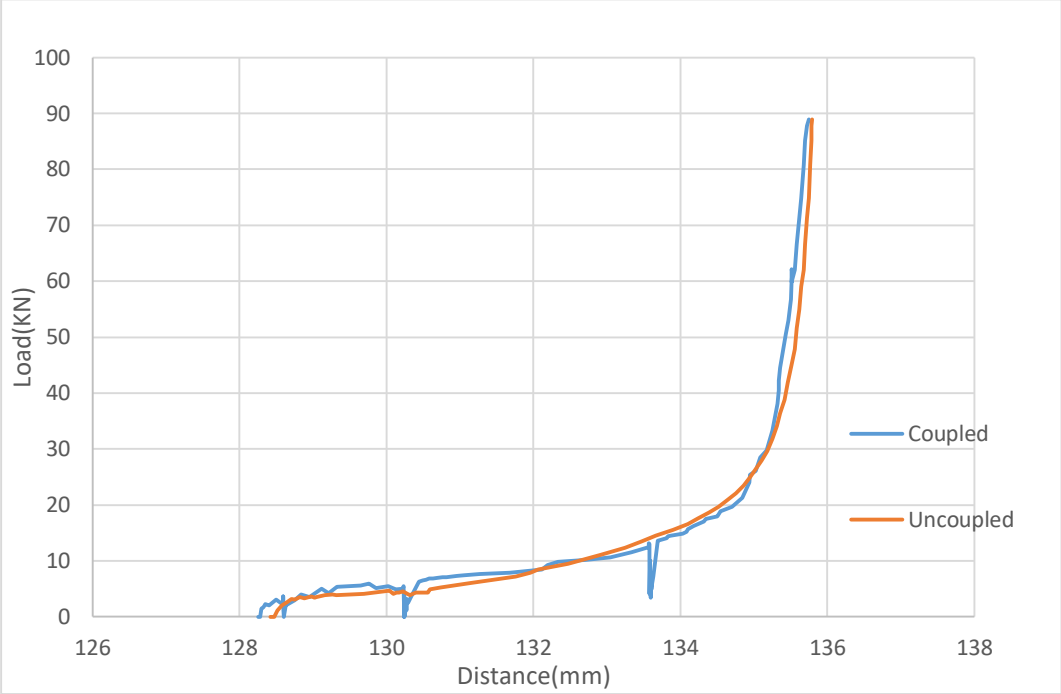
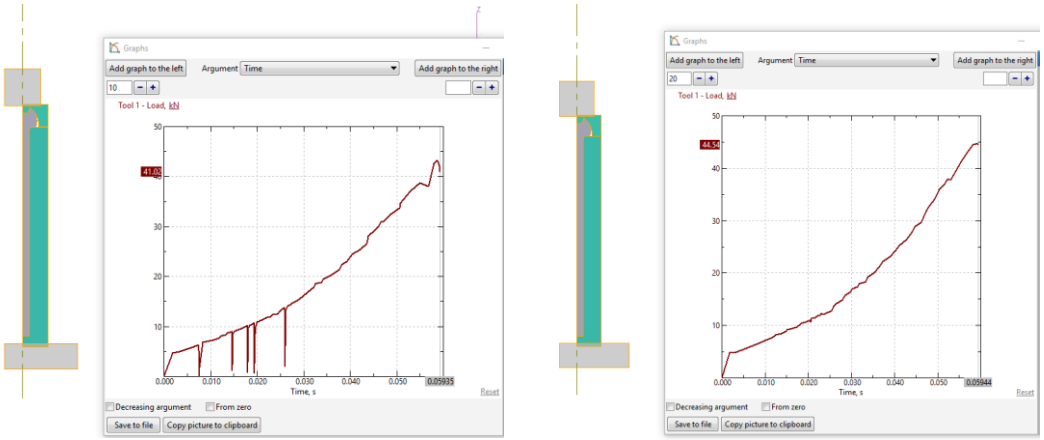


Figure 3-18 Comparative load variation curve for coupled and uncoupled conditions of plus head screw.

Case 2: Analysis of 2-D and 2-D parts with uncoupled and coupled simulations

Another analysis done on the other type of part, and same characteristics could be noticed. In this analysis both the models are kept in 2-D to analyze the consistent behavior of the uncoupled and coupled irrespective of model type being 2-D or 3-D as shown in Figure 3-19.



2-D Coupled **2-D Uncoupled**

Figure 3-19 2-D Models comparison.

The analysis done on cases in realtive is shown in Figure 3-20, showing downfall peaks like cracks, which is very similar to the real signal data obtained from sensors. This type of curves confirms the efficent behaviour of simulation, which could be used as one of the method of analysis to establish fundamentals required to build monitoring methods easily with less time and expence, which won't be possible to analyze if dependent on the real data analysis of the machinery.

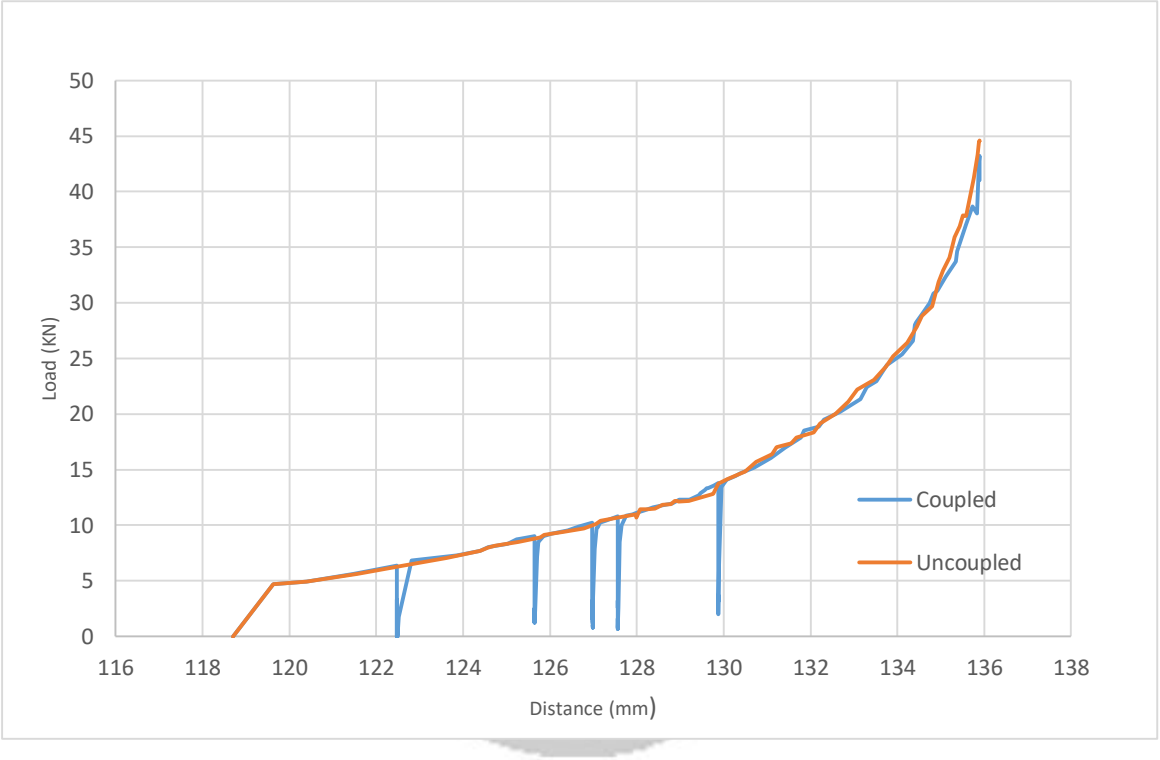


Figure 3-20 Comparitive load curve for 2-D models of ASSY screw.

The method discussed above could help in the analysis leading to results as follows:

(1) The above analysis proves that the data of load variation if obtained by analyzing either 2-D or 3-D analysis would not be vary much and hence both data could be used for load analysis.

(2) The other important contribution of this analysis is to study one of the boundary conditions criteria named as coupled and uncoupled leading to very realistic condition jerks.

(3) The analysis proves that other than realistic sharp pointed falls and rise which could be the jerks or impulses caused due to the interaction of the die and punch the load follows the same pattern.

3.1.4 Curve summation analysis for estimations of peaks and falls.

3.1.4.1. Plus head screw summation curve: In the Figure 3-21 the load curve obtained in all four stations are added to obtain the resultant curve. It's the overall estimation of load that could be observed at each instance of forging stroke length.

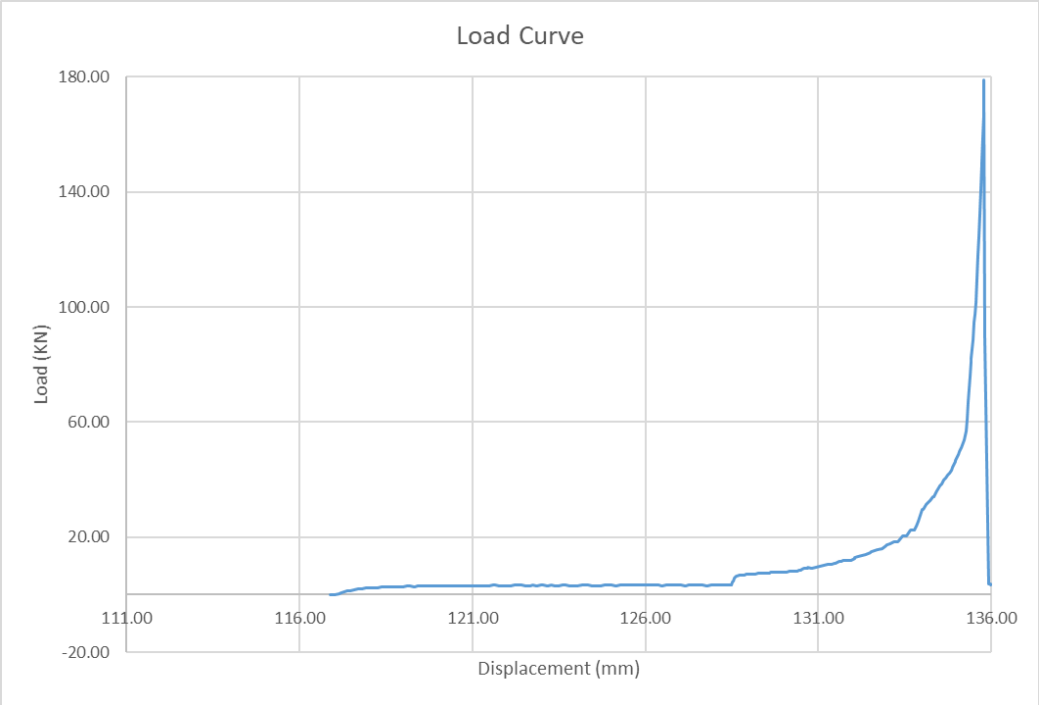


Figure 3-21 Load curve of plus head screw (4-process).

3.1.4.2. ASSY head screw summation curve: It has 2 pre-forms hence two substation assembly.The summation curve didn't fall in this case because of its both sub-stations has peaking curve till the end of the final stage forging as can be seen and analyzed from the curve in Figure 3-22.

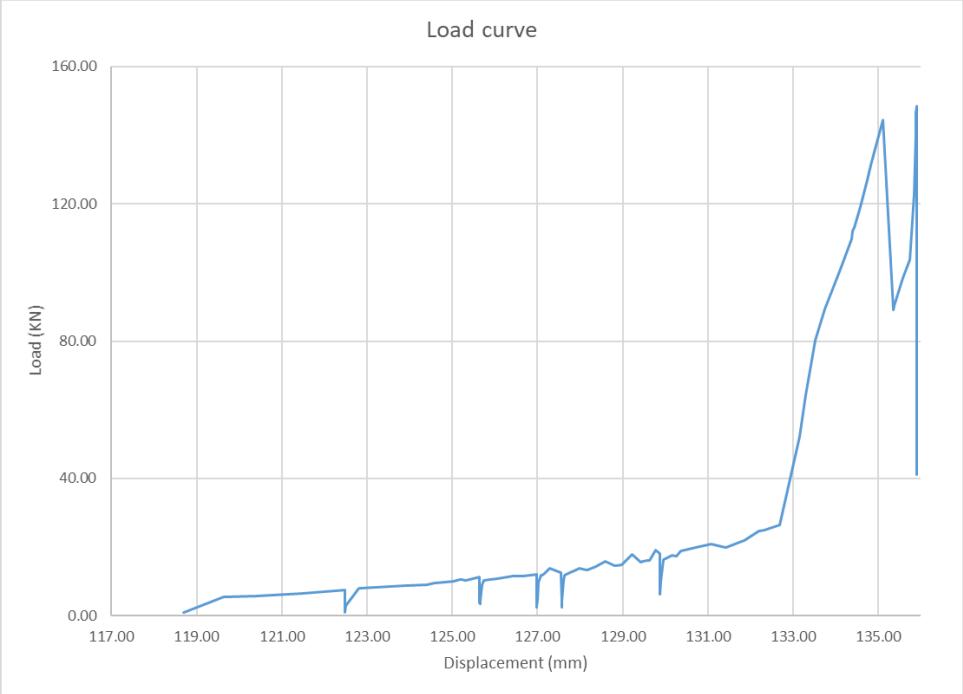


Figure 3-22 Load curve of ASSY head screw (2-process).

The result of the summation curve analysis are :

(1) The summation curve overall indicates the behaviour of fall and rise in load pattern.

(2) The load curve obtained above can be used in comapritive analysis with the signals obtained from sensors.

(3) In this way the summation load curve could be used as a guide to known the signals associated with signals corresponding to the loads at defined stroke length.

3.2 Methods established with processed sensor data

The methods of signal analyzing and processing as described in methodologies is followed with result and analysis of the processed data. The result analysis of the machinery by sensors in this section is done with consideration of few other sourced data obtained from the actual machinery workplace. The other research has been looked up for confirming the scenarios, which proves our working methods as universal and commonly find to similar work done by the authors as described by them in their paper. The type of phenomenon observed and common to other research done on monitoring of the forging machinery is the working of the machine in cold phase.

Cold phase of machinery- It was analyzed during the cold working phase of the cycle that's the time when machine didn't reach its steady state and temperature the parts produced were not acceptable and considered to be bad. The reason according to various research's done on similar type of forging machinery states that the temperature of the dies affects the quality of parts manufactured. In the paper "Measuring & control systems in industrial die forging processes" the above scenario is stated and proved with concluding remark stating, that in industrial forging the principal parameters are the forging forces as a function of time, displacement, and the temperature of the tools and the preforms [11]. The analysis ahead is based on conditions such as empty working, cold working and warm working of machinery in two different parts for checking the trend. The conditions are made in order to find the hidden signals which would be little difficult to analyze without this comparative analysis.

3.2.1 Time waveform analysis

The time waveform is the analysis where the signals are analyzed by its phase with respect to time. The time wavefrom analysis is checked to see the variation of phase which are the combination of sin waves originated from the jerks indicated by the accelerometer installed on the sliding block.

Case 1: RMS and mean value analysis of waveforms

The variation of each cycle by its RMS and mean value has been used to anlyze the periodic nature over defined time as shown in Figure 3-23.

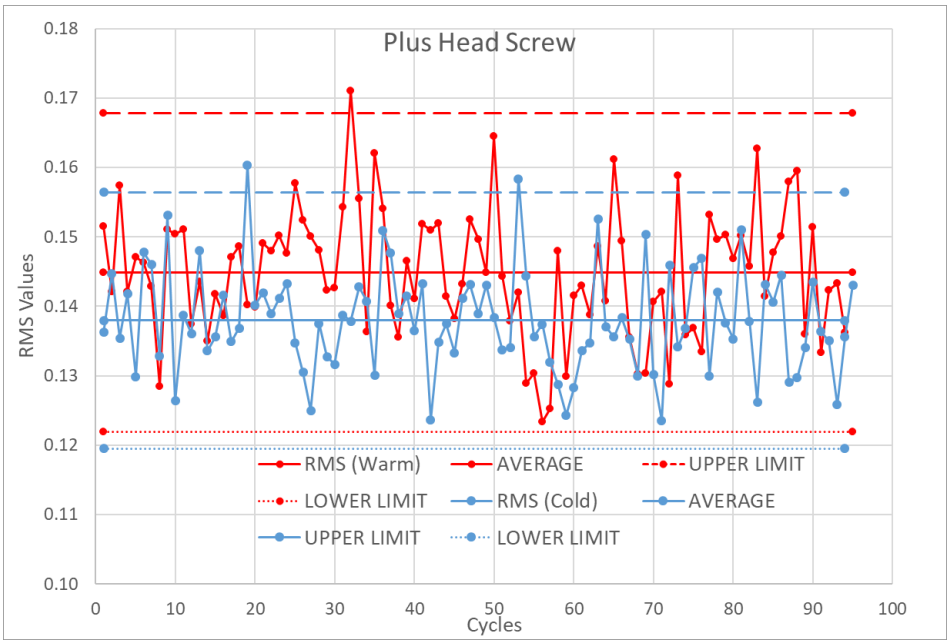


Figure 3-23 RMS value for 95 cycles of plus head screw warm and cold.

The RMS and mean value of cycles can help in check for the variation in cycles with respect to upper and lower limit range which could be set on the various parameters. The limit range set over here has the tolerance of 2.5 sigma, which could be a good tolerance range in the industry. Many similar observations in response to different conditions were done to typically examine the behavior with the described statistical analysis to monitor the behavior of dynamic machinery in various conditions.

The mean or the average of the warm is shifted up if compared with the cold phase, which in result affects the limits. The same type of analysis is done on the other type of the part with similar conditions to track the conditional pattern.

It can be observed from both warm and cold analysis that the RMS value fluctuate around the average value, the increase in RMS value could be the cause of random noise from other connected elements but if it reaches more than the upper limit then it can't only be the noise inside the cycle as it's not that frequent and hence could be embedding the vibrational signal which would cause catastrophic breakdown. Hence this data along with certain conditions is observed to analyze the pattern and its cause.

Another part named with ASSY head on other machinery is analyzed with the RMS and average value with same warm and cold condition. The warm cycle is shown in Figure 3-25.

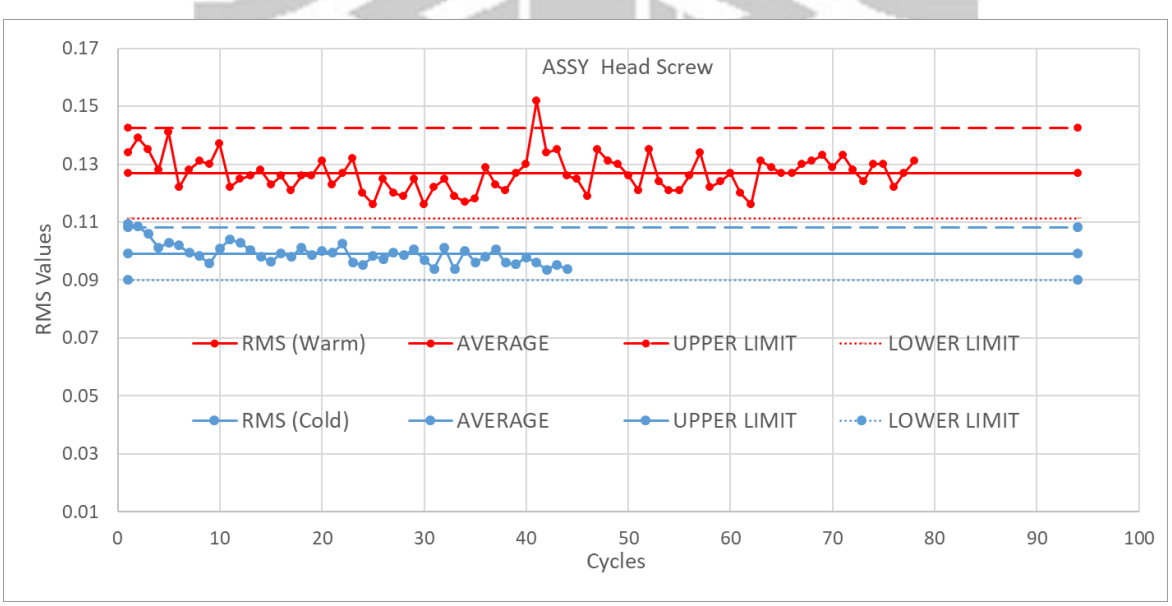


Figure 3-25 RMS value for 78 cycles of ASSY warm and cold cycles.

The plot for the warm cycle phase consists of 78 cycles recorded over a time. It is compared with the cold cycle phase. The

The data for the cold cycle consists of 44 cycles with RMS and the mean fluctuating between the limits. The results of the whole RMS analysis of two distinct parts with different conditions is stated below in results.

(1) For the analyzed cases the RMS value of each cycle of the machinery, the cold cycle shows the decreasing trend when compared with warm.

(2) As the cold parts are not considered to be good and the trend for this condition is indicated by decreasing nature of mean value .

(3) Thus this fluctuating trend of average of the RMS value along with RMS could be used as a indicator to monitor the raw signals obtained directly by setting the limits.

Case 2: The normalized RMS of the forged cycle w.r.t whole cycle

After estimation of the RMS value of the whole cycle the normalized RMS of forging period and of whole period is checked to analyze the severity of forging phase between warm and cold. The result of warm condition plotted is shown in Figure 3-27.

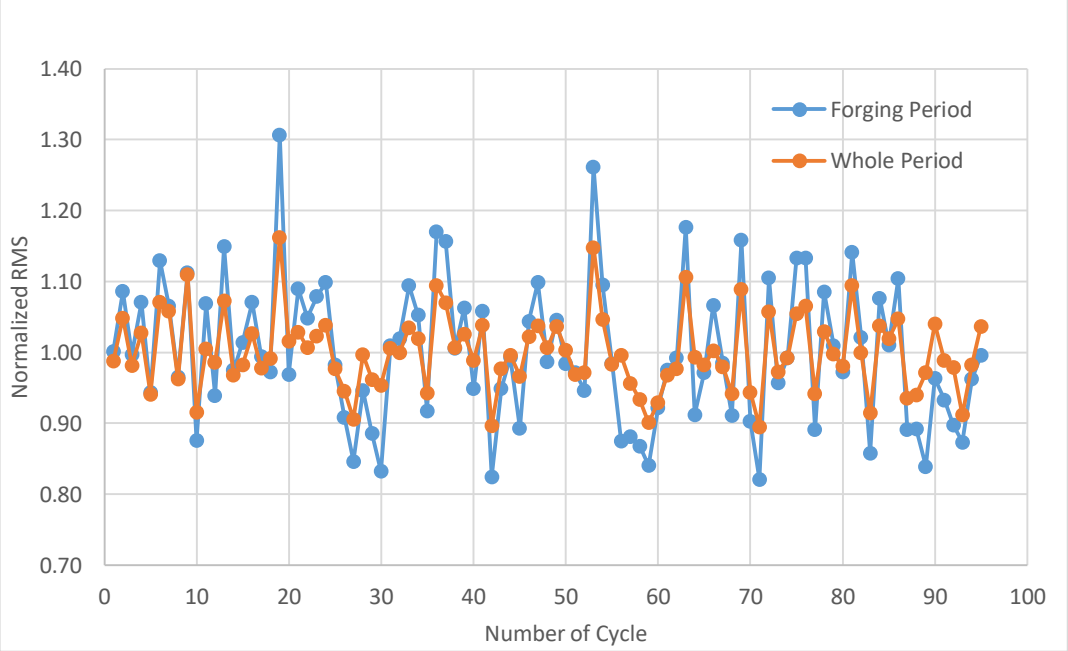


Figure 3-26 Normalized RMS of warm cycle for plus screw head.

The RMS plot of the forged warm cycles shows increase in peak comparitive to whole cycle. The same trend is followed depicting same features of RMS curve as shown in warm forging.The analysis done on the cold cycle is shown in Figure 3-27.

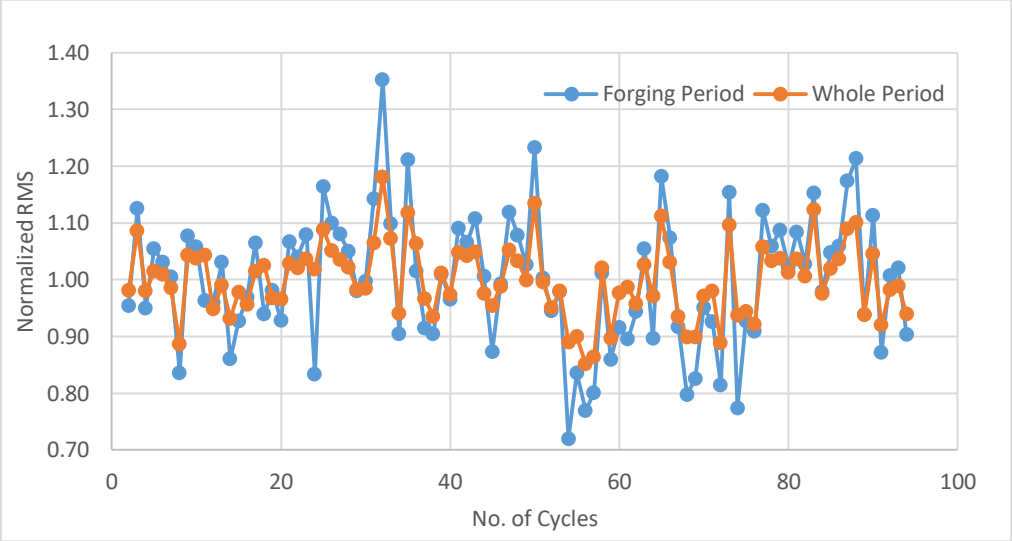


Figure 3-27 Normalized RMS of cold cycle for plus screw head.

The result for the normalized cold cycle also shows very similar trend of high peaks in case of forged period.The results of the analyzed condition are:

(1) The fact that the most of the signals constituted inside the whole cycle consists of forging influenced signals, hence the forged and whole cycle peaks looks very similar but with higher amplitude.

(2) This statistical cases like RMS and mean values proves the driving nature of the forging signals and its monitoring importance in case of bulk forming machineries.

3.2.2 Spectral analysis

The spectral analysis using the fast fourier transform (FFT) on the signals from time series is good for finding the major frequencies of the vibrations occurred in the machine cycles. Figure 3-28 shows the spectra of the three cycles from the aquired data of empty, cold and warm machine, respectively, producing the cross head products. The python code for this FFT is listed in Appendix D.

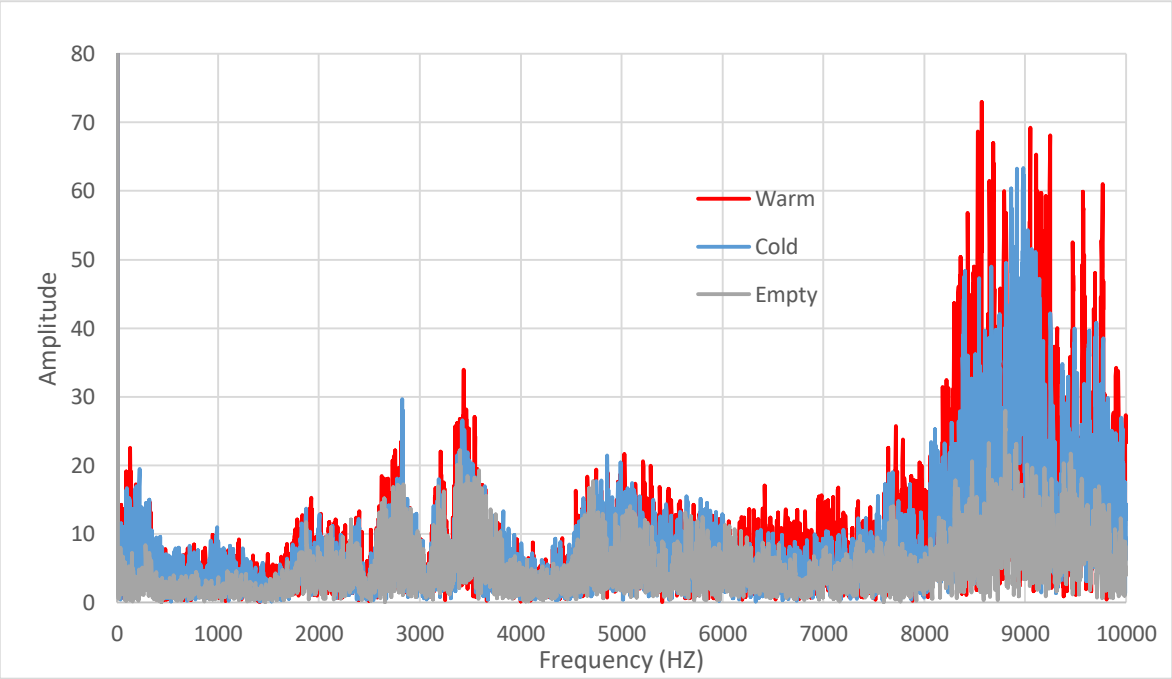


Figure 3-28 FFT analysis of plus head screw.

The results show the comparitive relation between the warm, cold and empty cycles. The features appearing in the spectra of all three cycles indicate that the sources of the features are originaed from the machine itself. It is evident from the plot that the spectrum with the highest amplitude is that of the warm cycle while the lowest amplitude is that of the empty cycle. The similar analysis with ASSY head part is done for the analysis of the trend in variation of parts. The spectra of the three kinds of cycles are plotted Figure 3-29.

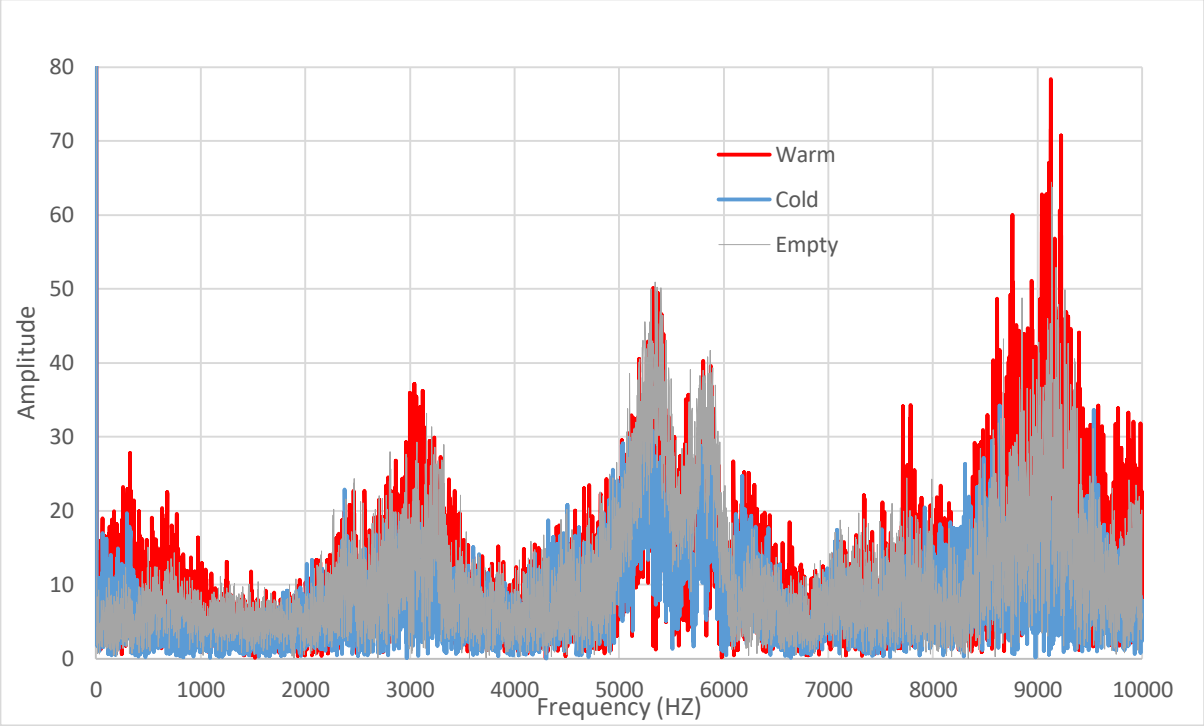


Figure 3-29 FFT analysis of ASSY head.

The FFT analysis of the ASSY head also shows the higher amplitude frequencies with warm but the empty signal in this case doesn't show the low amplitude. The FFT of the signal obtained above give a very fair representation of the power intensity assosciated with each frequencies along with comparitive analysis in different condition.

To exactly know the reason and see in detail the position of high amplitude frequencies and their origin, the spectrogram is studied ahead for further analysis. The spectrogram is computed by short time fourier transform of the time varying signals with help of moving window and moving step for its detailed information.

3.2.3 Time-Position-Amplitude-Frequency Signal Analysis

The location of forging is very crucial to know in order to exactly find the information embedded in the signal while forging. The conditions checked for it is again on the basis of warm, cold and empty stroke condition especially to analyze the location of the forging stroke with comparision study as shown below in the Figure 3-30.

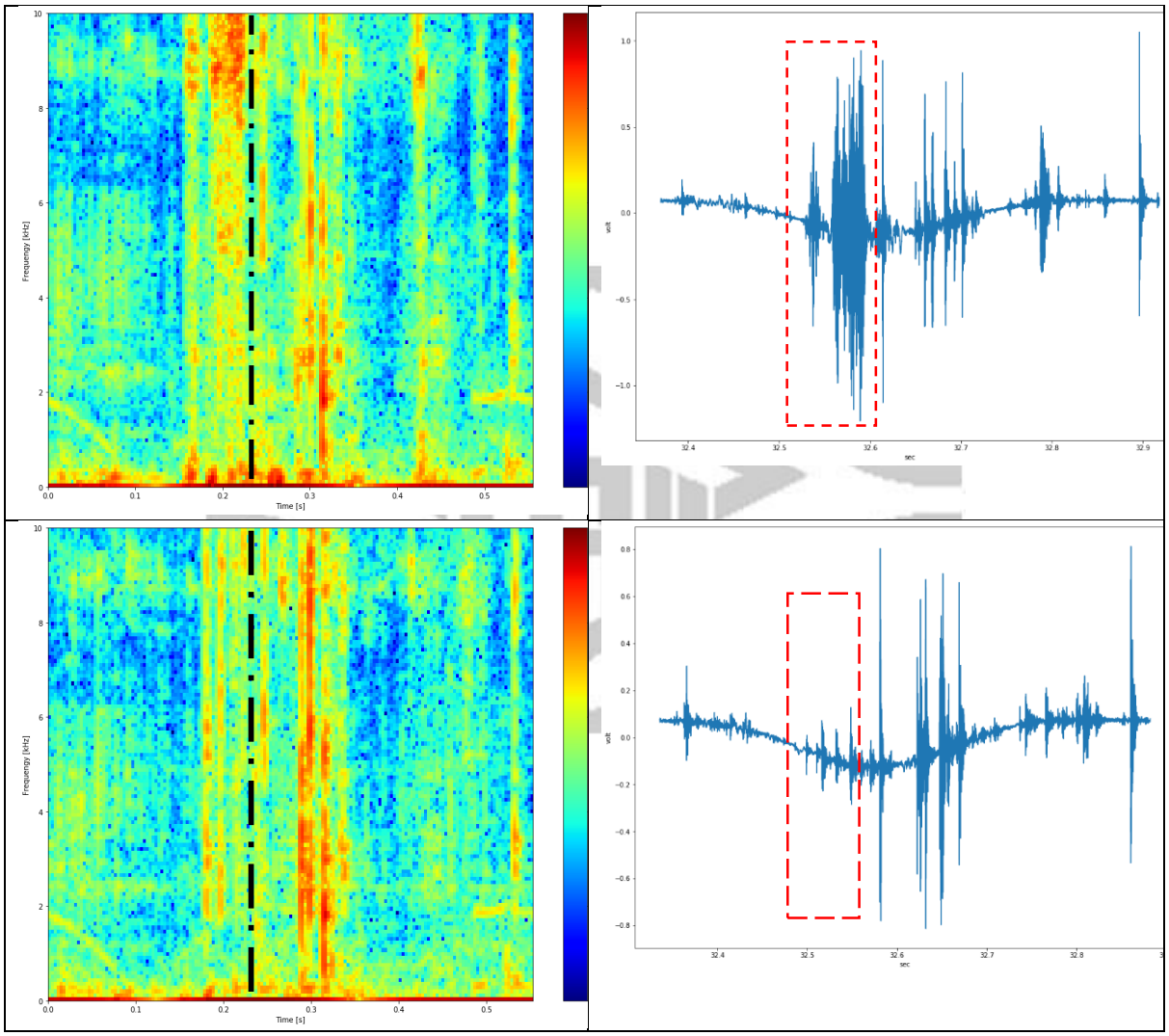


Figure 3-30 Spectrogram (left) and Time-waveform analysis (right) of plus screw head for loaded cycle (top) and empty cycle (bottom).

After getting the confirmation of the location of the forged area, in order to have the location of it, the analysis done for the position could be used as considered and used ahead with Figure 3-31.

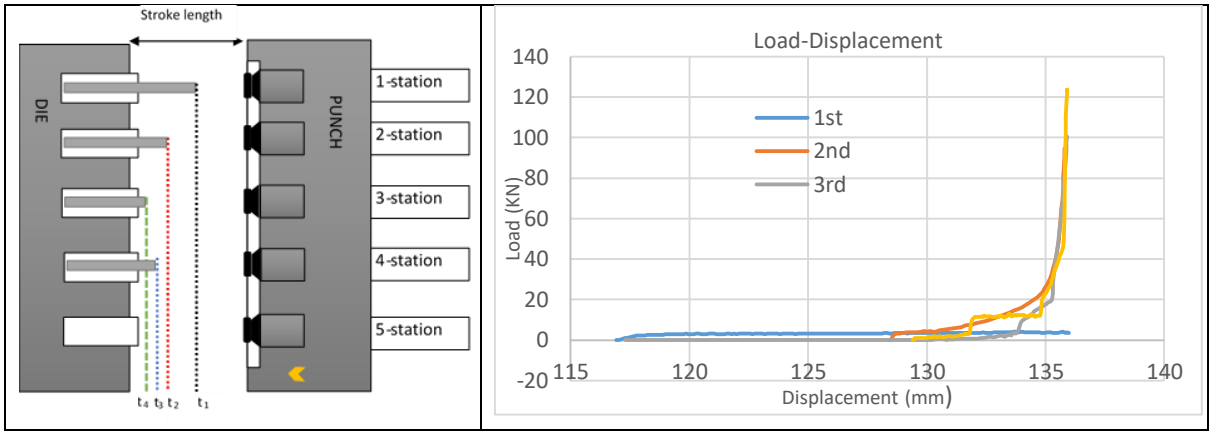


Figure 3-31 Load curve for estimating the forging station.

The dashed colored lines (t_1, t_2, t_3, t_4) shown in machinery top view schematics Figure 3-32 are time order used to show the relation with dashed line box besides the Figure 3-33 for comparitive analysis.

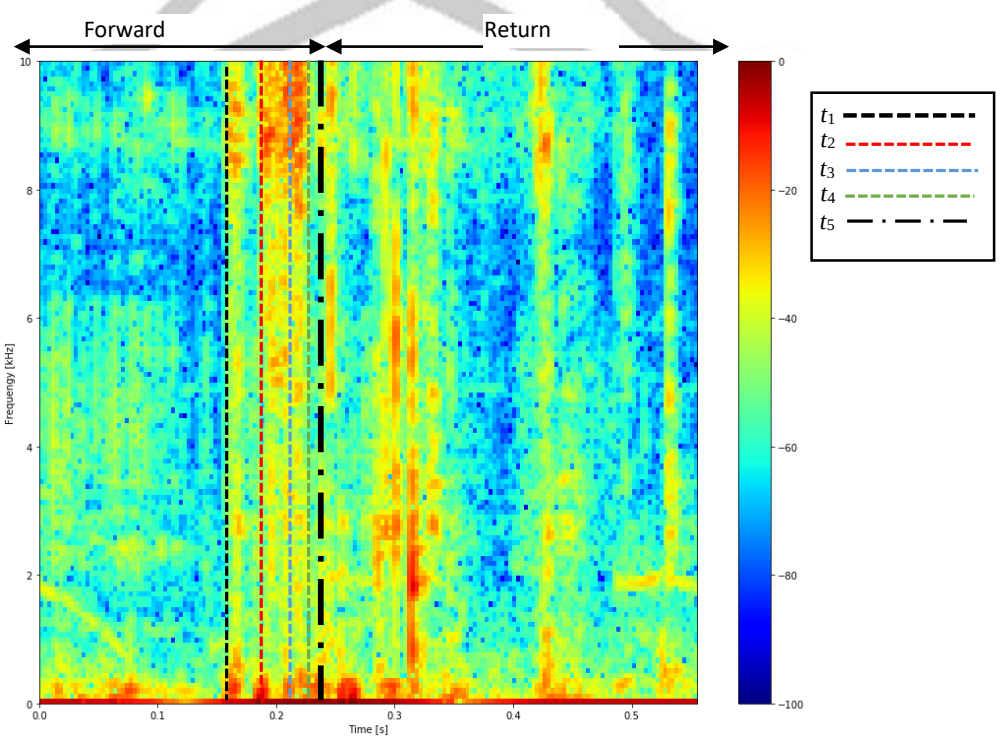


Figure 3-32 Time frequency analysis (Spectrogram).

The spectrogram in Figure 3-32 is an result of the averaged cycles, which is used here to see the relation of forging frequency at each time instance with the sliding block of machinery top view schematic Figure 3-31. The load-

Displacement curve relates the postion of preforms at which they are channeled in the forging stations.The analysis like this could be used very efficient for monitoring analysis of the forging position along with time, frequency and intensity assosiated with each forge stations.The steps to have a clear view and understanding of the procedure followed above is discussed below:

(1) At first the with help of load-displacement data obtained from the simulation can be looked up to see the position of preform and after getting the estimation of the position the stations could be marked and denoted with time order in which they occur.

(2) After acknowleging the location relative to time, it would be eassy to look in the spectrogram for the sequentional stations according the time order to find the frequency and intensity assoscited with it as shown above with dashed lines in the spectrogram.

The results that could be taken up from the analysis are as follows:

(1) The method described above can help relate with 4 parameters at instance from real time working data and can fetch many variations by which the the monitoring can be done.

(2) The location of 1st preform is at 19 mm, 2nd at 7.5 mm, 3rd at 2.93 mm and final at 4.6 mm from the stationary die. The location are very precise and obtained through simulation curve.

(3) The trend follwed in selection of prefrom is not in seqence order with the four lined preform station in this plus head screw case.

(4) The tracing of signal with signature pattern become easier due to combination of time and displacement featuring techinque.

The other part ASSY head screw is also checked in same way to find the information embedded with respect to its poistion and region of forging area.

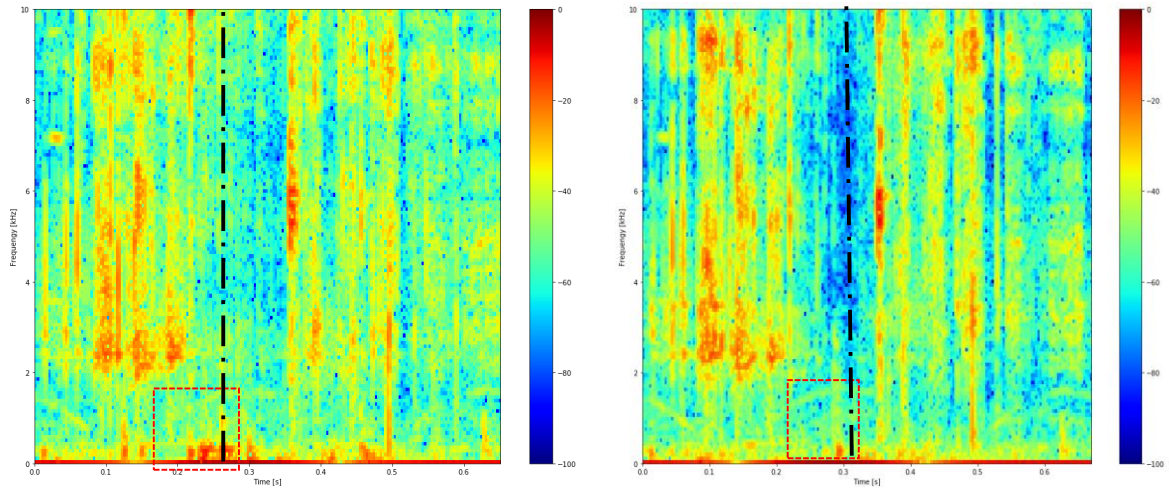


Figure 3-33 The spectrogram for one cycle in the process of production ASSY head screws. (Left: a loaded cycle; Right: an empty cycle).

The analysis of empty and loaded data as discussed above helped in estimation of the forging area and location with comparision Figure 3-33. The position is then checked with simulated data in Figure 3-34.

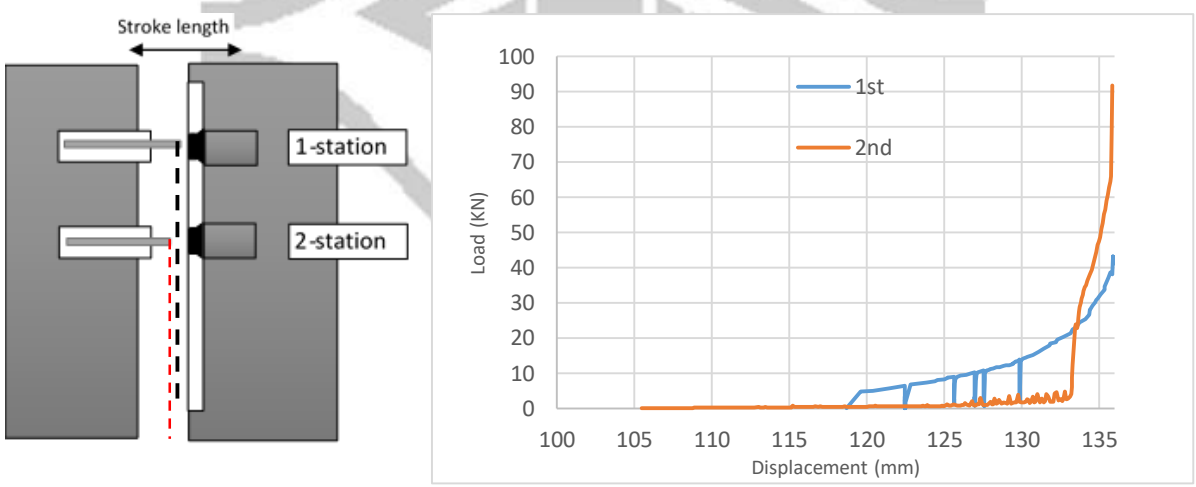


Figure 3-34 Load curve for estimating the forging station of ASSY head screws.

The same method analysis is followed here with (t_1, t_2, t_3), as here there are only two process station.The t_1 and t_2 denotes to forging station time with respect to position and t_3 denotes the combination juction denoted with grey colors .

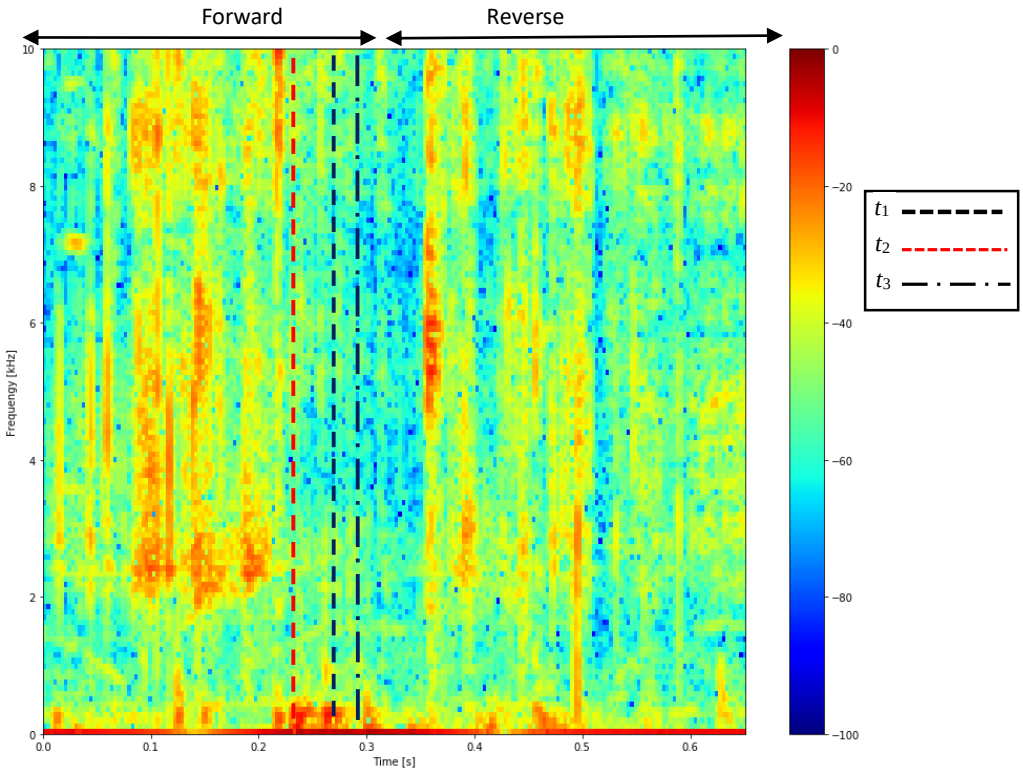


Figure 3-35 Spectrogram.

The spectrogram as shown in Figure 3-35 is then looked to observe the amplitute and the frequency variation with the same method as stated above in description for the analysis of plus head screw.The results for ASSY type part is as follows:

(1) Comparitive to the Plus screw head it has only 2-stations and the frequency with high intesity is spread out more comparitive to the plus screw head.

(2) The 1st preform is at 17.18 mm and the second is at 3.24 mm from the stationary die and follows the sequential order.

The high and low frequency power spectrum of forging period from spectrogram is extracted to examine the behaviour of the pattern which could be used as a signature curve for monitoring of forging period.

3.2.4. Power spectrum of forging stroke

Case 1: Lower frequencies power spectrum

The absolute value of the short fourier transform signal with low frequency ocurred during the forging stroke is summed up to obtain the curves as shown in the Figure 3-36.

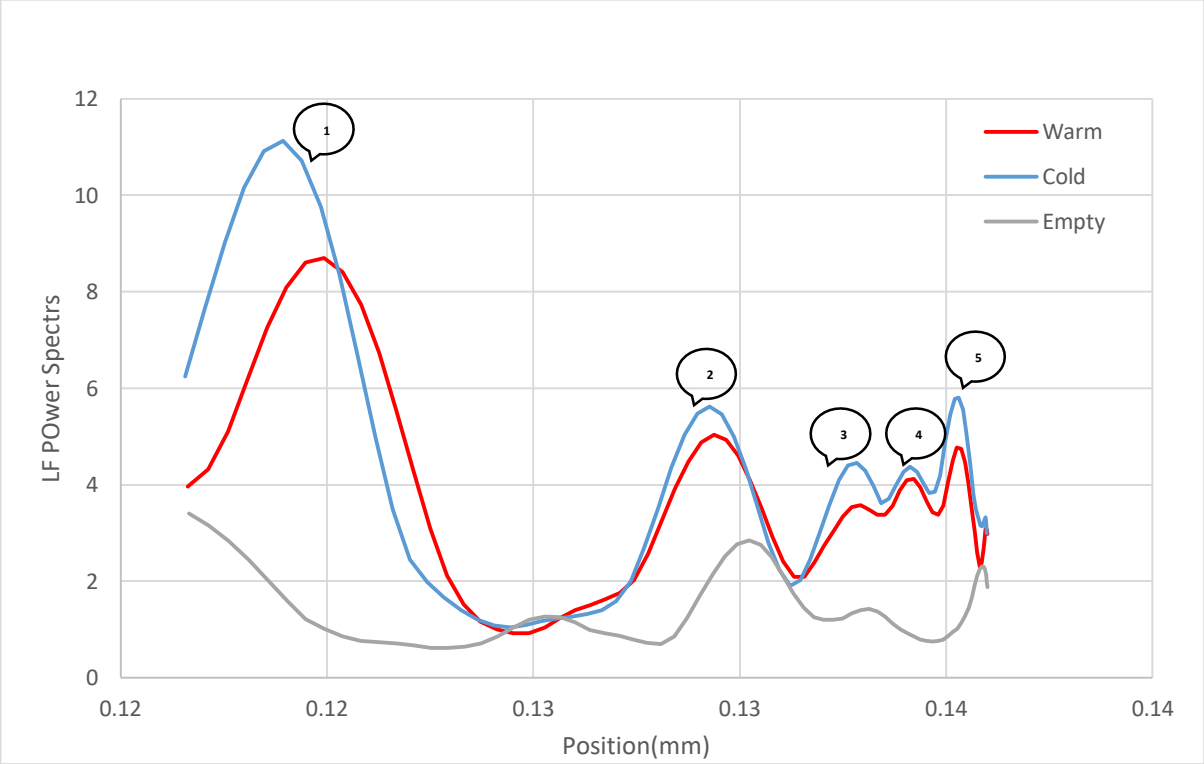


Figure 3-36 Power spectrum of forging stroke for plus screw head.

The spectral power has $n + 1$ peaks where n denotes the number of preforms. The additional peak is the result of the junction where all stations meet to finish the operation. In time frame, each preform stations has a unique time inetrval along with the junction where all the station end. When comapared with simulated load-curve the peaks with preform numbers can be seen as shown in Figure 3-37.

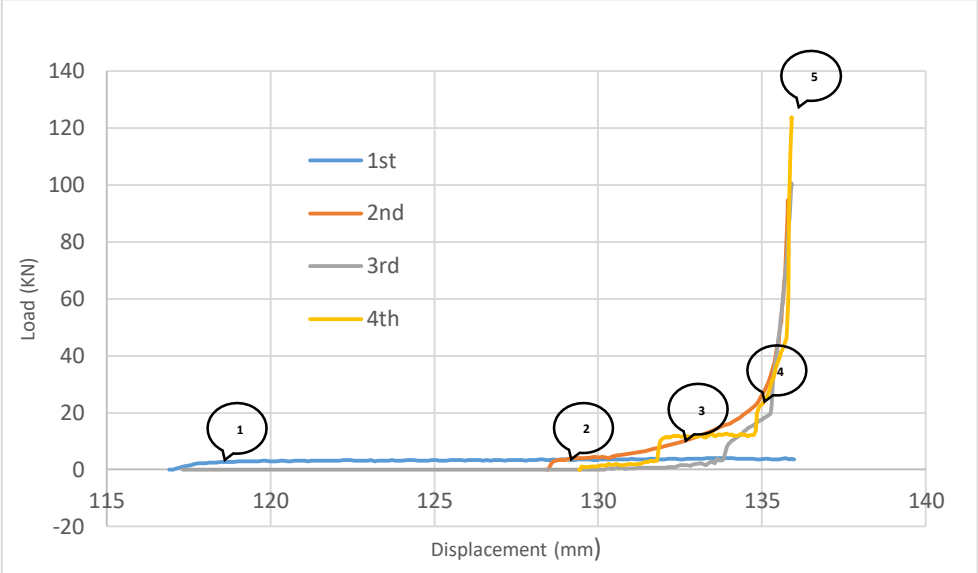


Figure 3-37 Load-displacement curve with preform location.

The power spectrum of the ASSY head type screw is similarly checked to see the nature of both the warm and cold cycles with variation of position. The load digram along with power spectrum is used as above to see the peaks position and estimate the preform station that's cause of the spectrum peak.The computed power spectrum of the ASSY head is shown in Figure 3-38.

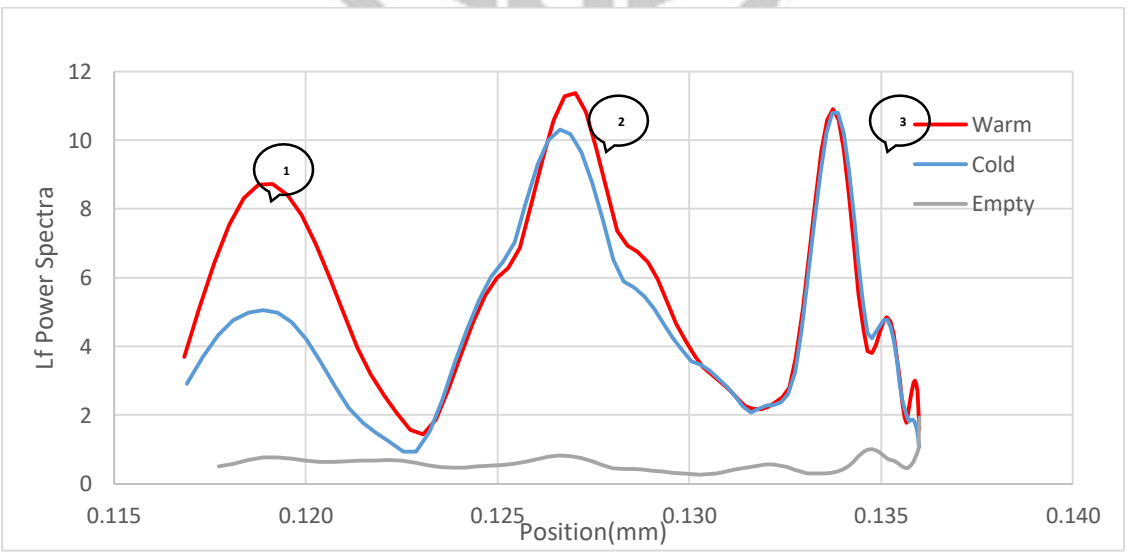


Figure 3-38 Power Spectrum of forging stroke for ASSY head.

The load curve for the part obtained through simulation is shown in Figure 3-39.

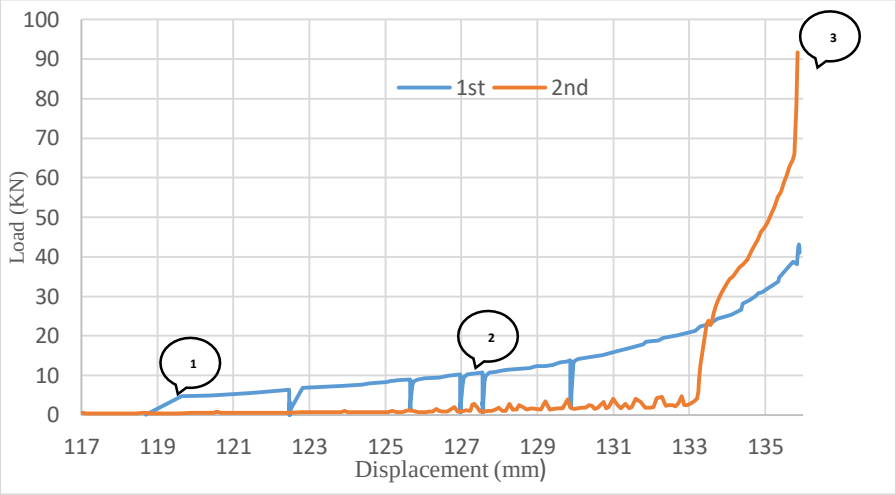


Figure 3-39 Load-displacement curve.

These waveform can be used as the signature of the low frequency power spectrum to correlate the differences and could be used for monitoring the process. Its observed during the spectrogram analysis that in Plus head screw frequency-time domain Figure 3-32 plot there is a presence of higher frequency and intense bands, which is not significantly noticed in the ASSY head. In order to see the nature of frequency of higher frequency bands associated with Plus head screw high frequency spectrum is captured from the absolute value of FFT values.

Case 2: Higher frequency power spectrum

After the extraction of the sum of frequency bands, it was observed that the high frequency signals embedded with high frequency noise doesn't comprise of forging signal peaks assosciated with realtive stroke length as shown in Figure 3-40.

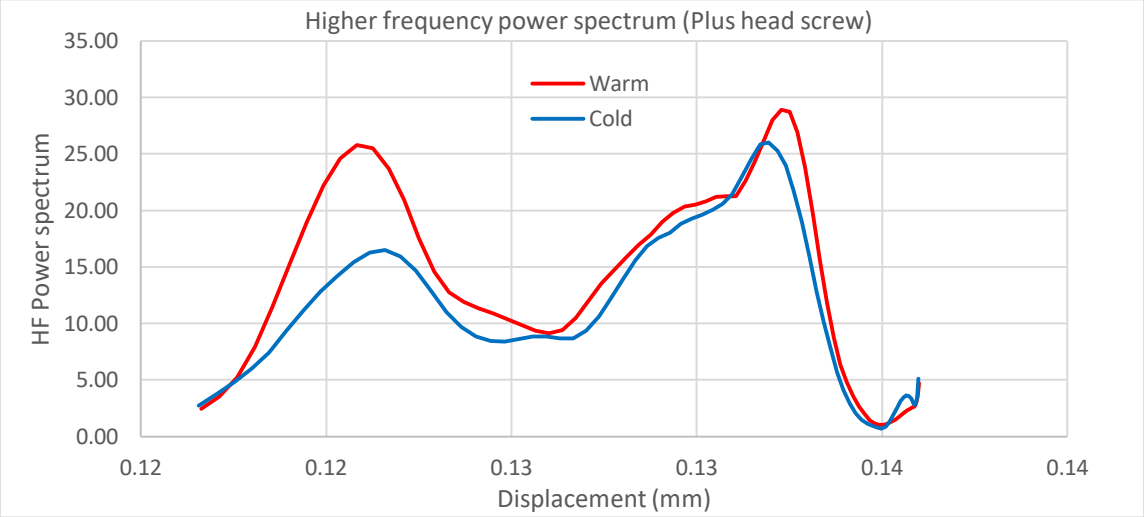


Figure 3-40 High frequency power spectrum of plus screw head.

The spectrum doesn't contain the peaks of the forging, which emphsizes that the signal with low frequency data can be trusted for the pattern assoscited to the behaviour of the forging. With the analysis of the above two cases few results obtained in formulation of methods are described below:

(1) The lower frequency bands could be a precise measure of the forging period rather than the higher frequency bands with most of noise content.

(2) The power pattern of low frequency could be compared with simulated data as a post check to re-check the signal pattern with preform peaks.

(3) The power spectrum obtained by the summation of low frequency band could be used as a feature extracting techniques to monitor the peculiar detailes associated with the forging periods with outlook of each preforms.

3.2.5 Specific kinetic energy absorbed

In this type of study the digital discreatized form of signals according to the sampling frequency is obtained from the machinery and converted into actual values as discussed above in the methods of signal processing from accelerometer. The variables obtained from the accelerometer signal includes the acceleration, velocity and position which is then used to further evaluate specific kinetic energy of forward and return stroke of the sliding block.These all variables can be monitored with on-line machinery with limits and any changes recorded could be taken as signal of alarm for the monitoring of the machinery. The one of the other considerations done for the in-depth machinery analysis in this section is the calculation of the absorbed specific kinetic enegry between the cold and warm cases. As it's already discussed above with the RMS value signals, there is a difference between cold and warm and cold could be considered faulty as observed through visual inspection during production process.The same critiria would be used to see the difference in energy absorbed, with the cases of cold, warm and empty cycle. The result of the full length cycles of the plus head screw is shown in Figure 3-41.

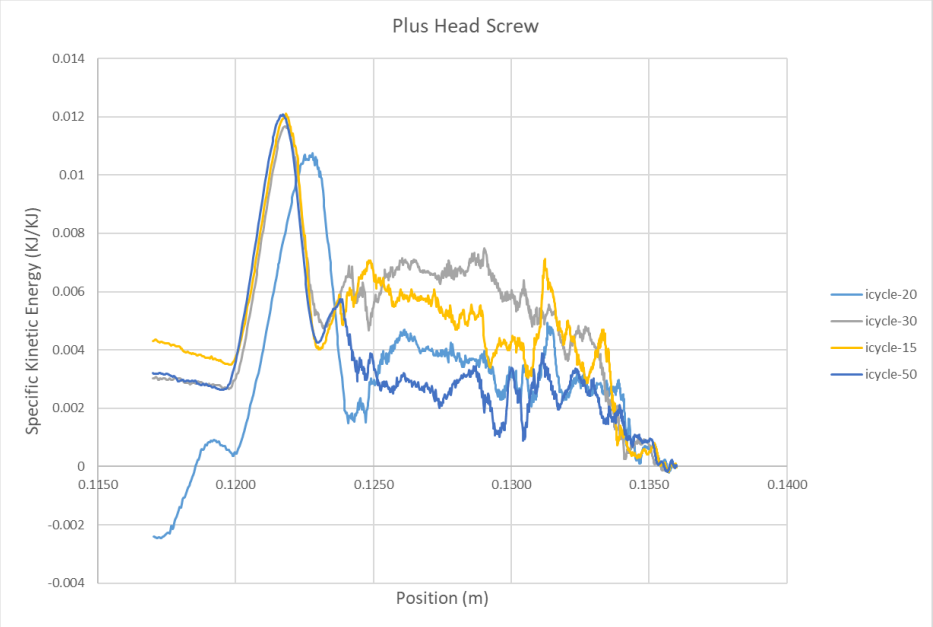


Figure 3-41 Forward cycles of plus screw head.

The warm, cold and empty conditions are observed in cycles of forged stroke in relation to displacement is computed to obtain the specific kinetic energy absorbed and comparitively studied to see the nature of each cases as shown in Figure 3-42.

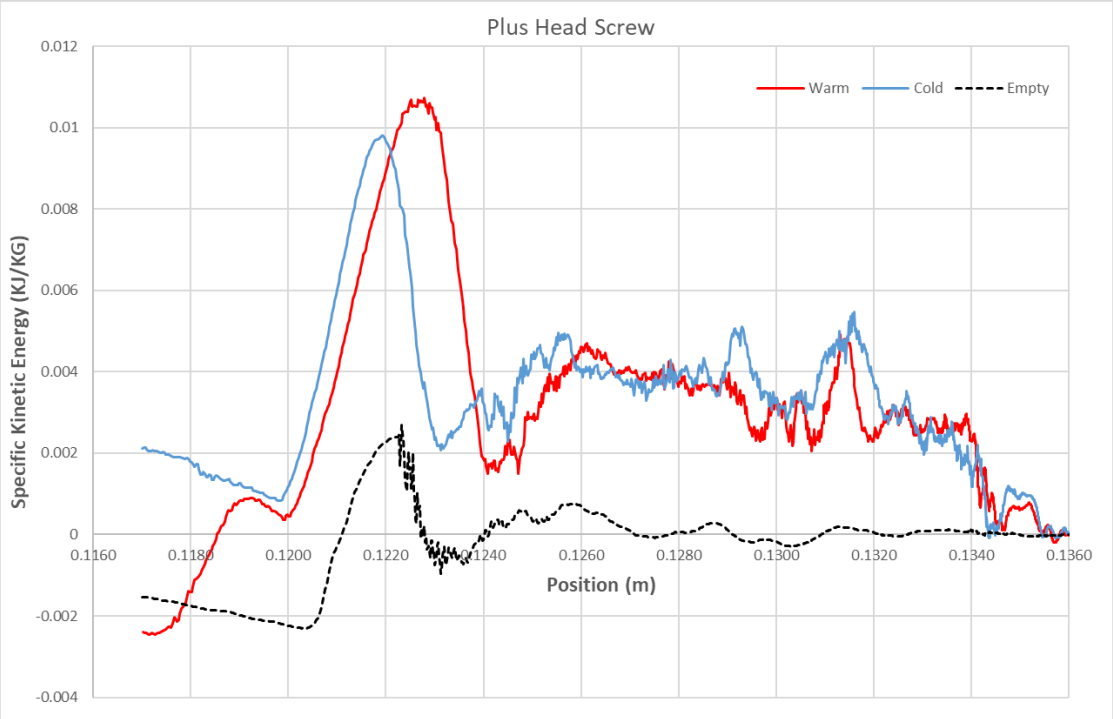


Figure 3-42 Analysis of plus screw head.

The curve of warm cycle varying with position shows higher values of absorbed kinetic energy with respect to cold and empty in most of the regions. The specific kinetic energy is half of the squared value of velocity. As the specific kinetic energy depending upon the velocity, the increase or decrease in velocity could be denoted with fluctuation related to changes in specific kinetic energy. As the plot depict the amount of kinetic energy absorbed, that is the difference of the ideal and actual kinetic energy implying lower the value of kinetic energy higher will be velocity and vice-versa.

From the above described content and the trend of the of warm, cold and empty curves it could be said that as the cold case is not considered efficient and

effecitive for manufactured part, the signature of this could be used as pattern for monitoring.

To check applicability of the phenomenon trend of kinetic energy in other parts, ASSY head screw design is also analyzed. The computed values of absorbed energy correspoding to varying position is displayed in Figure 3-43.

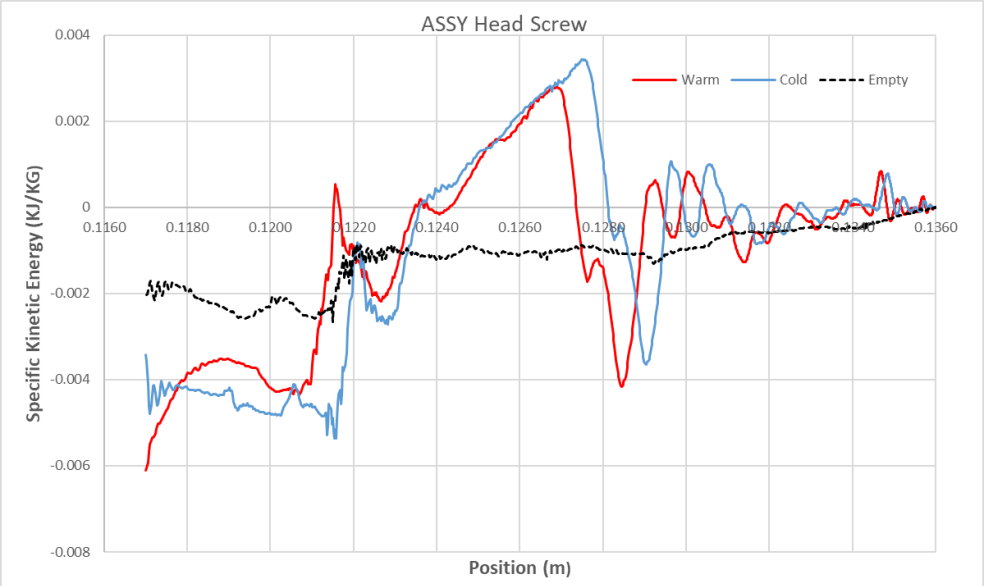


Figure 3-43 Analysis of ASSY head screw.

The trend is similarly follwed in ASSY head screw with first peak initiated by warm cycle and cold cycle always leading warm in both the scenarious concluding the signature move for the warm cycle.The result for the condition is discussed below:

(1) The signature trend of warm and cold cycle in both Plus head screw and ASSY head screw follows the same trend proving the feasiblity of the feature.

(2) It can be observed from the data that the warm cycle is delayed in both the cases with cold and empty.

(3) If analyzed for the specific part described in this paper, for both the parts the first peaks of warm seems to have high amplitude comparitive to cold

3.2.6 Comparative load analysis of acquired data vs simulated data

It can be illustrated from the plot in Figure 3-44 that simulation curve and experiment curve do not follow very precisely but follows the visible load variation pattern. The reasons of variation between actual and simulated plot are the factors like oscillations which exists in real cases but were not considered in numericaly simulated model. The significant load variation after 127 mm is clearly visible proving a trustable depicting trend for montoring analysis for a dynamic machinery. Similarly ASSY screw head is shown in Figure 3-45.

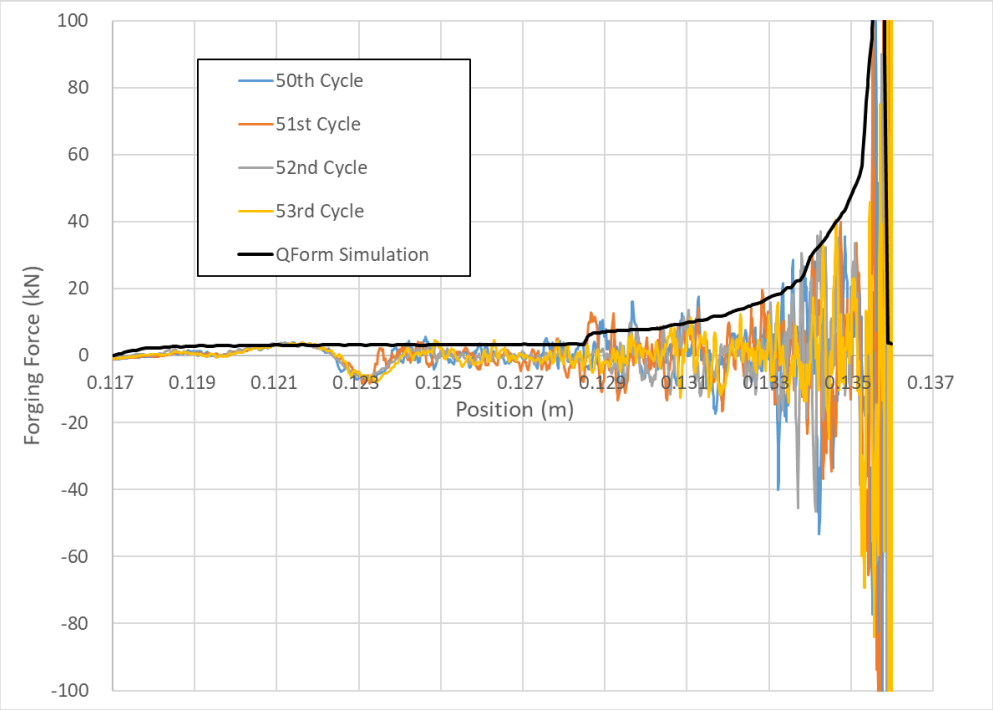


Figure 3-44 (a) Experimental Data vs (b) Simulated data.

The significant load variation after from beginning of steady load increase from 0.19 m is clearly visible proving a trustable depicting trend for monitoring with some oscillations and noises.

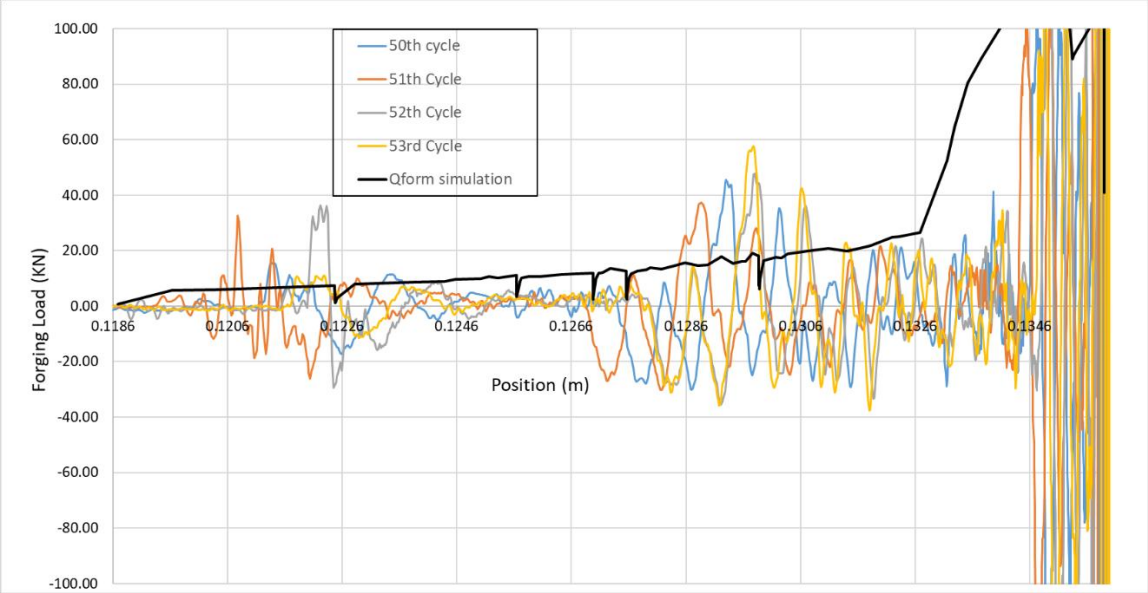


Figure 3-45 Experimental Data vs Simulated data

3.2.7 Comparative study between the repaired vs general machine

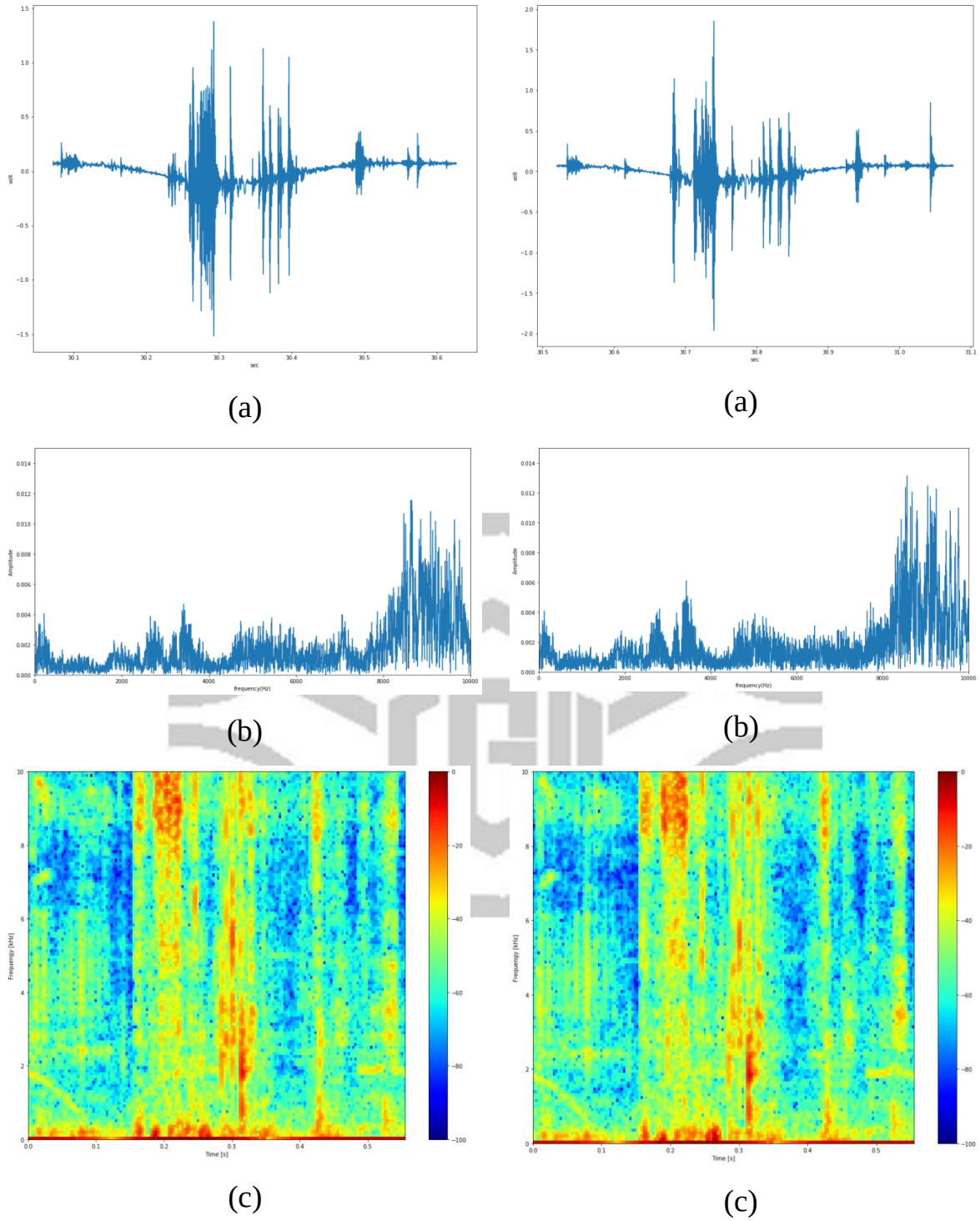
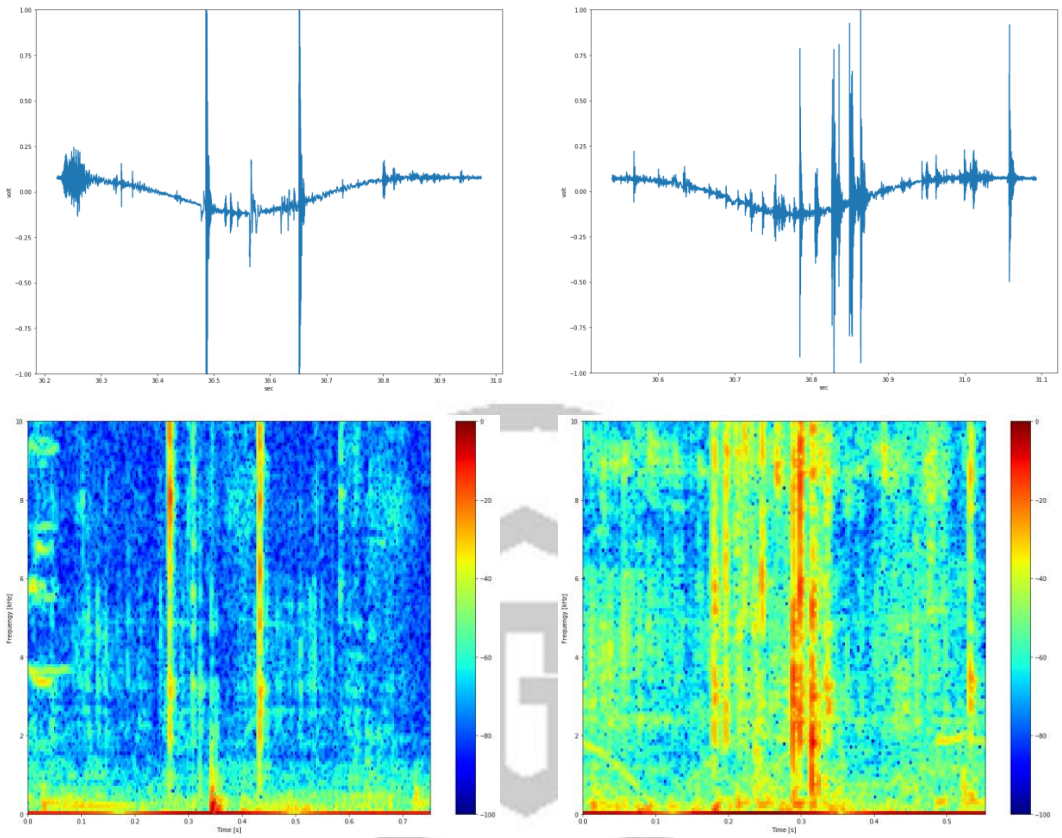


Figure 3-46 Plots for repaired (left) and general (right) machinery; (a) Time-waveform, (b) Spectrum, (c) Spectrogram.

The variation in time waveforms of repaired and the general machine is significantly visible in the cycle but when averaged cycle is computed for spectrogram the plot of both type of machineries looks almost same.

3.2.8 The analysis between healthy vs unhealty machinery



(a) (b)

Figure 3-47 The comparison between the plots of (a) healthy and (b) unhealthy machine. Time waveforms are shown in the upper row; spectrogram in the lower row.

The spectrogram and timewaveform data of unused brand new healthy machinery signal looks clean, wheras the spectrogram and timewaveform data of the unhealty machine always shows the noises along with the intense peaks which could be easily indictaed by the spectrograms.In this way spectrogram with the data of healthy machinery can be used as comparitive tool to see the variation of dynamic nature of machinery.

4. Conclusions

The research comprises of both experimental and simulation approach of analysis done on machineries named CBF-83S and CBP-65S with 4 and 2 forging stations producing two very distinct parts. The experimental analysis consists of vibrational analysis of the dynamic machinery equipped with accelerometer and optical encoder on the sliding block, whereas the simulation approach is done to get some characteristics behavior of the manufacturing process and parts. With the help of both approaches the methods which could be used to monitor the machinery and the production processes was developed. The parameters, trends and patterns resulting from the monitoring methods has been analyzed. The analysis leads to the following conclusions:

1. The RMS and mean vibrational value of each cycle follows a trend when analyzed for warm and cold condition of the machinery.
2. The spectrum and spectral plot of the vibrational data shows the forging feature that could be used for the tracking and tracing of the ongoing variation of the forging signals.
3. The low-frequency power spectrum obtained by short time Fourier transform (STFT) of acceleration varying with time shows obvious peaks associated with each forging stations.
4. The absorbed specific kinetic energy and force of the sliding block computed by velocity-displacement plot of defined cycles shows good recognizing features for empty, warm, and cold condition machinery.
5. The simulated load curves when compared with various experimental data curves shows similar featuring anomalies that could be considered as the monitoring parameters.

The trends and pattern obtained from the experimental data like low and high frequency power spectrum, absorbed specific energy along with simulations results like load curve trends could be used as input values of neural networks consisting of designed hidden layers with a defined output specifying the good or bad health of the machinery [12].

With the methods and the results proving the feature extraction capability the discussed methods in future could be used for training various deep learning models with channelized algorithms for condition monitoring as well as prognosis.



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Appendix A

The Python code for processing encoder signals is listed below:

```
import csv
import matplotlib.pyplot as plt
import numpy as np
from skimage import util
time=[]
acce=[]
posi=[]
with open('20180814EmptH-01.csv') as csvfile:
    readCSV = csv.reader(csvfile, delimiter=',')
    for row in readCSV:
        time.append(float(row[0]))
        acce.append(float(row[1]))
        posi.append(float(row[2]))
po = []
pi=[]
vmin = -2.53
dv = (0 - vmin) / 15
lp = len(posi)
for i in range(lp):
    pi=int((posi[i]-vmin) / dv + 0.5)
    #print('pi',pi)
    if pi > 15:
        pi = po[i-1]
    po.append(pi)
print(po)
pp = []
edge = []
ppmin=100
istart=0
pp.append(po[0])
j = 0
count = 0
for i in range(1,lp):
    dp = po[i]-po[i-1]
    if dp != 0:
        if count == 1:
            edge.pop()
            po[i-1]=po[i-2]
            pp[i-1]=pp[i-2]
        dp = po[i]-po[i-1]
        if dp != 0:
            edge.append(i)
            if dp < -5:
                j += 1
                po[i] = 0
            elif dp > 5:
                j -= 1
                po[i] = 15
            elif po[i+1] == po[i-1]:
                po[i]=po[i-1]
        count = 0
    pp.append(po[i]+j*16)
    count += 1
```

Appendix B

The python code for processing accelerometer signals is listed below:

```
ppmin = min(pp)
nedge = len(edge)
for i in range(lp):
    pp[i] -= ppmin
cycle_edge=[]
ncycle = 0
res = 18 * 3.1415926 / 16 / 1000
R = 68.0 / 1000
Nedge=int(2 * R / res) * 2
x0 = 0.019
xedge = 1
for i in range(nedge):
    if pp[edge[i]] == xedge and pp[edge[i-1]] == (xedge - 1):
        cycle_edge.append(i)
        if i != 0:
            ncycle += 1
        print(i,edge[i],ncycle)
print(Nedge, ncycle)
print(cycle_edge[0],cycle_edge[ncycle-1])
sumacce = 0
j = 0
for i in range(edge[cycle_edge[0]],edge[cycle_edge[ncycle-1]]):
    sumacce += acce[i]
    j += 1
sumacce /= float(j)
for i in range(lp):
    acce[i] -= sumacce
lcycle=int((edge[cycle_edge[ncycle-1]]-edge[cycle_edge[0]])/ncycle)+M
```

Appendix C

The python code for finding actual acceleration from the biased and noisy acceleration signal:

```
icycle = 1
p=acce[edge[cycle_edge[icycle]]:edge[cycle_edge[icycle+1]]]
t=time[edge[cycle_edge[icycle]]:edge[cycle_edge[icycle+1]]]
nt = len(t)
pmean = 0.0
for i in range(nt):
    pmean += p[i]
pmean /= float(nt)
for i in range(nt):
    p[i] -= pmean
ub1 = 0.0
ub2 = 0.0
xc = []
xc.append(0)
for i in range(1,nt):
    ub1 = ub2
    ub2 += (p[i]+p[i-1])/2*(t[i]-t[i-1])
    xc.append(xc[i-1]+(ub1 + ub2)/2*(t[i]-t[i-1]))
T = t[nt-1] - t[0]
q= -xc[nt -1] / T
xp = []
xp.append(0)
for i in range(1,nt):
    xp.append(q*(t[i]-t[0])+xc[i])
xpmin = min(xp)
xpmax = max(xp)
gamma = 2*R/(xpmax - xpmin)
u0 = q * gamma
x = []
u = []
x.append(0)
u.append(u0)
for i in range(1,nt):
    u.append(u[i-1] + gamma * (p[i] + p[i-1]) / 2 * (t[i] - t[i-1]))
    x.append(x[i-1] + (u[i] + u[i-1]) / 2 *(t[i] - t[i-1]))
xmin = min(x)
xmax = max(x)
nt1 = nt/2
for i in range(nt):
    if x[i] == xmax:
        nt1 = i
    x[i] -= xmin
fig, ax = plt.subplots(figsize=(15,12))
ax.plot(x,u)
ax.set_xlabel('Position [m]')
ax.set_ylabel('Velocity [m/s]')
```

Appendix D

The Python code for spectrum and spectrogram:

```
M=256
win = np.hanning(M + 1)[:-1]
st = 64
rate = 20000
icycle = 0
t=time[edge[cycle_edge[icycle]]:edge[cycle_edge[icycle]]+lcycle]
p=acce[edge[cycle_edge[icycle]]:edge[cycle_edge[icycle]]+lcycle]
p1=np.array(p,dtype=float)
N = p1.shape[0]
L = N / rate
ti=np.array(t,dtype = float)
ai=np.array(p,dtype = float)
from scipy.fftpack import fft #FFT of time waveform
fs=20000
t=ti
p=ai
n=np.size(t)
fr=(fs/2)*np.linspace(0,1,n/2)
plt.subplot(2,1,1)
plt.plot(t,p)
plt.xlabel('time(s)'),plt.ylabel('acc(volts)')
x=fft(p)
x_m=abs(x[0:np.size(fr)])
slices = util.view_as_windows(p1, window_shape=(M,), step = st)
slices = slices * win
slices = slices.T
spectrum = np.fft.fft(slices, axis = 0)[:M // 2 +1:-1]
for icycle in range(1,ncycle-1):
    p=acce[edge[cycle_edge[icycle]]:edge[cycle_edge[icycle]]+lcycle]
    p1=np.array(p,dtype=float)
    slices = util.view_as_windows(p1, window_shape=(M,), step = st)
    slices = slices * win
    slices = slices.T
    spectrum1 = np.fft.fft(slices, axis = 0)[:M // 2 +1:-1]
    spectrum = np.add(spectrum,spectrum1)
factor = 1.0/(ncycle-1)
spectrum = np.multiply(spectrum,factor)
S = np.absolute(spectrum)
Smax = np.max(S)
print(np.max(S))
S1 = 20*np.log10(S/10)
f, ax = plt.subplots(figsize=(15,12))
im = ax.imshow(S1, origin='lower',
     cmap=plt.cm.jet,extent=(0, L, 0, rate/2/1000 ),vmin=-100,vmax=0)
ax.axis('tight')
ax.set_ylabel('Frequengy [kHz]')
ax.set_xlabel('Time [s]')
f.colorbar(im, ax=ax)
```

Appendix E

The Python code for summing the simulated load curve:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
ld = pd.read_csv("add aug load curve1.csv", sep=',')
ld.head()
dist1=ld['dist 1'].values.tolist()
load1=ld['Tool 1 - Load, kN'].values.tolist()
dist2=ld['dist 2'].values.tolist()
load2=ld['Tool 1 - Load, kN.1'].values.tolist()
addlist=dist1+dist2
leng=len(addlist)
dac=sorted(addlist)
lengt=len(dac)
print("leng_dac",lengt)
print("leng_addlist",leng)
dla1=dict(zip(dist1,load1))
dla2=dict(zip(dist2,load2))
d=[]
d.append(dla1)
d.append(dla2)
leng=len(d)
print(leng)
load_v = np.array([],dtype = int)
da_v=np.array([],dtype=int)
for i in range  (len(dac)):
    value=0.0
    for j in range(len(d)):
        for k in d[j]:
            if k != 0:
                kj=kk
                kk=k
                da=(dac[i])
                if kj<da and da<=kk:
                    value=value+(d[j][kk]-d[j][kj])/(kk-kj)*(da-kj)+d[j][kj]
            kk=k
      da_v=np.append(da_v,da)
      load_v=np.append(load_v,(value))
```